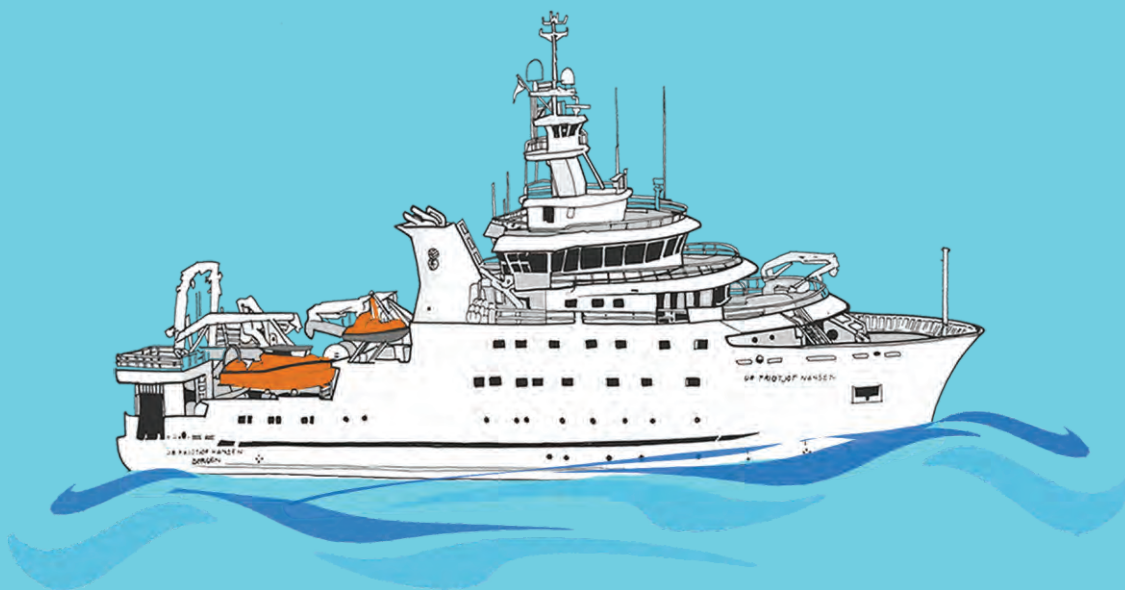


NORAD-FAO PROGRAMME
GCP/INT/690/NOR

CRUISE REPORTS *DR FRIDTJOF NANSEN*
EAF-Nansen/CR/2021/07



Cabo Verde ecosystem survey

20 November to 15 December 2021



Instituto Do Mar, Cabo Verde
Instituto Español de Oceanografía, Spain
Centre De Recherche Océanologique, Côte d'Ivoire

Institute of Marine Research, Bergen, Norway

THE EAF-NANSEN PROGRAMME (2017–2021)

The EAF-Nansen Programme “Supporting the Application of the Ecosystem Approach to Fisheries Management considering Climate and Pollution Impacts” supports partner countries and regional organizations in Africa and the Bay of Bengal to improve capacity for the sustainable management of their fisheries and other uses of marine and coastal resources through the implementation of the ecosystem approach to fisheries (EAF), taking into consideration the impacts of the climate and pollution.

The Programme is executed by the Food and Agriculture Organization of the United Nations (FAO) in close collaboration with the Institute of Marine Research (IMR) of Bergen, Norway, and funded by the Norwegian Agency for Development Cooperation (Norad). This Programme is the current phase (2017–2021) of the Nansen Programme which started in 1975.

The aim of the Programme is that sustainable fisheries improve food and nutrition security for people in partner countries. It builds on three pillars, Science, Fisheries Management and Capacity Development, and supports partner countries to produce relevant and timely evidence-based advice for management, to manage fisheries according to the EAF principles and to further develop their human and organizational capacity to manage fisheries sustainably. In line with the EAF principles, the Programme adopts a broad scope, taking into consideration a wide range of impacts of human activities and natural processes on marine resources and ecosystems including fisheries, pollution, climate variability and change.

A new state of the art research vessel, the *Dr Fridtjof Nansen*, is an integral part of the Programme. A comprehensive science plan, covering a broad selection of research areas, and directed at producing knowledge for informing policy and management decisions, guides the Programme’s scientific work.

The Programme works in partnership with countries, regional organizations, other United Nations agencies as well as other partner projects and institutions.

LE PROGRAMME EAF-NANSEN (2017–2021)

Le Programme EAF-Nansen «Soutenir l’application de l’approche écosystémique pour la gestion des pêches Le programme EAF-Nansen « Soutenir l’application de l’approche écosystémique pour la gestion des pêches compte tenu des impacts du climat et de la pollution » appui les pays partenaires et les organisations régionales en Afrique et dans le golfe du Bengale pour améliorer leur capacité de gestion durable de leurs pêcheries et d’autres usages de la mer ainsi que les ressources côtières, grâce à la mise en oeuvre de l’Approche écosystémique des pêches (AEP), en tenant compte des impacts du climat et de la pollution.

Le programme est exécuté par l’Organisation des Nations Unies pour l’alimentation et l’agriculture (FAO) en étroite collaboration avec l’Institut de recherche marine (IMR) de Bergen, en Norvège, et financé par l’Agence norvégienne de coopération au développement (Norad). Ce programme est la phase actuelle (2017-2021) du programme Nansen qui a débuté en 1975.

L’objectif du programme est que la pêche durable améliore la sécurité alimentaire et nutritionnelle des populations des pays partenaires. Il s’appuie sur trois piliers, la science, la gestion des pêches et le développement des capacités, et aide les pays partenaires à produire des avis pertinents et opportuns fondés sur des données factuelles pour la gestion, à gérer les pêcheries conformément aux principes de l’AEP et à développer davantage leur capacité humaine et organisationnelle à gérer durablement les pêches. Conformément aux principes de l’AEP, le programme adopte une large vision, prenant en considération un large éventail d’impacts des activités humaines et des processus naturels sur les ressources et les écosystèmes marins, y compris la pêche, la pollution, la variabilité et le changement climatique.

Un nouveau navire de recherche de pointe, le *Dr Fridtjof Nansen*, fait partie intégrante du programme. Un plan scientifique complet, couvrant un large éventail de domaines de recherche et visant à produire des connaissances pour éclairer les décisions de politique et de gestion, guide les travaux scientifiques du programme.

Le programme travaille en partenariat avec des pays, des organisations régionales, d’autres agences des Nations Unies ainsi que d’autres projets et institutions partenaires.

Required citation: **Michalsen, K., Ramos, V., Tiedemann, M., Zaera-Pérez, D., Bruck, S.A., Zamora-Terol, S., Cervantes, D., Jimenez, F.A.S. & N’Guessan, B.** 2022. *Cabo Verde ecosystem survey, 20 November to 15 December 2021*. NORAD-FAO PROGRAMME GCP/GLO/690/NOR. Cruise Reports *Dr Fridtjof Nansen*. EAF-Nansen/CR/2021/07.

CRUISE REPORTS *DR FRIDTJOF NANSEN*

Cabo Verde Ecosystem Survey

Leg 2

Cabo Verde

20 November to 15 December 2021

by

Kathine Michalsen¹, Vito Ramos², Maik Tiedemann¹, Diana Zaera-Pérez¹, Sarah Ann Bruck¹, Sara Zamora-Terol¹, David Cervantes¹, Francisca A. Salmeron Jimenez³, Benjamin N'Guessan⁴.

¹ Institute of Marine Research, PO Box 1870 Nordnes, N-5817 Bergen, Norway

² Instituto Do Mar, Cova D'Inglesa, C.P. 132, Mindelo, São Vincente - Cabo Verde

³ Instituto Español de Oceanografía, Spain

⁴ Centre De Recherche Océanologique, 29, Rue des pêcheurs, BPU 18, Abidjan-Treichville, Côte d'Ivoire

**Institute of Marine Research
Bergen, 2023**

CONTENTS

Abbreviations and acronyms	7
EXECUTIVE SUMMARY	9
RÉSUMÉ	11
1 INTRODUCTION	13
1.1 The EAF-Nansen Science Plan	13
1.2 The survey area	13
1.3 Survey objectives	15
1.4 Participation	15
1.5 Narrative	16
1.6 Survey design and effort	16
2 METHODS	21
2.1 Meteorology	21
2.2 Oceanography	21
2.2.1 Underway hydrographic sampling	21
2.2.2 Fixed hydrographic sampling	21
2.2.3 Water column ocean currents	22
2.2.4 Ocean acidification parameters	22
2.2.5 Nutrient samples	22
2.3 Plankton	22
2.3.1 Phytoplankton	22
2.3.2 Zooplankton	22
2.3.3 Ichthyoplankton	22
2.3.4 Jellyfish	23
2.4 Fishery resources	23
2.4.1 Pelagic resources	23
2.4.2 Demersal resources	25
2.5 Microplastics	27
2.6 Marine debris	27
2.7 Bottom habitat mapping	27
2.8 Vulnerable marine ecosystems	27
2.9 Top predators	27
2.10 Food safety	27
2.11 Sediment sampling	27
2.12 Additional sampling	27

3 RESULTS	29
3.1 Meteorology	29
3.2 Oceanography	29
3.2.1 Underway hydrographic sampling	29
3.2.2 Fixed hydrographic sampling	32
3.3 Plankton	34
3.3.2 Zooplankton	34
3.3.3 Ichthyoplankton	34
3.3.4 Jellyfish	35
3.4 Fishery resources	35
3.4.1 Pelagic resources	35
3.4.2 Demersal resources	37
3.4.3 Diversity	40
3.5 Microplastics	40
3.6 Marine debris	40
3.7 Bottom habitat mapping	40
3.7.1 Seafloor maps	40
3.7.2 Sediment composition	40
3.7.3 Benthic fauna	41
3.8 Vulnerable marine ecosystems	41
3.9 Top predators	41
3.10 Food safety	41
3.11 Sediment analysis	42
3.12 Additional sampling	42
3.13 Capacity building	44
3.14 Data recorded and collected samples	44
4 CONCLUDING REMARKS	45
5 REFERENCES	47
ANNEX 1. SAILING ORDERS	48
Appendices A1.1 to A1.14	48
ANNEX 2. DESCRIPTION OF INSTRUMENTS AND FISHING GEAR	69
ANNEX 3. LENGTH DISTRIBUTION OF MAIN SPECIES	73
ANNEX 4. MATURITY AND SEX	74
ANNEX 5. CAPACITY BUILDING	75
ANNEX 6. SAMPLE OVERVIEW	76
ANNEX 7. DATA COLLECTED AND OVERVIEW OF PROCEDURES FOR DATA HAND-OVER TO PARTNERS	79

Abbreviations and acronyms

ADCP	acoustic Doppler current profile
CTD	conductivity, temperature and depth
CRO	Centre De Recherche Océanologique (Centre for Ocean Research)
CVOO	Cabo Verde Ocean Observatory
DIC/TA	dissolved inorganic carbon/total alkalinity
EAF	ecosystem approach to fisheries
FAO	Food and Agriculture Organization of the United Nations
IEO	Instituto Español de Oceanografía (Spanish Institute of Oceanography)
IMar	Instituto Do Mar (Institute of the Sea)
IMR	Institute of Marine Research
LADCP	lowered acoustic Doppler current profiler
LSSS	Large-Scale Survey System
NASC	nautical area scattering coefficient
NEC	North Equatorial Current
NECC	North Equatorial Counter Current
NM	nautical mile
Norad	Norwegian Agency for Development Cooperation
OMZ	oxygen minimum zone
TS	target strength
TSG	thermosalinograph
UTA	Universidade Técnica do Atlântico (Atlantic Technological University)

EXECUTIVE SUMMARY

The survey plan for 2021 had to be substantially reduced because of the COVID-19 pandemic and only two surveys were conducted by the research vessel *Dr Fridtjof Nansen* in 2021. These were a dedicated bottom habitat environmental monitoring survey off Mauritania and Senegal and an ecosystem survey in Cabo Verde. The survey in Cabo Verde covered shallow water (about 20 m depth) to the upper slope (about 1 000 m depth). The survey started and ended in Las Palmas, Spain.

Cabo Verde is an area with coastal upwelling, high biodiversity and part of the large-scale oxygen minimum zone band/area. This survey had multiple objectives and consequently it was multidisciplinary. Physical, chemical and biological (plankton) oceanographic sampling, as well as sampling for microplastics, was carried out at a series of fixed stations, while at the same time oceanographic data were also collected underway. Fish resources were assessed using acoustic technology complemented by trawling. The survey used standardized methods to ensure comparability. This report presents the methodology and preliminary results from samples and data that were analysed on board the survey off the coast of Cabo Verde from 19 November to 15 December 2021.

The survey tracks and the sampling frequency were based on past demersal and pelagic surveys conducted by the RV *Dr Fridtjof Nansen* in the area. Demersal trawling was carried out during daytime on predetermined stations and depth strata. Stations were placed at approximately the same positions as during the survey in 2011, to ensure comparability of catch rates of demersal fish between the two surveys.

Midwater trawling for acoustic target identification was conducted in areas of increased echo backscattering, as determined from the echo sounders. For acoustic biomass estimation results to be comparable to the survey in 2011, two areas of the Cabo Verde island complex were used, namely Area 1 and Area 2.

Altogether, 35 midwater trawl hauls were carried out to identify acoustic targets during the survey. Twenty-seven were bottom trawls equipped with floats to allow fishing in the water column in shallow depths, and 18 were with pelagic trawls. A total of 101 conductivity, temperature and depth (CTD) deployments were carried out to describe the hydrography of the survey area, and 128 plankton stations were sampled with various types of plankton sampling gears to study the planktonic community. Samples of commercially important fish species were collected for food nutrition and toxicology tests, while *Sargassum* spp. aggregations were sampled for genetic analyses.

The information presented below is a summary of the results of the data analysed during the survey. Some samples and data have been transported to research institutes in the region, and farther afield (notably IMR in Bergen, Norway). Samples will be analysed in close cooperation with partner institutions and the results will be reported separately. The resulting datasets will support research conducted in the framework of the EAF-Nansen Programme's Science Plan.

The survey in Cabo Verde had multiple objectives, of which all were successfully achieved, following the Sailing Orders. Due to some adverse weather conditions some stations had to be dropped and some additional sampling was carried out, to obtain better biological and oceanographic information in specific areas.

The acoustic recordings were low for the whole area coverage and – as in 2011 – no clupeids were found. On a single occasion, the trawl was deployed based on strong acoustic signals. This turned out to be a very dense aggregation of mesopelagic fish. The biomass of pelagic fish is assessed at 6 430 tonnes, exclusively belonging to the PEL2 group (specific taxa within the Carangidae, Scombridae and Sphyrænidae families). This is more than double the roughly 3 000 tonnes found during the 2011 survey. The geographical distribution of some of the pelagic fish species did vary slightly between the two surveys. In 2021, pelagic fish mainly occurred south of the northern islands while they were

distributed north of the same islands in 2011. The recordings of PEL2 fish south of Boa Vista occurred in both surveys. Species dominance varied in the survey area. Species like mackerel scad (*Decapterus macarellus*), round scad (*Decapterus punctatus*) and false scad (*Caranx rhonchus*, previously *Decapterus rhonchus*) accounted for the main pelagic species found during the 2011 Nansen survey in Cabo Verde. During the 2021 survey, only a few individuals of *Decapterus punctatus* were caught and none belonging to the other two species. However, another commercially important species, *Selar crumenophthalmus* was found in some of the catches. The catch rates (density in tonnes per NM²) of the demersal trawl catches were similar in 2021 and 2011, however the composition of the catches was different, with fewer commercially important species and fewer macroinvertebrates like shrimps, scallops and molluscs. Only two (*Pseudupeneus prayensis* and *Lithognathus mormyrus*) of the demersal species monitored by the Instituto do Mar (Institute of the Sea, IMar) were caught during the 2021 survey, of which one was represented by only a few juvenile specimens. This is in accordance with the trends observed in the commercial catch statistics recorded by IMar, where long-lived species are almost absent from catches while species with a shorter life expectancy seem to dominate now. The change in the composition of demersal species points out the need for both close monitoring of the landings and the introduction of appropriate management measures to ensure the future of the stocks.

Microplastic-like items were found in five of the 26 stations where the Manta net was deployed and these were sent to IMR for further analysis.

Marine debris was registered at 11 trawl stations, mostly with only one piece of litter per station. Marine debris (remains from fishing equipment and other human activities) was found in trawl catches as deep as 150 m.

Two commercially important pelagic species were preselected for nutritional content and food safety (microplastics and parasite) analysis, namely *Auxis thazard* and *Selar crumenophthalmus*, and collected for this purpose when caught. None of these species were caught at more than one station and therefore there are no replicates, as requested in the Sailing Orders. Additionally, three species (*Antigonias capro*, *Ariomma melanum* and *Maurolicus* spp.) with high ecosystem importance were opportunistically sampled at one station each.

High biodiversity of fish was recorded in some of the catches. Each of the two trawl hauls at 800 m contained more than 80 distinct taxa. In total, over 400 taxa were identified to species level during the survey and more than 7 700 individuals were measured and weighed in the fish lab.

RÉSUMÉ

Le plan de campagne pour 2021 a dû être considérablement réduit en raison de la pandémie de Covid19 et seules deux campagnes ont été réalisées par le navire de recherche (N/R) *Dr Fridtjof Nansen* en 2021. Il s'agissait d'une campagne de suivi environnemental de l'habitat benthique au large de la Mauritanie et du Sénégal et d'une étude de l'écosystème au Cabo Verde. La campagne réalisée au Cabo Verde a sondé les eaux peu profondes (environ 20 m de profondeur) jusqu'au talus supérieur (environ 1 000 m de profondeur). La campagne a débuté et s'est terminée à Las Palmas, en Espagne.

Le Cabo Verde est une zone d'upwelling côtier, de grande biodiversité et fait partie de la bande d'oxygène minimale à grande échelle. Cette campagne avait des objectifs multiples et était donc multidisciplinaire. Un échantillonnage océanographique physique, chimique et biologique (plancton), ainsi qu'un échantillonnage des microplastiques, ont été réalisés dans une série de stations fixes, tandis que des données océanographiques étaient également collectées en mer. Les ressources halieutiques ont été évaluées à l'aide de la technologie acoustique complétée par le chalutage. Des méthodes standardisées ont été utilisées afin de garantir la comparabilité des résultats. Ce rapport présente la méthodologie et les résultats préliminaires des échantillons et des données qui ont été analysés à bord du navire au large de la côte du Cabo Verde du 19 novembre au 15 décembre 2021.

Les trajectoires et la fréquence d'échantillonnage ont été basées sur les campagnes démersales et pélagiques antérieures menées par le N/R *Dr Fridtjof Nansen* dans la zone. Le chalutage démersal a été effectué pendant la journée sur des stations et des strates de profondeur prédéterminées. Les stations ont été placées approximativement aux mêmes endroits que lors de la campagne de 2011, afin de garantir la comparabilité des taux de capture des poissons démersaux entre les deux campagnes.

Le chalutage pélagique pour l'identification des cibles acoustiques a été effectué dans les zones de rétrodiffusion accrue de l'écho, telles que déterminées par les échosondeurs. Pour que les résultats de l'estimation acoustique de la biomasse soient comparables à ceux de la campagne de 2011, deux zones des îles du Cabo Verde ont été utilisées, à savoir la zone 1 et la zone 2.

Au total, 35 traits de chalut pélagique ont été effectués pour identifier les cibles acoustiques au cours de la campagne. Vingt-sept étaient des chaluts de fond équipés de flotteurs pour permettre la pêche dans la colonne d'eau à faible profondeur, et 18 étaient des chaluts pélagiques. Un total de 101 déploiements de la sonde CTD (mesure de la conductivité, température et profondeur) ont été effectués pour décrire l'hydrographie de la zone d'étude, et 128 stations de plancton ont été échantillonnées avec différents types d'engins d'échantillonnage du plancton afin d'étudier la communauté planctonique. Des échantillons d'espèces de poissons commercialement importantes ont été prélevés pour des tests de nutrition et de toxicologie, tandis que des agrégations de *Sargassum* spp. ont été échantillonnées pour des analyses génétiques.

Les informations présentées ci-dessous sont une synthèse des résultats des données analysées au cours de la campagne. Certains échantillons et données ont été envoyés à des instituts de recherche de la région ou vers d'autres laboratoires plus éloignés (notamment l'IMR à Bergen, en Norvège). Les échantillons seront analysés en étroite collaboration avec les institutions partenaires et les résultats seront présentés séparément. Les ensembles de données qui en résulteront contribueront à la recherche menée dans le cadre du plan scientifique du Programme EAF-Nansen.

La campagne réalisée au Cabo Verde avait de multiples objectifs, qui ont tous été atteints avec succès, conformément aux ordres de navigation. En raison de conditions météorologiques défavorables, certaines stations ont dû être abandonnées et un échantillonnage supplémentaire a été réalisé afin d'obtenir de meilleures informations biologiques et océanographiques dans des zones spécifiques.

Les enregistrements acoustiques ont été faibles pour l'ensemble de la zone couverte et, comme en 2011, aucun clupéidé n'a été trouvé. À une seule occasion, le chalut a été déployé sur la base de signaux

acoustiques forts. Il s'est avéré qu'il s'agissait d'une agrégation très dense de poissons mésopélagiques. La biomasse de poissons pélagiques a été évaluée à 6 430 tonnes, appartenant exclusivement au groupe PEL2 (taxons spécifiques des familles Carangidae, Scombridae et Sphyraenidae). Ce chiffre représente plus du double des quelque 3 000 tonnes trouvées lors de la campagne de 2011. La répartition géographique de certaines espèces de poissons pélagiques a légèrement varié entre les deux campagnes. En 2021, les poissons pélagiques se trouvaient principalement au sud des îles du nord, alors qu'ils étaient répartis au nord de ces mêmes îles en 2011. Les enregistrements de poissons PEL2 au sud de Boa Vista ont eu lieu dans les deux campagnes. Les espèces dominantes ont varié dans la zone d'étude. Des espèces telles que la comète maquereau (*Decapterus macarellus*), la comète quiaquia (*Decapterus punctatus*) et la comète coussut (*Caranx rhonchus*, anciennement *Decapterus rhonchus*) sont les principales espèces pélagiques trouvées lors de la campagne Nansen de 2011 au Cabo Verde.

Au cours de la campagne de 2021, seuls quelques individus de *Decapterus punctatus* ont été capturés et aucun appartenant aux deux autres espèces. Cependant, une autre espèce commercialement importante, *Selar crumenophthalmus*, a été trouvée dans certaines des captures. Les taux de capture (densité en tonnes par NM²) des captures au chalut démersal étaient similaires en 2021 et 2011, mais la composition des captures était différente, avec moins d'espèces commercialement importantes et moins de macroinvertébrés comme les crevettes, les coquilles Saint-Jacques et les mollusques.

Seules deux des espèces démersales (*Pseudupeneus prayensis* et *Lithognathus mormyrus*) suivies par l'Instituto do Mar (Institut de la mer, IMar) ont été capturées lors de la campagne de 2021, dont l'une n'était représentée que par quelques spécimens juvéniles. Ceci est conforme aux tendances observées dans les statistiques de captures commerciales enregistrées par l'IMar, où les espèces à longue durée de vie sont presque absentes des captures tandis que les espèces à plus courte espérance de vie semblent dominer maintenant. L'évolution de la composition des espèces démersales met en évidence la nécessité d'un suivi étroit des débarquements et de l'introduction de mesures de gestion appropriées pour assurer l'avenir des stocks.

Des objets de type microplastiques ont été trouvés dans cinq des 26 stations où le filet Manta a été déployé et ils ont été envoyés à l'IMR pour une analyse plus approfondie.

Des débris marins ont été enregistrés dans 11 stations de chalutage, la plupart du temps avec un seul déchet par station. Des débris marins (restes d'engins de pêche et d'autres activités humaines) ont été trouvés dans les prises de chalut jusqu'à 150 m de profondeur.

Deux espèces pélagiques d'importance commerciale ont été présélectionnées pour l'analyse du contenu nutritionnel et de la sécurité sanitaire (microplastiques et parasites), à savoir *Auxis thazard* et *Selar crumenophthalmus*, et collectées à cette fin lors de leur capture. Aucune de ces espèces n'a été capturée sur plus d'une station et il n'y a donc pas de répliqués, comme demandé dans les ordres de navigation. En outre, trois espèces (*Antigonia capro*, *Ariomma melanum* et *Maurolicus* spp.) très importantes pour l'écosystème ont fait l'objet d'un échantillonnage opportuniste dans une station chacune.

Une grande biodiversité de poissons a été enregistrée dans certaines des captures. Chacun des deux traits de chalut à 800 m contenait plus de 80 taxons distincts. Au total, plus de 400 taxons ont été identifiés au niveau de l'espèce au cours de la campagne et plus de 7 700 individus ont été mesurés et pesés dans le laboratoire des poissons.

1 INTRODUCTION

1.1 The EAF-Nansen Science Plan

The research activities under the EAF-Nansen Programme are guided by and provide direct input to the EAF-Nansen Science Plan. The Science Plan is intended to ensure good scientific use of the wealth of data generated by the RV *Dr Fridtjof Nansen* and other related data, addressing key research questions in support of the ecosystem approach to fisheries management (EAF). The Science Plan covers 11 research themes, presented in Figure 1.

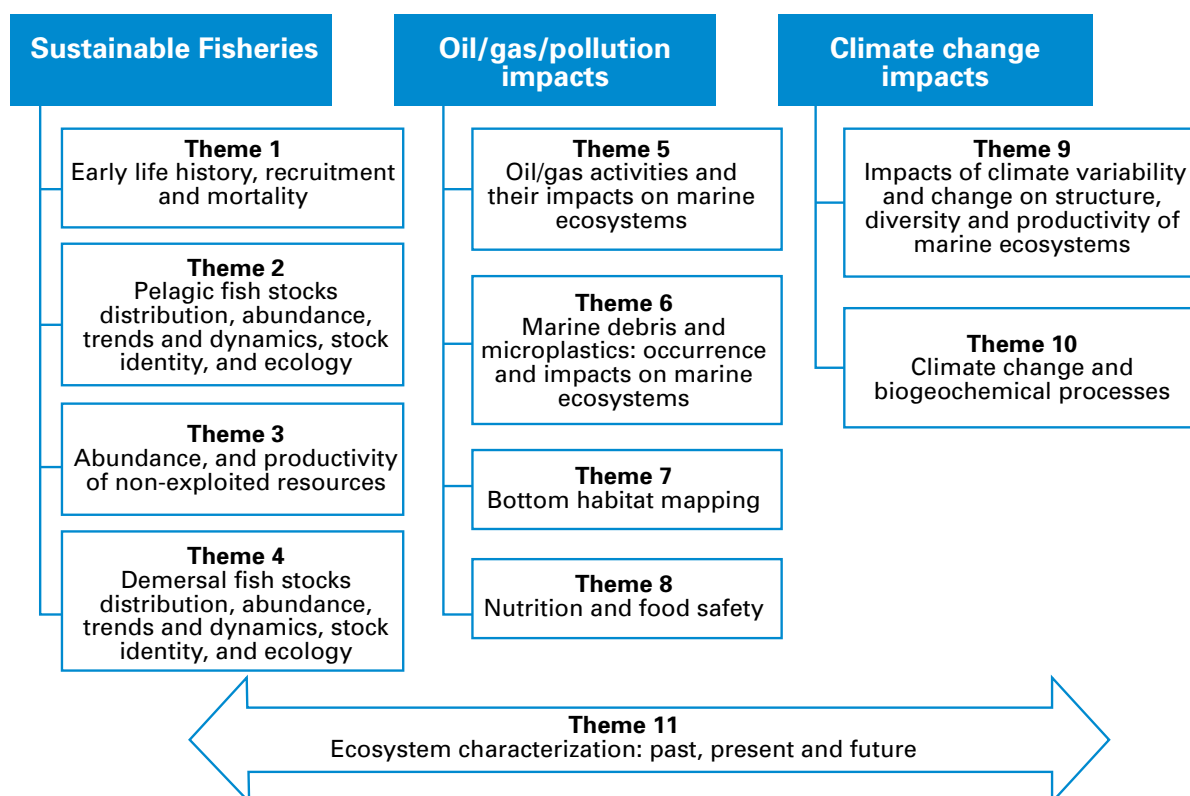


Figure 1. Research themes of the EAF-Nansen science plan

1.2 The survey area

The survey plan for 2021 had to be substantially reduced because of the COVID-19 pandemic and only two surveys were conducted by the RV *Dr Fridtjof Nansen* in that year. The survey programme for 2021 included a dedicated bottom habitat environmental monitoring survey outside Mauritania and Senegal and an ecosystem survey of Cabo Verde. The survey in Cabo Verde covered waters from shallow depths (about 20 m) to the upper slope (about 1 000 m) (Figure 2). The survey started and ended in Las Palmas, Spain.

The Cabo Verde archipelago is situated on the west African coast about 550 km west of Senegal. It consists of ten islands and several islets. The island group has a surface area of 4 033 km² and a coastline of 1 020 km. The continental shelf is small (5 394 km²) and the country has an exclusive economic zone of 734 265 km². There is some disagreement on the origins and age of Macaronesian¹ islands, but the eastern islands Sal and Boavista are older and much of the volcanic topography has been eroded by wind

¹ Macaronesia: biogeographical province or region commonly accepted in ecology comprising the Azores, Cabo Verde, Canaries, and Madeira islands.

and time, resulting in more sandy sediments. Cabo Verde is considered a biodiversity “hotspot” with a high degree of endemism (e.g. Merino *et al.*, 2002; Roberts *et al.*, 2002). The wide shelf regions can be found around Boavista and between Boavista and Maio.

The biogeographical unit Macaronesia is widely accepted in ecology, characterized by its relic laurisilva vegetation (Lloris *et al.*, 1991). In Cabo Verde, this area is considered a “branch” of the Equato-Guinean province with some connection to the Madeira District (the Canaries and Madeira islands primarily) (Lloris *et al.*, 1991). As referred to above, this is in complete agreement with the expected influence of the Canary Current and equatorial currents (both the North Equatorial Current [NEC] and North Equatorial Counter Current [NECC]) in immigration and colonization.

Cabo Verde is characterized by low precipitation. Marine resources are therefore of great significance and the fishery has large social and economic effects. It guarantees employment for a large part of the population, contributes to food security, and fish is the major source of animal protein for most of the people living on the islands. The fishing industry also contributes to the acquisition of foreign currency through exports.



Figure 2. Area surveyed by the RV *Dr Fridtjof Nansen* during the Cabo Verde 2021 survey

1.3 Survey objectives

The overall objectives of the survey were to assess the demersal and pelagic resources of the continental shelf and upper slope by determining their distribution and abundance, while also studying the oceanographic conditions, distribution of microplastics and marine debris, and measuring nutrient and contaminant levels in commercial fish. All the above were carried out with a capacity building and hands-on training in scientific sampling and reporting perspective.

Based on the above, the following activities were conducted (in a prioritized order) in accordance with the Sailing Orders (ANNEX 1):

- Demersal trawl hauls for assessing the distribution and biological parameters, and to provide catch rates of demersal resources.
- Pelagic trawling (alongside acoustic registrations for pelagic resources) for assessing the distribution of resources and biological parameters, and providing biomass estimates of pelagic resources.
- Standard biological sampling of priority species (including length, weight, sex and maturity).
- Sampling for nutrients, contaminants and the presence of microplastics for priority fish species.
- Standard sampling of environmental variables (salinity, temperature, dissolved oxygen, fluorescence, dissolved nutrients, pH, total alkalinity and chlorophyll *a*).
- Ocean current measurements.
- Sampling for phytoplankton, zooplankton, ichthyoplankton and microplastics.
- Underway pCO₂ measurements in surface water.
- Standard sampling for meteorological variables (wind direction and speed, air pressure, relative humidity, air temperature and solar radiation).
- Additional sampling at the Cabo Verde Ocean Observatory (CVOO) for hydrographic condition and plankton.

1.4 Participation

A total of 26 researchers and technicians from Cabo Verde, Côte d'Ivoire, Spain and Norway participated in the survey. The full list of the participants and their affiliations is provided in Table 1.

Table 1. List of participants, roles, affiliations and duration of onboard stay

No	Leg	Participant	Role	Sex	Affiliation	Country	Period
1	Leg 2	Kathrine Michalsen	Cruise leader	F	IMR	Norway	19.11–15.12
2	Leg 2	Sarah Ann Bruck	Fish Team leader	F	IMR	Norway	19.11–15.12
3	Leg 2	Diana Zaera-Pérez	Fish Team leader	F	IMR	Norway	19.11–15.12
4	Leg 2	Sara Zamora Terol	Plankton Team leader	F	IMR	Norway	19.11–15.12
5	Leg 2	David Cervantes	Chemical oceanography	M	IMR	Norway	19.11–15.12
6	Leg 2	Olaf J. Sørås	Chief instruments engineer	M	IMR	Norway	19.11–15.12
7	Leg 2	Jori Neteland-Kyte	Instruments engineer	F	IMR	Norway	19.11–15.12
8	Leg 2	Vito Ramos	Co-cruise leader	M	IMar	Cabo Verde	19.11–15.12
9	Leg 2	Sandra Correia	Fish Team	F	IMar	Cabo Verde	19.11–15.12
10	Leg 2	Alciany Luz	Fish Team	F	IMar	Cabo Verde	19.11–15.12
11	Leg 2	Carla Santos	Fish Team	F	IMar	Cabo Verde	19.11–15.12
12	Leg 2	Ailton Rocha	Fish Team	M	IMar	Cabo Verde	19.11–15.12
13	Leg 2	Anibal Medina	Chemical oceanography	M	IMar	Cabo Verde	19.11–15.12
14	Leg 2	Katelene Delgado	Fish Team	F	IMar	Cabo Verde	19.11–15.12
15	Leg 2	Rui Freitas	Fish Team	M	UTA	Cabo Verde	19.11–15.12

No	Leg	Participant	Role	Sex	Affiliation	Country	Period
16	Leg 2	Valéria Lopes	Fish Team	F	UTA	Cabo Verde	19.11–15.12
17	Leg 2	Pércles Silva	Chemical oceanography	M	IMar	Cabo Verde	19.11–15.12
18	Leg 2	Ivanice Silva	Chemical oceanography	F	IMar	Cabo Verde	19.11–15.12
19	Leg 2	Dario Évora	Physical oceanography	M	IMar	Cabo Verde	19.11–15.12
20	Leg 2	Elizandro Rodrigues	Biological oceanography	M	IMar	Cabo Verde	19.11–15.12
21	Leg 2	Keider Neves	Biological oceanography	M	Biosf	Cabo Verde	19.11–15.12
22	Leg 2	Marcia Costa	Biological oceanography	F	IMar	Cabo Verde	19.11–15.12
23	Leg 2	Chrislainne Alves	Biological oceanography	F	IMar	Cabo Verde	19.11–15.12
24	Leg 2	Francisca A. Salmeron Jimenez	Fish Team; taxonomist	F	IEO	Spain	19.11–15.12
25	Leg 2	Benjamin N'Guessan	Physical oceanographer	M	CRO	Côte d'Ivoire	19.11–15.12
26	Leg 2	Maik Tiedemann	Training (cruise leader)	M	IMR	Norway	19.11–15.12

Notes: Abbreviations: IMR – Institute of Marine Research, Norway; IMar – Instituto Do Mar, Cabo Verde, IEO – Instituto Español de Oceanografía (Spanish Institute of Oceanography), Spain; CRO – Centre De Recherche Océanologique (Centre for Ocean Research), Côte d'Ivoire; UTA – Universidade Técnica do Atlântico (Atlantic Technological University), Biosfera, Cabo Verde.

1.5 Narrative

The vessel departed from Las Palmas, Spain at 11.00 UTC on 19 November 2021 and sampling commenced at the first hydrographic station (CVOO) northeast of Cabo Verde on 22 November at 15.30. After completing sampling at the CVOO station, the vessel proceeded to the eastern tip of São Nicolau, where the acoustic coverage was initiated on 23 November, heading westwards towards the Nola Bank (consisting of two seamounts). The coverage consisted of an acoustic sampling grid that had a preset transect spacing, covering the shelf (>20 m) and slope up to the 1 000 m depth contour. In parallel with the pelagic coverage, environmental stations, demersal and pelagic trawl stations were also carried out, as described in the Sailing Orders. When arriving at the Nola Seamount on 24 November, a process study (see ANNEX 1) was conducted with sampling at predefined stations with the multinet Mammoth and CTD casts. At sunset of the same day, the acoustic coverage of the Nola Seamount was carried out together with CTD stations (with water sampling at predefined locations and to the maximum depth) and acoustic Doppler current profile (ADCP) recording (around the western Nola Seamount) until sunrise. Afterwards, the multinet sampling continued until the end of daylight on 25 November. The vessel headed towards the southern part of Area 2 (Santo Antão, São Vicente, Santa Luzia and São Nicolau) to continue the acoustic coverage and sampling of the demersal resources. On 26 November, a first attempt to calibrate the echo sounders was carried out, before resuming the sampling again. Area 2 was fully covered on 27 November.

The vessel then moved east, towards Sal. On the way, three environmental stations were carried out. The acoustic coverage of Sal started on 28 November and Area 1 (Sal, Boa Vista, Maio, Santiago, Fogo and Brava) was covered. The last environmental transect between Area 1 and 2 started on 6 December and ended a day later. The vessel then headed towards Nola Bank again, to conduct a multinet sampling transect from southwest to northeast along the two Nola Seamounts during the night. One bottom trawl station and one haul with the beam trawl were also conducted on top of each of the two Nola Seamounts (the bottom trawl station on the southwestern seamount and the beam trawl on the northeastern seamount).

On 9 December, all sampling was concluded and the vessel steamed to Las Palmas, where it arrived at 09.00 on 13 December. Days spent in the territorial waters of Cabo Verde are shown in Table 2. The weather was favourable during the first survey days, but when the vessel headed towards Area 1, the wind increased and the speed had to be reduced. Only one bottom trawl station (east of Sal) could not be conducted due to unfavourable weather conditions. In total, 2 049.2 NM were sailed.

1.6 Survey design and effort

The survey tracks and the sampling frequency followed the agreed survey design described in the Sailing Orders and was based on past demersal and pelagic surveys of the R/V *Dr. Fridtjof Nansen* in the area.

Demersal trawling was carried out during day on predetermined positions within predetermined depth strata (Figure 4). Trawls were placed at approximately the same positions as they were during the survey in 2011 to ensure, at a certain level, comparability of catch rates of demersal fish between the two surveys.

Acoustic trawls for target identification were conducted using pelagic trawls in areas of increased echo backscattering, as determined from the echo sounders. The acoustic densities were then allocated to main pelagic species groups. The design for the pelagic fish abundance estimate consisted of transects extending from about 20 m to 1 000 m. In an effort to compare with the survey in 2011, Area 1 and 2 of the Cabo Verde island complex were used. To avoid overestimation of acoustic backscatter, randomized zig-zag transects were carried out on each side of island complexes with the distance between the acoustic transects defined based on the equal average coverage probability method (Harbitz, 2019), taking into account the desired time to be spent on each stratum, the speed of the vessel and the optimal degree of coverage aimed for. In total, 14 strata were surveyed (Figure 2). The abundance of pelagic fish in each of the strata has been summed to provide one estimate for Area 1 and one for Area 2. The software StoX (Johnsen *et al.*, 2019; www.hi.no/en/hi/forskning/projects/stox) was used for the calculation of the abundance indices of pelagic resources.

Hydrographic variables were measured at every bottom trawl station and at predefined environmental stations (Figure 5) called super-stations. At these stations, more elaborate sampling was carried out, including CTD with water samples at standard depths for chemical and nutrient analyses (Figure 3). Plankton (phytoplankton, zooplankton and ichthyoplankton) samples were collected using various sampling gear (phytoplankton net, WP-2 standard net, Bongo net and Manta trawl), employing vertical, horizontal or oblique tows (depending on the gear). Microplastic samples were also taken by means of the Manta trawl (Figure 3). A CTD and a vertical multinet (type Midi) deployment were carried out at the Cabo Verde monitoring station CVOO.

Plankton sampling was conducted at 30 predefined super-stations throughout the survey area (figures 6 and 7). Stations were located at similar positions as in 2011, ranging from 30 m to 4 000 m. South of São Vicente an extra station was sampled to establish a reference point that could be used to monitor potential changes in the area due to planned industrial activity (tuna farming). Depth stratified sampling of ichthyoplankton was also carried out using the multinet (type Mammoth) at selected stations of special hydrographic interest around the Nola Bank (Figure 7).

The survey effort in each area considered (Area 1 and 2) is summarized in Table 2.

Environmental transect sampling

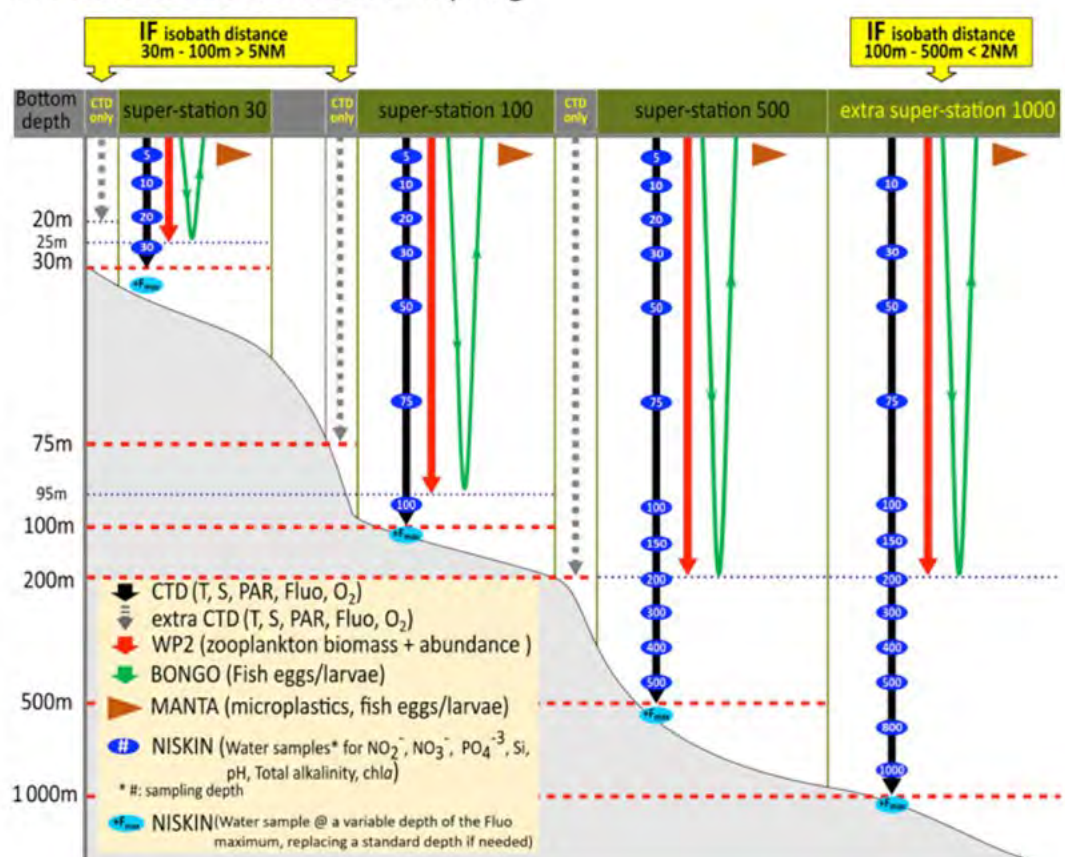


Figure 3. Sampling along environmental transects at predefined stations, called super-stations

Table 2. Survey effort in number of sampling stations (total and by subarea). Number of: BT (demersal), PT (pelagic) and BM (beam) trawl hauls; CTD; WP2; Bongo; phyto, multinet and Manta gears used for plankton and microplastics sampling. The distance sailed (NM) and the time spent in each region are also provided.

Region	Activity	Period	BT	PT	BM	CTD	Manta	Bongo	Multinet Mammoth	Multinet Midi	WP2	Phyto	Distance (NM)	Days
Cabo Verde	steaming	2021-11-21 to 2021-11-22 2021-12-08 to 2021-12-09											392.3	2.1
Cabo Verde	survey	2021-11-22 to 2021-12-08											2 049.2	15.9
Cabo Verde islands, east	entire survey		21	9	1	63	22	28	12		31	4		
Cabo Verde islands, northwest	entire survey		6	9	1	38	4	4	17	1	4	1		
Total		2021-11-21 to 2021-12-08	27	18	2	101	26	32	29	1	35	5	2 441.5	18.0

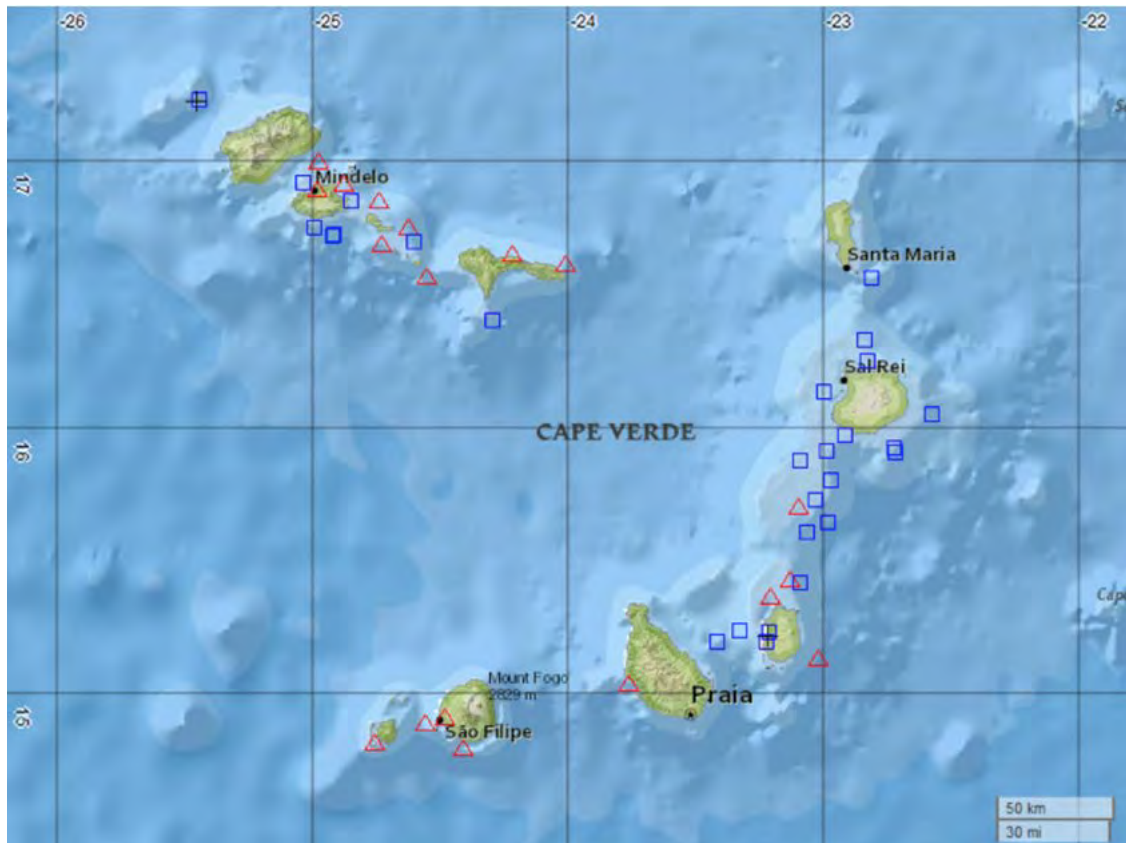


Figure 4. Map of trawl station locations (bottom: squares; pelagic: triangles; beam: cross)

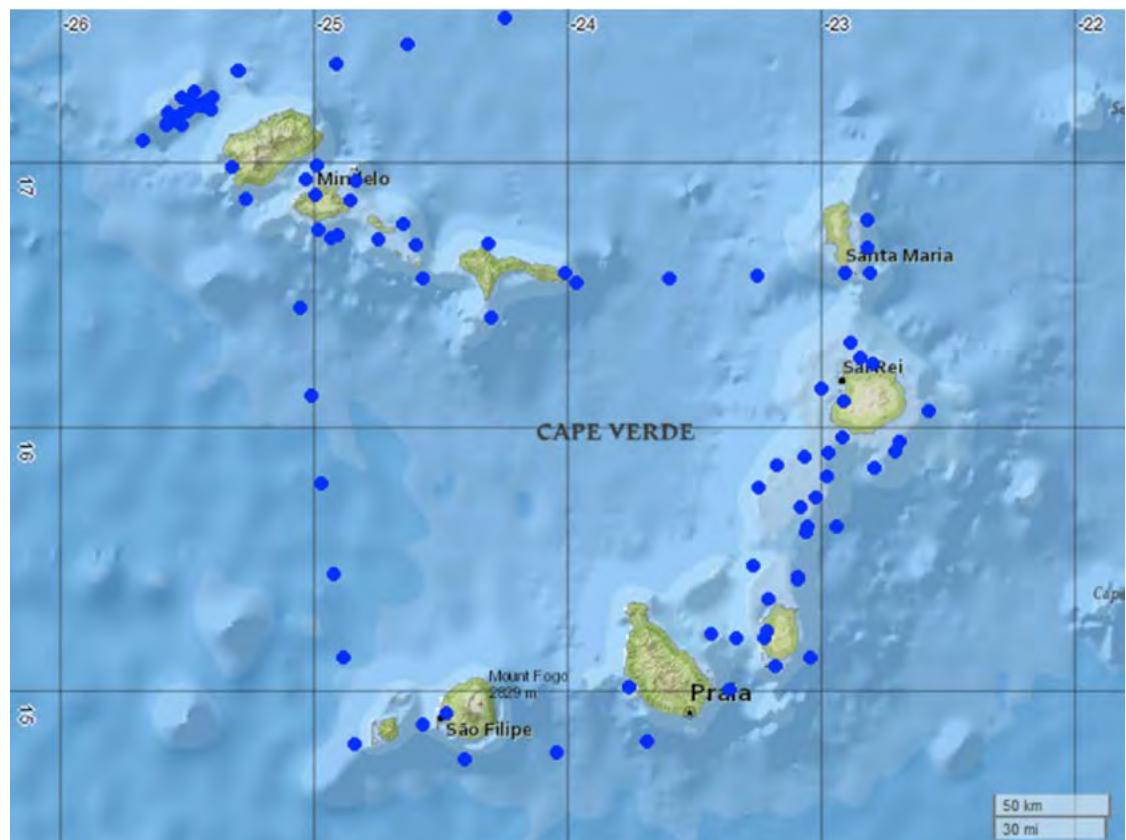


Figure 5. Map of hydrographic sampling (CTD) station locations

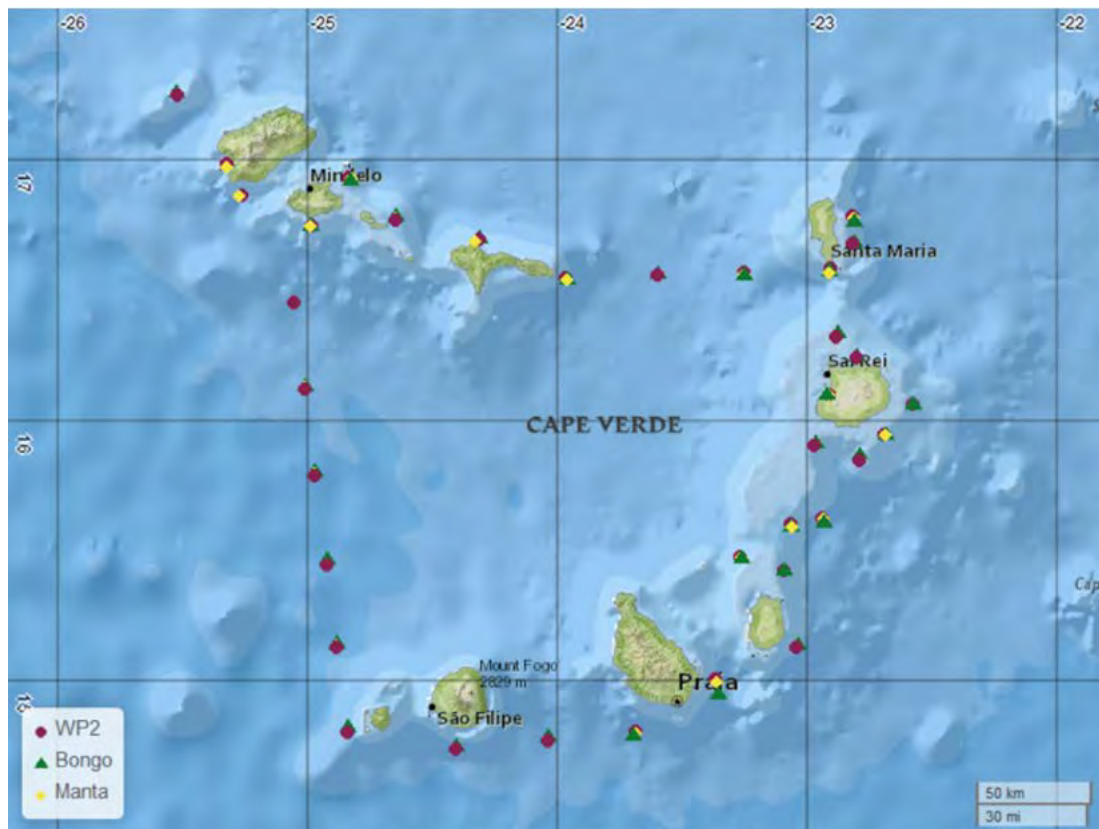


Figure 6. Map of plankton station locations. WP2: circles; Bongo: triangles; and Manta: diamonds

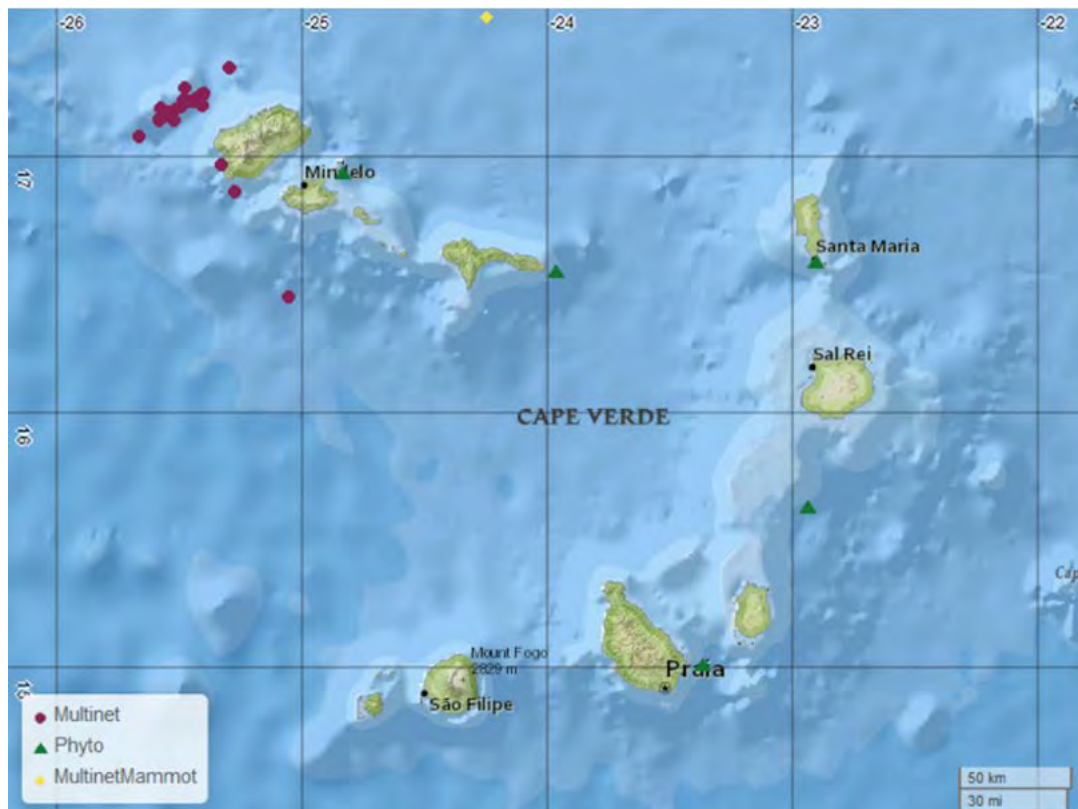


Figure 7. Map of plankton station locations. Multinet Midi: circles; phytoplankton: triangles; multinet Mammoth: diamonds

2 METHODS

2.1 Meteorology

Meteorological data were logged continuously from the AANDERAA Smartguard meteorological station and included wind direction and speed, air pressure, relative humidity, air temperature, and solar radiation. All data were stored on the vessel's cruise logger database.

2.2 Oceanography

2.2.1 Underway hydrographic sampling

2.2.1.1 Ocean currents

Ocean current data were collected with a vessel-mounted Teledyne RDI Ocean Surveyor ADCP, operating at 150 kHz. The second vessel mounted ADCP, operating at 75 kHz, was undergoing maintenance and was not available for the survey. The ADCP ran in narrow band mode and averaged data in 8 m vertical bins. Heading, pitch, roll and positional data were acquired by a Kongsberg Marine SEAPATH unit. Teledyne's VmDAS software was used to collect the raw current data, whereas software created at IMR (QCSYS) was used to process the data.

2.2.1.2 Sea surface temperature, salinity and fluorescence

TSG 4m C3 and 6m WETStar

Both SBE 21 SeaCAT Thermosalinographs (TSG) were needed during the survey due to instrument malfunctions. The survey began with the 4 m intake TSG, but due to problems with the conductivity sensor, the second 6 m drop keel intake TSG was put into operation from 20 November onwards. The 6 m intake TSG ran continuously during the survey, obtaining samples to measure seawater salinity and temperature every 10 seconds. The 4 m TSG was originally equipped with a Sea-Bird WETStar fluorometer but that was moved to the 6 m TSG on 3 December.

2.2.1.3 $p\text{CO}_2$

Water from the vessel's 4 m intake was pumped through the flow head of a CONTROS HydroC[®] CO₂ FT sensor. Dissolved gases diffused through a composite membrane into the internal gas circuit leading to a detector chamber, where the partial pressure of CO₂ is then determined via IR absorption spectrometry. Concentration-dependent IR light intensities are converted into the output signal from calibration coefficients stored in firmware and data from additional sensors within the gas circuit.

2.2.2 Fixed hydrographic sampling

2.2.2.1 CTD sensors

A Sea-Bird 911plus CTD containing the sensors identified in Annex 2 were mounted to a 12-bottle rosette for use at every fixed hydrographic station. Sensor data logging and profiling were performed using Sea-Bird's Seasave software.

For conductivity sensor validation, water was collected from the rosette water sampler at the bottom of each non-shallow CTD deployment when the rosette water bottles were in use. Water was also collected from the entire water column from selected deep CTD stations for conductivity and dissolved oxygen sensor calibration. For this purpose, a Guildline Portasal Salinometer 8410A was used to measure the samples collected for conductivity, whereas the dissolved oxygen water samples were measured using a Metrohm 916 Ti-Touch potentiometric titrator performing automated Winkler titrations (Grasshoff *et al.*, 1983; Langdon, 2010).

2.2.3 Water column ocean currents

The lowered acoustic Doppler current profiler (LADCP) was not used during the survey.

2.2.4 Ocean acidification parameters

pH & total alkalinity

Seawater samples for the analysis of pH and total alkalinity were collected at pre-selected hydrographic stations in 250 ml borosilicate glass bottles with silicone tubing. After collection, samples were placed directly into a covered 25 °C water bath in preparation for analysis. Samples were measured immediately after reaching 25 °C because no preservative was added to the samples. pH was determined using an Agilent Cary 8454 UV-Vis Diode Array spectrophotometer and a 2 mM m-cresol purple indicator dye solution. The indicator dye was measured every 24 hours (or as needed) during the survey to calculate the correction factor appropriate for the measured pH values (Clayton and Byrne, 1993; Chierici *et al.*, 1999). All pH spectrophotometric measurements were performed in triplicate on board. Total alkalinity was determined via an open-cell potentiometric titration using a 0.05 M HCl solution with a sodium chloride background as the titrant (Dickson *et al.*, 2007). A Metrohm 888 Titrando equipped with an Aquatrode plus pH electrode with an integrated Pt1000 temperature sensor was used in combination with Metrohm's tiamo™ software to measure the change in pH for total alkalinity titrations. Certified CO₂ in seawater reference material of known total alkalinity from Scripps Institution of Oceanography was measured every 24 hours (or as needed) during the survey to calculate the correction factor appropriate for the measured total alkalinity values. All total alkalinity titrations were performed in triplicate on board.

2.2.5 Nutrient samples

Seawater samples for the analysis of nitrite, nitrate, silicate and phosphate were collected in 20 ml polyethylene vials at every plankton sampling station and frozen for preservation as requested by our partners from Cabo Verde. When ready for analysis, samples will be thawed in a 50 °C water bath for 40 minutes and then allowed to cool to room temperature for 45 minutes (Becker *et al.*, 2019). The nutrient samples are planned to be measured in February 2022 on the vessel with a SEAL QuAAtro39 continuous segmented flow analyser (QuAAtro Methods:Q-070-05 Rev. 7; Q-068-05 Rev. 12; Q-064-05 Rev. 8; Q-038-04 Rev. 4 [multitest MT3B]).

2.3 Plankton

2.3.1 Phytoplankton

Phytoplankton samples were collected using a phytoplankton net with 10 µm mesh size towed at 0 m to 30 m at five selected super-stations (Annex 1) for the analysis of the phytoplankton community. Samples were preserved in 2 percent acidic Lugol's solution and stored in 100 ml dark glass bottles.

2.3.2 Zooplankton

Zooplankton samples were collected at 35 selected stations, with vertical tows of a WP2 net (180 µm). Sample collection and processing followed the Sailing Orders of the survey. Specifically, the net was towed within 5 m from the bottom to the surface, or from 200 m depth to the surface at deep stations. Each sample was halved into parts with a Motoda splitter. One half was size fractionated through 2 000 µm, 1 000 µm and 180 µm mesh sizes, and dried in the oven (60 °C) in pre-weighed aluminum trays. The second half was preserved in 4 percent borax buffered formaldehyde solution.

2.3.3 Ichthyoplankton

Ichthyoplankton was collected with double oblique tows of a Bongo net (405µm). Samples were collected at the 35 environmental stations within 5 m to 10 m from the bottom to the surface, or a maximum depth of 200 m to the surface at deep stations. At station 528 the Bongo net was lost and for the last the stations the multinet Mammoth (405 µm) was used in its place.

In all cases, once the Bongo was on board the sample from the two nets was treated as follows:

- a) One of the nets, the Bongo V, was sieved on a 180 µm sieve and transferred to a 100 ml bottle (or bigger) and preserved immediately in 4 percent formaldehyde.
- b) The other net, the Bongo H, was examined under the microscope and ichthyoplankton was sorted. Sorting was performed at all Bongo net stations. The sorted larvae were photographed and preserved in absolute ethanol in small-labelled scintillation vials or cryovials indicating the part of the sample used (i.e. 50 percent), the preservative, station, etc. When sorting was completed, the bulk sample was preserved in absolute ethanol for genetic analysis in small-labelled scintillation vials indicating which net was used for sorting, the preservative, station, etc. In a few cases, during of bad weather, an alternative approach was to fix the total sample for ichthyoplankton without separating larvae from the sample onboard.

Samples collected with the multinet Mammoth were divided into two parts using a Motoda splitter and each half was preserved as mentioned above.

2.3.4 Jellyfish

Not relevant for this survey.

2.4 Fishery resources

2.4.1 Pelagic resources

The area between 20 m and 1 000 m bottom depths were surveyed for pelagic fish abundance estimation using the acoustic method.

Acoustic data were recorded using a Simrad EK80 scientific split beam echo sounder equipped with keel-mounted transducers at nominal operating frequencies of 18 kHz, 38 kHz, 70 kHz, 120 kHz and 200 kHz. Calibration of the echo sounders (18 kHz) was first conducted in Flamengo Bay on 26 November and then followed up with a second calibration (38 kHz) south of Maio on 2 December 2020. The gain was adjusted for recordings throughout the survey. In Annex 2 the details of the acoustic settings used during the survey are provided. Acoustic data were processed on board and the backscatter was assigned for biomass estimation to the acoustic categories listed in Table 3.

Pelagic trawl hauls were conducted whenever strong acoustic signals were displayed on the echosounder and “blindhauls” were undertaken close to the surface (in the blind zone of the echo sounder) at night.

Table 3: Allocation of acoustic densities to species groups. The mean integrator value in each sampling unit (S_A -values) was divided between the listed standard categories/groups of fish. Note that for the groups sardinella and horse mackerel, all encountered species are listed, while only examples are listed for the remaining groups.

Acoustic category	Group	Taxon	Species
SARD	Sardinella	<i>Sardinella</i> spp.	<i>Sardinella maderensis</i> , <i>Sardinella aurita</i>
HMACK	Horse mackerel	<i>Trachurus</i> spp.	<i>Trachurus trecae</i> , <i>Trachurus picturatus</i>
MACKE	Mackerel	<i>Scomber</i> spp.	<i>Scomber colias</i>
PEL1	Pelagic species 1	Clupeiformes ^[1]	No records of other clupeids found in Cabo Verde

Acoustic category	Group	Taxon	Species
PEL2	Pelagic species 2	Carangidae ^[2]	<i>Caranx crysos</i>
			<i>Decapterus punctatus</i>
			<i>Pseudocaranx dentex</i>
			<i>Selar crumenophthalmus</i>
			<i>Selene dorsalis</i>
			<i>Seriola dumerili</i>
			<i>Seriola fasciata</i>
			<i>Seriola rivoliana</i>
		Scombridae	<i>Auxis thazard</i>
			<i>Sphyraena guachancho</i>
		Sphyraenidae	<i>Lithognathus mormyrus</i>
BOTT	Demersal species	Sparidae ^[3]	<i>Dentex macrophthalmus</i>
			<i>Galeoides decadactylus</i>
			<i>Antigonia capros</i>
			<i>Dactylopterus volitans</i>
		Other taxa	<i>Glossanodon leioglossus</i>
			<i>Lethrinus atlanticus</i>
PLANK	Plankton	Plankton, euphausiids, mesopelagic fish	
OTHER	Other pelagic fish	Other taxa	

Notes: ¹Except specimens from the *Sardinella* and *Engraulis* genera; ²Except specimens from the *Trachurus* and *Scomber* genera; ³Main taxa group

2.4.1.1 Trawling and biological sampling

Standard biological sampling, including length, weight, sex and maturity registrations were carried out for all priority species at all trawling stations. In addition, length–weight measurements of up to 30 individuals per station were taken from all other fish species caught, as well as genetic samples for selected species (Annex 1) based on the standard sampling protocol for genetics (https://nansen-surveys.imr.no/doku.php?id=biological_sampling_procedures).

2.4.1.2 Acoustic abundance estimation

Acoustic data were logged and post-processed on board using the latest acoustic data post-processing software, the Large-Scale Survey System (LSSS) Version 2.7.

In cases where the integrated echo contained more than one category of fish (see Table 3), the mean s_A -value allocated to each category was in the same ratio as their contribution to the abundance in trawls in that area.

The following target strength (TS) function was applied to convert s_A -values (mean integrator value for a given species or group of species in a specified area) to number of fish:

$$TS = 20 * \log L - 72dB$$

which can be converted (see Toresen *et al.*, 1998 for details) to the area form (scattering cross sections of acoustic targets):

$$C_{Fi} = 1.26 * 10^6 L^{-2}$$

where L is total length in 1 cm length group i and C_{Fi} (m^2) is the reciprocal back scattering strength, or so-called fish conversion function. In order to split and convert the allocated s_A -values (m^2/NM^{-2}) to fish densities (numbers per length group per NM^2), the following formula was used:

where

Q_i = density of fish in length group i

s_A = mean integrator value

p_i = proportion of fish in length group i

$\sum_{i=1}^n \frac{p_i}{C_{Fi}}$ = the relative back scattering cross section (m^2) of the length frequency sample of the target species, and

C_{Fi} = reciprocal back scattering cross section (σ_{bs}^{-1}) of a fish in length group i .

The integrator outputs were split into the fish groups listed in Table 3 using a combination of behaviour pattern as deduced from echo diagrams, the LSSS analysis and catch composition.

The acoustic backscatter was scrutinized daily and allocated to the various acoustic categories. For Pelagic I, Pelagic II and “other species” -50 to -55 dB was used. To identify mesopelagic layers a threshold of -60 dB was used.

For a stratum representing a distribution of a target group, the following basic data are needed for the estimation of abundance:

- the average s_A -value for the region;
- the surface area (usually square nautical miles, NM^2); and
- a representative length distribution of the fish in the region.

If the acoustically targeted fish was a mixture of more than one species, a representative distribution of all the species within the stratum, as shown in the trawl catches, was used. A mean length of 23 cm was used as a proxy for estimating the proportion of the total biomass of this group of species. While it is recognized that catch is a poor indicator of relative abundance, especially for pelagic fish, no other method is easily available from the data available.

The process followed was therefore to:

- divide the s_A -value between groups of fish and/or species;
- use a mean length of a specific group and/or species in the above equation; and
- calculate the biomass estimates for a region, using the following procedure:
 - o The length-frequency samples of the species in the category were respectively pooled together with equal importance (normalized).
 - o The mean back scattering strength (σ/s_A) of each length frequency distribution of the target group/species was calculated and summed.
 - o The pooled length distribution was used, together with the mean s_A -value, to calculate the density (numbers per NM^2) by length groups and species, using the above formula. The total number by length group in the area was obtained by multiplying each number by the area. The numbers were then converted to biomass using the estimated weight at length.

2.4.2 Demersal resources

2.4.2.1 Trawling and biological sampling

Demersal trawl stations were carried out on the shelf at approximately the same positions as during the survey in 2011, to ensure comparability of catch rates between the two surveys.

The potential to carry out additional bottom trawl stations was explored. Trawls were carried out during daylight hours (06.30 to 18.30 UTC), from 20 m to 1 000 m.

Trawl duration was standardized to 30 minutes, however, trawls with a duration of longer than 15 minutes (and more than 0.5 NM) were included in the estimates. The trawling start time is determined by using a SCANMAR sensor to detect the landing of the trawl on the bottom, and the stop-time is defined as the time when the wires start to haul the net in. In some cases, the towing was interrupted before 30 minutes due to poor bottom conditions. Hence, some stations were coded as invalid in the database due to flawed sampling. During the demersal sampling, one station was flagged as invalid (Station 257) due to short trawling time, while two other stations (Station 285 and 286) were flagged for species identification only, and as such, excluded from the demersal catch rates. A detailed description of instruments and fishing gear is given in Annex 2.

Once the catch was on deck, it was assessed, and, if necessary, subsampling was carried out. At all trawl hauls, the catch was sorted and length and weight measurements were taken for all fish species using an electronic fish meter and its customized data-acquisition software (Fish2Data). Registrations were then transferred and stored to the Biotic Editor software. In addition, further biological sampling was conducted for pre-agreed priority species.

The Sailing Orders provided guidelines on the detailed sampling protocols that were used during the survey. As preparation for the work conducted in the fish lab, a training course was provided on the use of the electronic fish meter, Fish2Data and Biotic Editor software before the survey started.

During this survey, trawling was restricted to certain areas. Large parts of the shelf in Cabo Verde consist of hard and rough grounds that prevented trawling. The catch rates are indicative only and reference to biodiversity and catch rates is relevant for only the areas covered.

2.4.2.2 Swept-area abundance estimate

Due to the bottom topography around the islands of Cabo Verde the catches cannot be considered representative for the species and abundance of fish in areas that were not sampled (not accessible to trawling). Therefore, swept area abundance estimates have not been provided, but the catch rates were calculated and compared with those from 2011.

2.4.2.3 Other trawl-related sampling

The protocol that was followed for cartilaginous taxa was the same as for demersal and pelagic resources described in Annex 1. Specimens for taxonomic courses were collected following the protocol in Appendix A1.10 of Annex 1. Taxa that could not be identified were preserved in formalin or frozen and sent to taxonomic experts for identification following the protocol provided in Appendix A1.10 of Annex 1.

IMar preserved individuals of crustaceans and molluscs (except for cephalopods), for curation purposes (see Appendix A1.13 [in Annex 1]). With respect to fish, cephalopods and sponges, the samples were sent to IEO. However, in some cases IMar also preserved samples of specific fish taxa (preserved in formalin or ethanol). These samples were off-loaded in Las Palmas, handed over to IEO and then sent to Cabo Verde. All frozen samples were sent from Las Palmas to Bergen, Norway (sea freight).

In addition, samples of uncertain and unidentified individuals of fish and crustaceans were preserved for identification on land at IEO. The protocol for preservation and labelling can be found in Appendix A1.13 (in Annex 1). IEO was responsible for the samples after the vessel arrived in Las Palmas.

For epibenthos sampling, specimens in the trawl catches were identified and photographed (one picture per trawl, all species included).

The identification of marine fish species found in Cabo Verde was based on the use of digital and online taxonomy guides, printed books and FAO identification guides.

For specimens of *Sargassum* spp., the protocol in the Sailing Orders (Annex 1) was followed.

2.5 Microplastics

Samples from Manta trawls were collected and processed according to the Sailing Orders. All samples were sorted on board for microplastics and ichthyoplankton. Sorted microplastics were photographed and placed in 2 mL eppendorf tubes with freshwater. Remaining fish larvae and eggs were sorted, photographed and preserved in absolute ethanol for genetic analysis in 20 mL scintillation vials. The bulk of neuston samples was preserved in methylated ethanol after sorting.

2.6 Marine debris

Marine debris were registered and classified at each station according to the standard sampling protocol for marine debris (https://nansen-surveys.imr.no/lib/exe/fetch.php?media=nansen-surveys:fishlab_marine_litter_categories.docx). All pieces of litter were weighed and counted and when possible, photographs were taken of the individual pieces for a future reference catalogue. The data and the reference photos were recorded using Fish2Data.

2.7 Bottom habitat mapping

The EM 710 multibeam echo sounder is a high-resolution seabed mapping system. The EM 710 is mounted on the drop keel and the operational depth of the EM 710 is from 3 m to 2 000 m. Across track coverage (swath width) is up to 5.5 times the water depth and may be limited by the operator either in angle or in swath width without reducing the number of beams. The operating frequencies are between 70 kHz and 100 kHz. There are 128 beams with dynamic focusing employed in the near field. The transmitting fan is divided into three sectors to maximize range capability and to suppress interference from multiples of strong bottom echoes. The sectors are transmitted sequentially within each ping and use distinct frequencies or waveforms. The along-track beam width is 1 degree. Ping rate is set according to depth. The receiving beam width is 2 degrees. Sound profiles were set manually in the system according to the area of operation. The EM 710 was not operational for most of the survey. Data from the EM 710 was logged to the on-board Olex plotting system in standard resolution mode.

2.8 Vulnerable marine ecosystems

Not relevant for this survey.

2.9 Top predators

Not relevant for this survey.

2.10 Food safety

For nutrition and food safety sampling the protocol in the Sailing Orders was followed Annex 1. Samples of selected species were taken and analysed for nutrient profiles, contaminants and microplastics. Samples were collected at different geographical coordinates. A total of 25 specimens from at least 3 different stations were frozen and will be analyzed for determination of nutrients and food safety parameters at IMR (Bergen, Norway).

2.11 Sediment sampling

Not relevant for this survey.

2.12 Additional sampling

2.12.1 Sampling at Cabo Verde Ocean Observatory and Nola Seamount

Oceanography

The 12-bottle rosette water sampler was used to support the joint monitoring project between Cabo Verde and GEOMAR at the CVOO monitoring station. In addition to the proposed sampling plan by Cabo Verde (Table 4), the CTD was deployed to 3 650 m depth utilizing all the mounted sensors and samples were also collected and measured for pH, total alkalinity, chlorophyll *a*, and salinity. Samples were also collected for dissolved nutrients and dissolved inorganic carbon/total alkalinity (DIC/TA). The dissolved nutrient samples were to be measured on board in February 2022 and the DIC/TA samples were sent to GEOMAR for analysis, with an estimated completion date of March 2022. Due to an

oversight during the CTD deployment, the first bottle was closed at 3 650 m instead of 450 m. The 10 m depth sample was then removed to make room for the other sampling depths in the sampling plan.

CTD deployments were performed in conjunction with every plankton sampling station in the Nola Bank area (Sailing Orders – Appendix A1.6 [in Annex 1]). Four of these CTD deployments were additionally used to collect water and two extra CTD water samplings were performed on either side of the two Nola Seamounts. All water bottle samplings followed the super-station format described in Figure 3. This procedure covered a transect from the southwest to the northeast of the seamounts, together with samples from the top of, and in between, the seamounts. Dedicated underway data collection with the 150 kHz ADCP was also carried out around each seamount. The southwestern seamount was covered close to the start and the northeast seamount was covered at the end of the survey. All oceanographic measurements performed in the Nola Bank area followed standard operating procedures described in Section 2.2.

Table 4. Executed water chemistry sampling at the CVOO hydrographic monitoring station. Except for samples taken at 3 650 m, 250 m and 150 m, all depths correspond to the R/V *Islândia* standard sampling depths.

Bottle	Depth	Oxygen	DIC TA	Nutrients	Chlorophyll <i>a</i>
1	3650 ^[1]	1	1	1	
2	450	1	2	1	
3	350	2	1	3	
4	250	1	1	1	
5	200	1	1	1	1
6	150	2	1	3	1
7	120	1	1	1	1
8	100	1	1	1	1
9	80	1	1	1	1
10	60	1	1	1	1
11	40	1	1	1	1
12	20	1	2	1	1

Notes: ¹Due to an oversight during the CTD deployment, the first bottle was closed at 3 650 m instead of 450 m. The 10 m depth sample was then removed to make room for the other sampling depths in the sampling plan.

Plankton

Depth stratified ichthyoplankton sampling was conducted at the CVOO during the night (Figure 2). Vertical stratified sampling was performed with a multinet Midi equipped with five 180 µm nets at depths strata of 1 000 m to 600 m, 600 m to 300 m, 300 m to 200 m, 200 m to 100 m, and 100 m to 0 m at a speed 0.5 m sec⁻¹. Each sample from each depth stratum was preserved in 4 percent borax buffered formaldehyde solution and sent to IMar for analysis.

Further stratified sampling was conducted to collect mesozooplankton and ichthyoplankton at selected stations in the area of the Nola Seamounts (Appendix A1.6 [in Annex 1]) using double oblique tows of a multinet Mammoth. The first net (N1) had a 180 µm mesh and six additional nets had a 405 µm mesh size (N2–N7). The multinet was deployed with the first net opened at surface to collect mesozooplankton only during the downcast, with a speed of 0.5 m sec⁻¹. During the upcast, the six other nets (N2–N7) were opened sequentially at the predefined sampling strata of 200 m to 150 m, 150 m to 100 m, 100 m to 75 m, 75 m to 50 m, 50 m to 25 m and 25 m to 0 m, with a retrieval speed of 0.5 m sec⁻¹. A total of 17 stations were sampled during the day and nine stations at night on different dates.

Samples from nets N2–N7 (405 µm) were preserved in absolute ethanol, while the samples from N1 (180 µm) were split with a Motoda splitter in two parts. One part was preserved in 4 percent formaldehyde solution and the other part in absolute ethanol.

2.12.2 *Sargassum* spp. sampling

For *Sargassum* spp., specimens of opportunistic samples collected with the fishing trawl and/or with the Manta trawl were photographed and stored following the protocol in the Sailing Orders (Annex 1).

3 RESULTS

3.1 Meteorology

The observed wind pattern showed prevailing northeasterlies over the islands, with speeds ranging between about 0.8 m s^{-1} to 20 m s^{-1} . Three areas of stronger winds were observed: (1) north of 16° north and over Sal and Boa Vista; (2) Santa Luzia to Santo Antão; and (3) over the region between Santiago and Brava. The strongest winds occurred south of Fogo, north of Santo Antão and east of Sal (Figure 8).

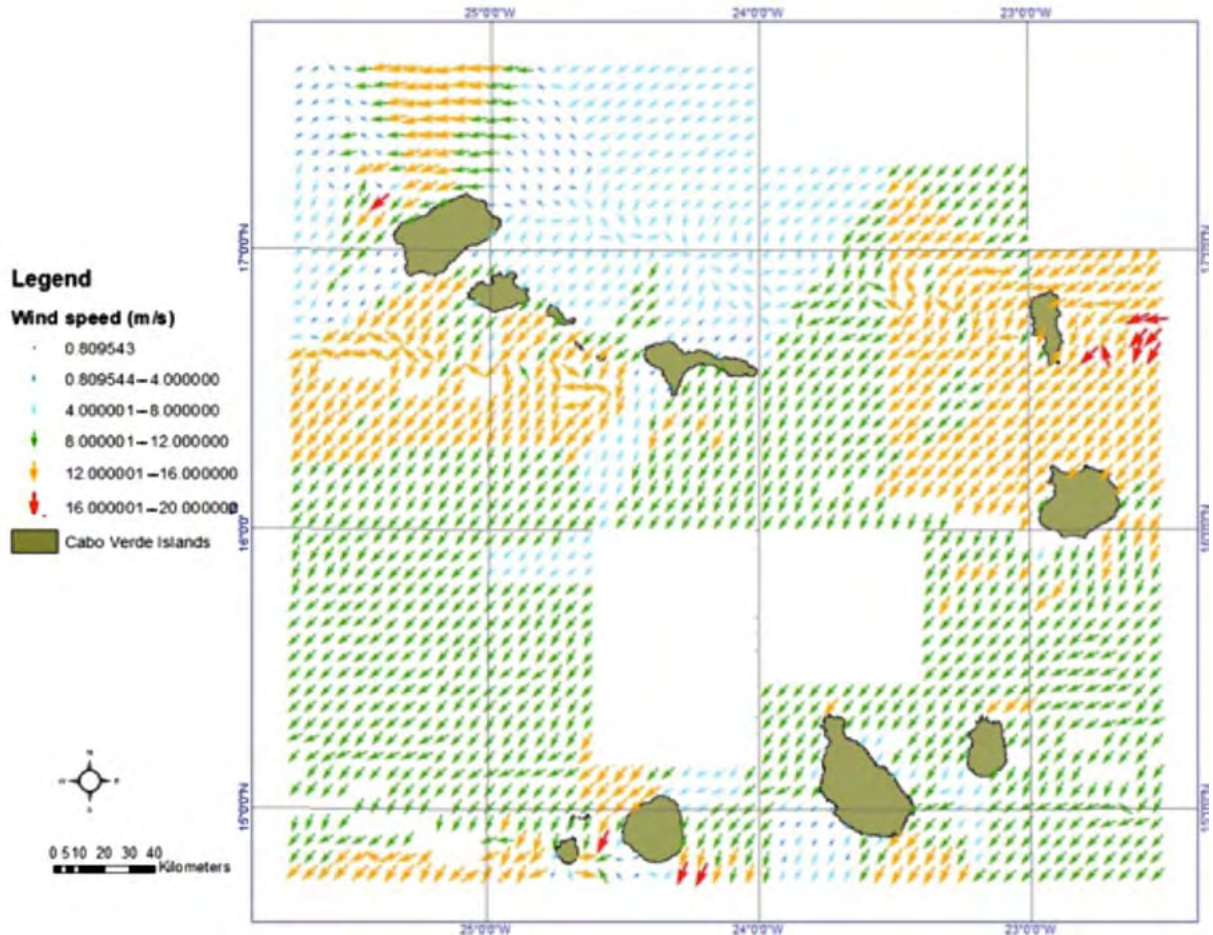


Figure 8. Six-kilometer resolution surface winds over the Cabo Verde archipelago from 17 November to 7 December 2021

3.2 Oceanography

Oceanographic results are presented for ocean currents, sea surface measurements and CTD deployments around the Cabo Verde archipelago from 22 November 2021 to 9 December 2021.

3.2.1 Underway hydrographic sampling

The 150 kHz ADCP was used throughout the survey to depict ocean current results with a maximum range of 400 m. Because of the malfunctioning salinity and fluorescence sensors on the TSGs, it was decided to use the surface readings from the 101 CTD deployments to depict the sea surface temperature, salinity and fluorescence results (Figure 9). Because the CTD was used, sea surface dissolved oxygen results are also presented.

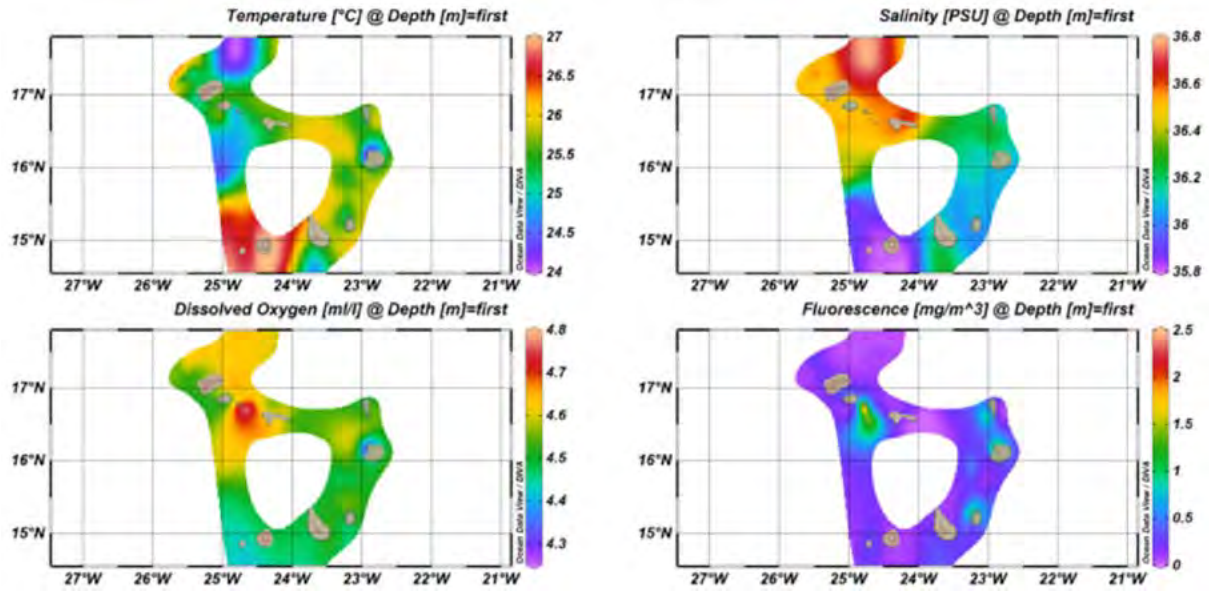


Figure 9. Sea surface horizontal distribution of temperature, salinity, dissolved oxygen and fluorescence from the 101 CTD deployments performed during Leg 2 of the survey

3.2.1.1 Ocean currents

The Cabo Verde islands are located in the southern vicinity of the Cabo Verde Frontal Zone delimited by the North Atlantic subtropical gyre and the Atlantic tropical gyre. The area surveyed during the cruise is also a domain of water mass transition mixing the South Atlantic Central Water and the North Atlantic Central Water which occupy the upper layers of the water column. Over the area, the large-scale surface circulation is modulated by the seasonal variability of the winds and the oscillation of the intertropical convergence zone, resulting in seasonality in the flow field with velocities dominated by cyclonic and anticyclonic eddies. Recently, a major current centered at 14° north with two branches, previously observed, was documented as the Cabo Verde Current System (Figure 10; Peña-Izquierdo *et al.*, 2015).

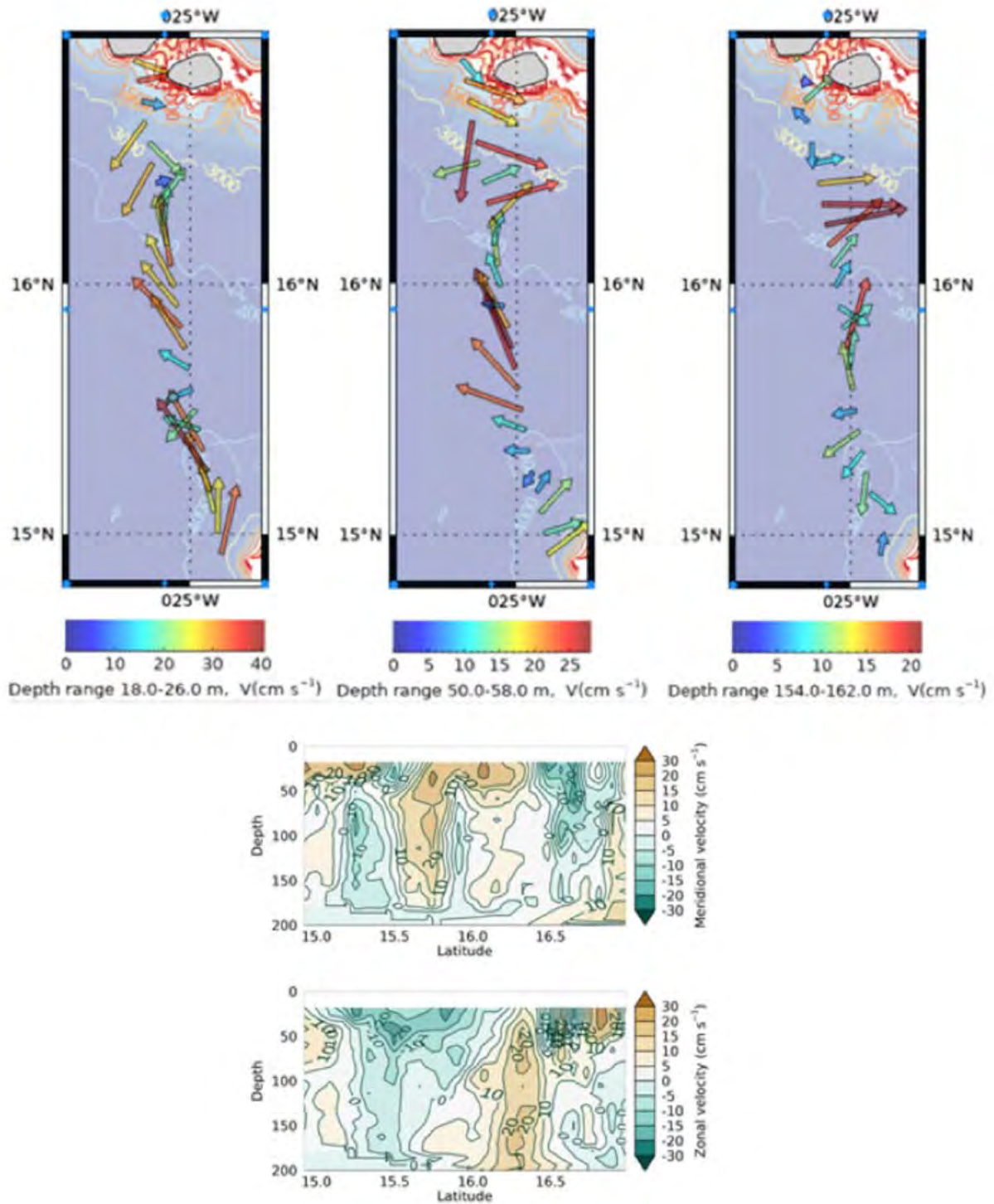


Figure 10. Brava to São Vicente ADCP transect averaged at 5 NM

3.2.1.2 Sea surface temperature, salinity and fluorescence

Temperature varied throughout the archipelago with the warmest waters above 26.5 °C observed in the southwestern region surrounding Brava (Figure 10). Colder waters below 25 °C were found between the CVOO monitoring station and the Nola Seamount area and south of São Vicente and north of Sal. The most saline surface waters (above 36.5 PSU) were found in the northern region of the archipelago and in between the CVOO monitoring station and the Nola Seamount. The strongest dissolved oxygen concentrations, above 4.7 ml l⁻¹, were recorded off Santa Luzia, which is the same area where the

strongest fluorescence signal was recorded and which is a marine protected area for Cabo Verde. Sea surface fluorescence signals were relatively higher north of Maio and Boa Vista and southwest of Sal than elsewhere (Figure 11).

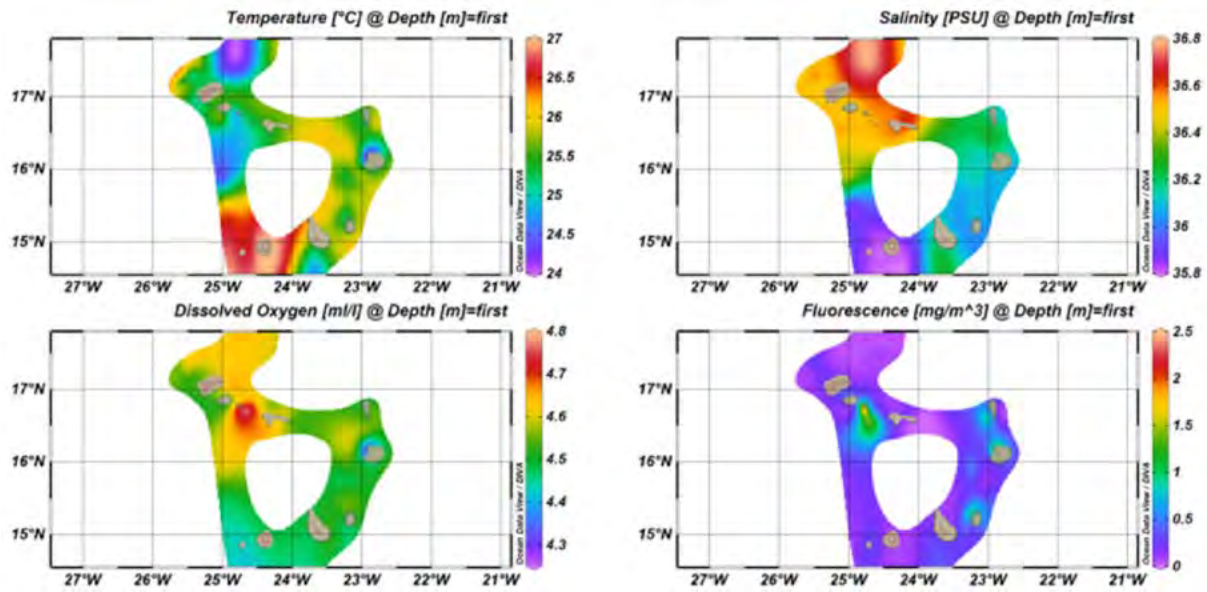


Figure 11. Sea surface horizontal distribution of temperature, salinity, dissolved oxygen and fluorescence from the 101 CTD deployments performed during Leg 2

3.2.1.3 pCO_2

3.2.2 Fixed hydrographic sampling

CTD deployments were performed at every bottom trawl station, plankton sampling station, and opportunistically at the Nola Seamount and CVOO monitoring station (Figure 1.5). Of the 1010 CTD deployments, 45 collected water samples for nutrients and chlorophyll *a* and 32 stations also collected samples for pH and total alkalinity. Some stations were as deep as 1 000 m and ~4 000 m (Figure 5).

3.2.2.1 CTD Sensors

Although surface waters cannot be used to identify water masses, they can be used to depict seasonal variability and subsurface mixing, as the surface waters of Cabo Verde show temperatures between 20 °C and 27 °C, and salinity from 35.58 PSU to 37.08 PSU. Beneath the surface water, however, the Cabo Verde Frontal Zone water masses in central and intermediate waters converge just north of the Cabo Verde archipelago (Figure 12). Below the mixed layer, down to approximately 500 m, the oxygen minimum zone (OMZ)- dominant SACW and slightly more saline NACW was detected (Pelegrí *et al.*, 2017). At intermediate depths, the colder Antarctic Intermediate Water was found mixing with the slightly more saline Eastern Atlantic Subarctic Intermediate Water (Pelegrí *et al.*, 2017). At the CTD stations deployed beyond 3 000 m between Brava and São Vicente, São Nicolau and Sal, and at the CVOO monitoring station, the relatively warmer North Atlantic Deep Water dominated the deep waters (Emery, 2001) (Figure 12).

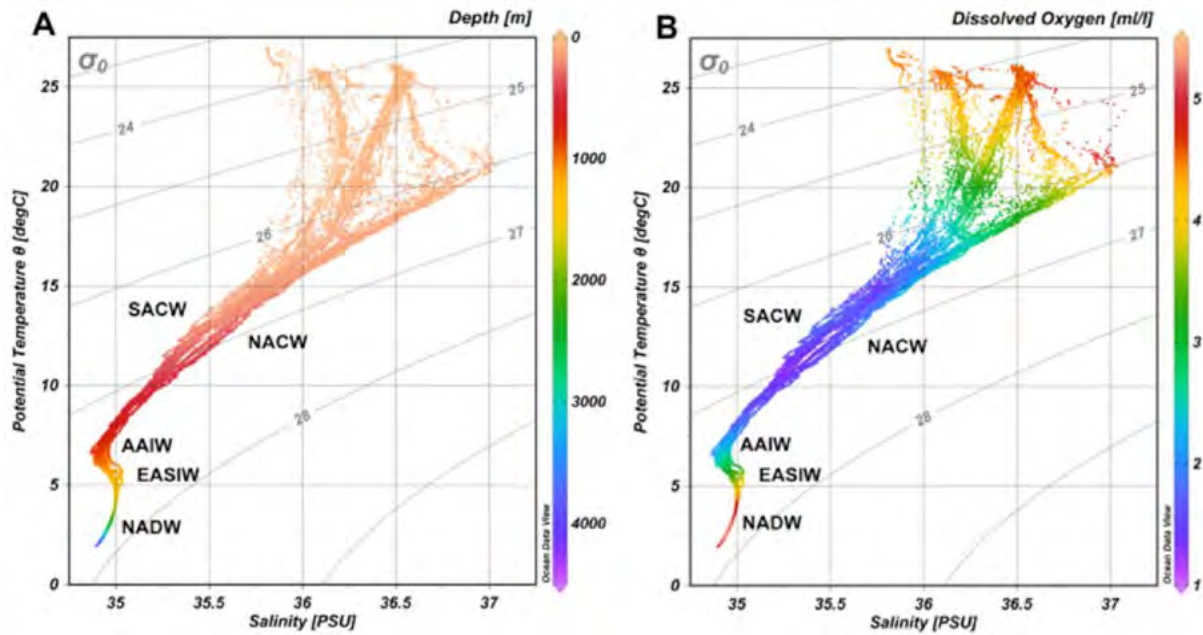


Figure 12. TS diagrams with water mass identifications. Image A has an added depth variable and image B has an added dissolved oxygen variable

Open ocean transect – Brava and São Vicente

To conclude the originally planned sampling, the vessel traversed north from Brava toward São Vicente with CTD deployments carried out on the way at depths ranging from 1 000 m to 4 200 m. Figure 13 depicts the temperature, salinity, dissolved oxygen and fluorescence sections from CTD deployments at the above mentioned transect down to 1 000 m depth. Surface waters were found to be relatively warm, over 25 °C with higher temperatures in the south and a deepening of the thermocline as early as 15.25° north from 50 m in the south to 80 m in the north. Less saline surface water below 36 PSU were observed in the south with a subsurface salinity maximum between 50 m to 70 m. The surface waters north of 15.5° north reached over 36.5 PSU and extended deeper into the water column, as far down as 120 m. Dissolved oxygen portrayed a similar surface water pattern to temperature because a tightly stratified subsurface oxycline was observed at 50 m in the south with a deepening also beginning at 15.25° north. At depths between 300 m and 600 m, the OMZ can be observed most prominently from below 16.25° north, which is less affected by the enclosure of the archipelago. Dissolved oxygen concentrations fell below 1.35 ml l⁻¹, with preliminary data showing a minimum of 1.19 ml l⁻¹. Fluorescence showed maxima between 50 m and 70 m until 15.75° north, before rising in the water column to ~30 m. Despite a decrease in signal at the deep station 529 before São Vicente, the fluorescence signal was revived at the coast at the shallow station 475 (Figure 13).

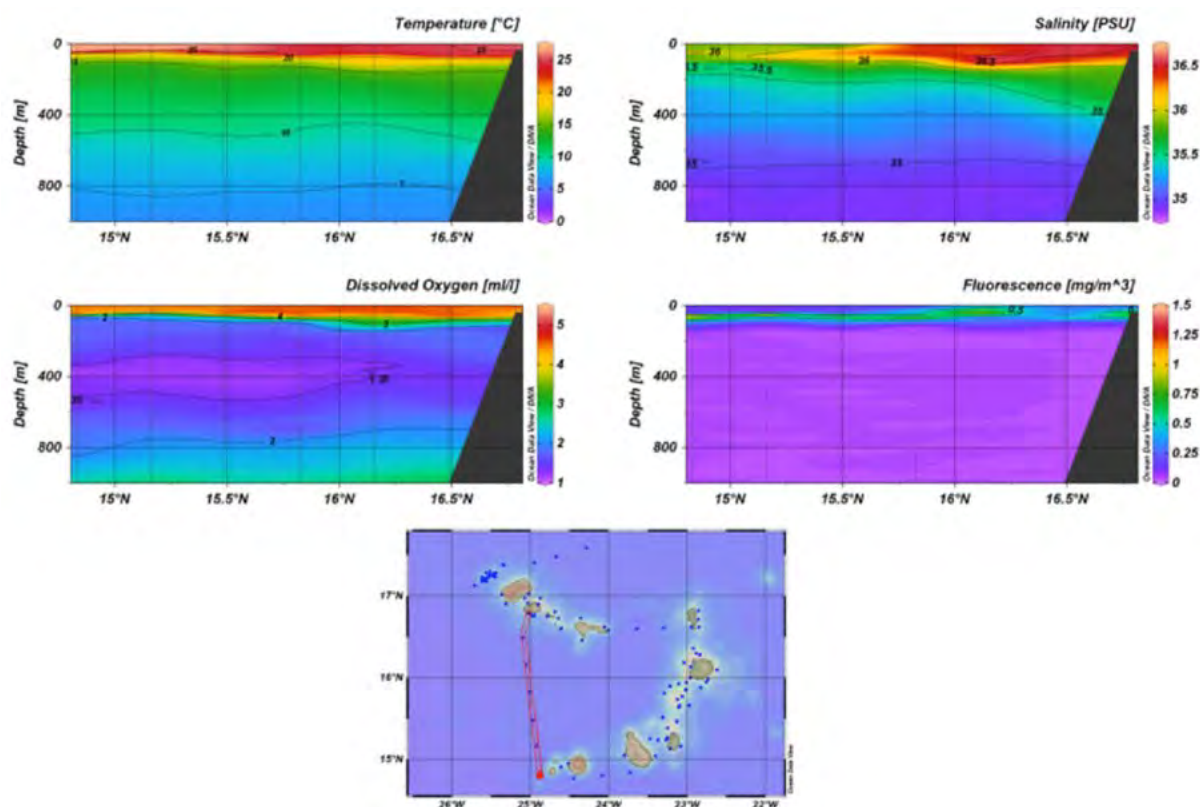


Figure 13. CTD transect from Brava toward São Vicente depicting temperature, salinity dissolved oxygen and fluorescence to 1 000 m

3.2.2.2 Water column ocean currents

Not relevant for this survey.

3.3 Plankton

A summary of the number of plankton stations and type of samples collected is presented in Table 2.

3.3.1 Phytoplankton

A total of five phytoplankton samples were collected for taxonomic identification. Samples were sent to IMR in to be analysed.

3.3.2 Zooplankton

A total of 101 aluminum trays from the survey were transferred to IMR for fine scale zooplankton dry weight measurements. A total of 13 samples (preserved in formaldehyde) belonging to nine stations were analysed with the FlowCam Macro on board the vessel. All WP2 samples preserved in formaldehyde (35 in total) were sent to IMar in Cabo Verde to be analysed with a ZooSCAN system.

3.3.3 Ichthyoplankton

A total of 70 ichthyoplankton samples were collected for taxonomic purposes, including 64 samples collected with a Bongo net and six samples collected with a multinet Mammoth. A total of six samples from the Bongo net, preserved in ethanol, were sorted for fish larvae and eggs and kept in 20 mL scintillation vials. The number of sorted fish larvae and fish eggs was 181 and 261, respectively. All samples were sent to IMR for further taxonomic and genetic analyses.

3.3.4 Jellyfish

Not relevant for this survey.

3.4 Fishery resources

A total of 47 valid trawl station were carried out, divided into 27 bottom, 18 pelagic and 2 beam trawls (Table 2). In the 17 pelagic trawl stations, 85 species were registered, of which 965 individuals were measured (length and weight). In the 27 bottom trawl stations, 331 unique species were registered (including fish, epibenthos, crustaceans and marine litter) and 6 746 individuals were measured (length and weight).

3.4.1 Pelagic resources

The hydroacoustic survey covered the shelf and slope from approximately 20 m to 1 000 m bottom depth. The very narrow shelf around Cabo Verde was covered by sailing in zig zag patterned transects within 14 different strata. The most common commercially important pelagic species recorded in the trawl catches were *Seriola fasciata* and *Caranx crysos* (frequency of occurrence), while the highest catches by weight were attributed to *Acanthurus monroviae*.

3.4.1.1 Biological characteristics

Length frequencies of pelagic priority species are given in Annex 3. The three main species presented in this report are *Pseudocaranx dentex*, *Selene dorsalis* and *Seriola rivoliana*, with the latter being the most numerous and largest species of the three. For the two commercially important species *Seriola fasciata* and *Seriola rivoliana*, the smallest individuals were recorded in northwest, in Area 2, while the larger individuals were caught in the eastern part (Figure 14). This give rise to the hypothesis that the main spawning areas for these species are in Area 2.

Sex as well as stage of maturity was determined for 17 species including sharks, rays and priority species with commercial interest (Annex 4). None of the pelagic species were mature.



Figure 14. Relative abundance of five length groups (represented by various colours) of *Seriola fasciata* (left) and *Seriola rivoliana* (right). The size of the bubbles represents the catch rate of the species (in tonnes per NM²) at each station.

3.4.1.2 Distribution and abundance estimates

The biomass of pelagic fish is assessed to be 6 430 tonnes, exclusively belonging to the PEL2 group (Table 3). Of this, 4 160 tonnes were found in Area 2. No aggregations of clupeid fish (PEL1, Table 3) were observed during the acoustic survey or in any of the trawl catches.

The PEL2 group (Carangidae, Scombridae, Sphyraenidae, Sparidae, etc. [Table 3]) was absent from the southern part of the survey area between Brava and Santiago. Some minor aggregations were found on the west side of Maio, while the main concentrations in Area 1 were found on the southern side of Boa Vista and around Sal. The largest concentrations of pelagic fish were found in shallow waters (<100 m) in Area 2 between São Vicente and Ilheu Raso. The distribution of the PEL2 acoustic group (mainly carangids and scombrids), as inferred from the acoustic backscatter of the fish (nautical area scattering coefficient, NASC) is presented in Figure 15. The concentrations corresponded with increased levels of fluorescence in that area.

The biomass estimate is based on an average fish length of 23 cm (see Section 2.4.1.2 and Figure 16). Due to the coastal geomorphology (steep continental shelf), the vessel was not able to assess the shallow waters in the most effective way. As such, the abundance of pelagic fish may be underestimated.

In the pelagic trawl hauls, other taxa such as various mesopelagic species were quite abundant, mainly along the steep slope areas. The species *Ariomma melanurum* was caught frequently and over a large area in the pelagic trawl hauls. Separate biomass estimates were not carried out for this acoustic category. In future surveys, the mesopelagic organisms should be allocated to a specific group and not be grouped together with plankton and euphausiids, as was done in this survey. For the purposes of the current survey, the acoustic category “OTHER” was also used to assign backscatter of organisms that could not be categorized with certainty as “PEL2” or “PLANK” acoustic groups (Table 3).



Figure 15. Distribution of the PEL2 acoustic category during the survey

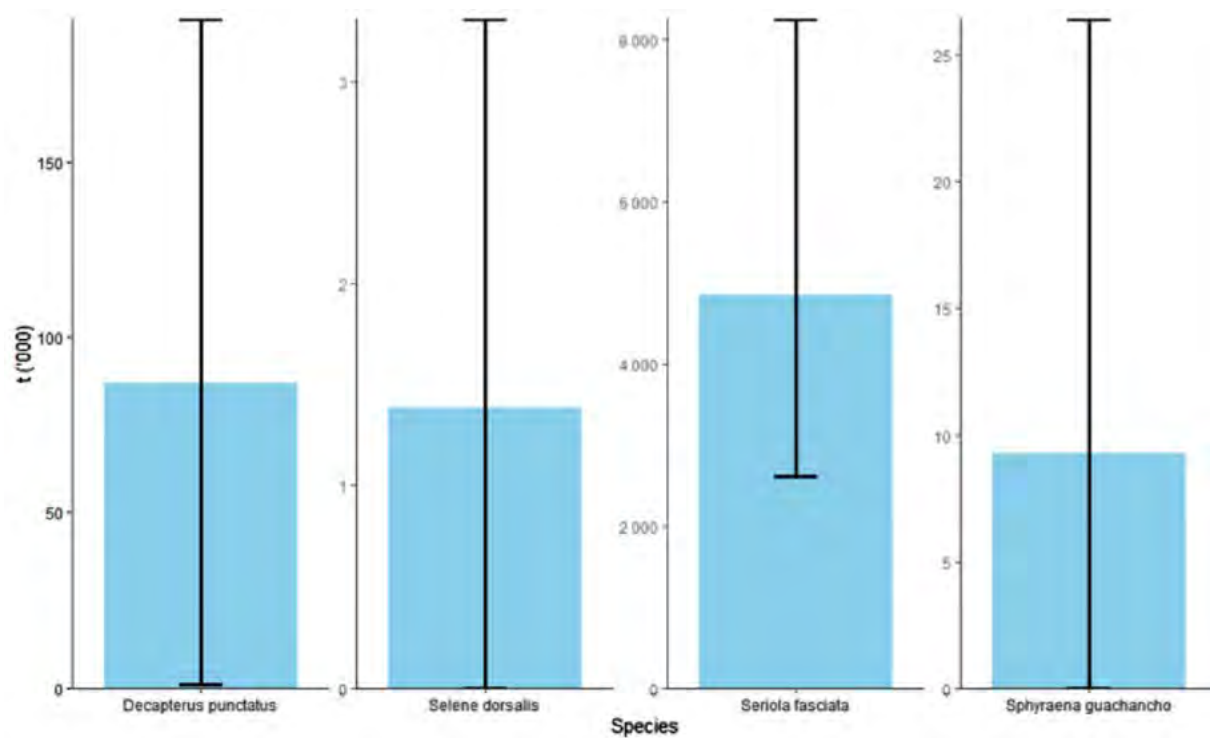


Figure 16. Biomass estimates for the main pelagic species, calculated using the StoX software. The error bars represent 5 percent and 95 percent confidence intervals.

3.4.2 Demersal resources

The topography of Brava, Fogo and Santiago generally prevents bottom trawling. No information on the fish resources around these islands was therefore obtained as part of the trawl survey. As these islands are characterized by a very narrow shelf, the number of demersal fish is expected to be low compared to the rest of the islands. This assumption is supported by the acoustic recordings where only minor backscatter strengths were recorded around the three islands.

Two separate regions were used for the analyses of trawl catch data, namely Area 1 (with a wide shelf area and the João Valente Bank) and Area 2 (where Santo Antão, São Vicente, Santa Luzia and São Nicolau share a common shelf). Both areas have a relatively shallow shelf, a mixture of hard and sandy substrate on the shelf and a very steep and rough slope to around 3 000 m or more. Only “valid” bottom trawls were included in the analyses, and catch rates were calculated for three depth regions: >30 m to 50 m, >50 m to 100 m and >100 m to 500 m. Two trawls at approximately 800 m depth were not included in the catch rate calculations. Total densities from demersal trawls are presented in Figure 17.

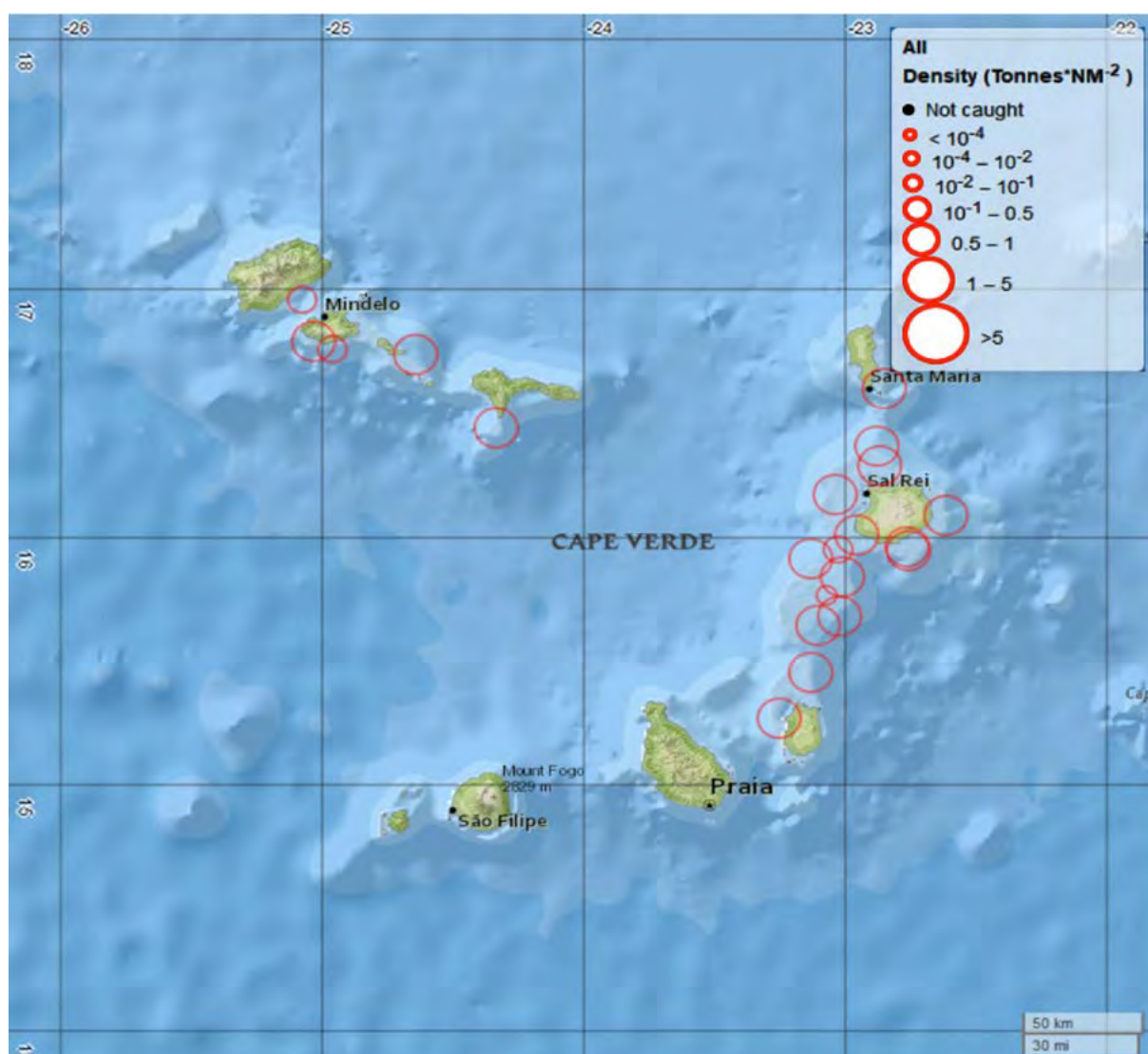


Figure 17. Catch rates (density in tonnes per NM²) of the demersal trawl hauls

3.4.2.1 Biological characteristics

Length frequencies of demersal species are presented in Annex 3. Most of the species seemed to be relatively young and did not consist of more than one or two year classes (Annex 4). Among these, the demersal species *Pseudupeneus prayensis* and *Lithognathus mormyrus*, were the only species where mature and spent individuals could be found.

3.4.2.2 Distribution and catch rates

A total of 21 valid trawl stations were conducted on the shelf between Maio and Sal, while another six stations were sampled in the area of the northwestern islands (Table 5). Around the eastern islands the highest average catch rate was recorded in the depth interval between >100 m and 500 m, with a value of 341 kg h⁻¹, compared to around 100 kg h⁻¹ in the two other depth intervals (Table 6). Around the northwestern islands the catch rates were much lower with values around 20 kg h⁻¹, in all three depth regions (Table 6).

Of the total average catch rate, pelagic species represented 4 percent (22.36 kg h⁻¹). Fifteen percent (88.911 kg h⁻¹) consisted of demersal species and 75 percent belonged to the group called “others” (species not included in any of the other groups, Table 6). The species in this group are often considered to be commercially less important. Some of the most abundant species within the group were various

mesopelagic species and *Antigonia capros*. In the two deep trawl stations (above 800 m) some big shrimps, especially *Aristaeopsis edwardsiana*, were caught.

The total average catch rate in the depth region between 50 m to <100 m was 298 kg h⁻¹ (Table 6). Of this, commercial demersal species constituted 5.7 percent (17 kg h⁻¹), pelagic species were caught with an average catch rate of 11 kg h⁻¹ (3.8 percent), sharks with 17 kg h⁻¹ (5.7 percent) and cephalopods with 0.2 kg h⁻¹. The catch rate for the group of other species was 253 kg h⁻¹, representing 84.8 percent of the overall catch rate.

The group of commercially important demersal species was subdivided into groupers, seabreams, snappers and others (Table 7). Seabreams were the only group within the depth range >30 m to 50 m around the eastern islands, that indicated some importance in the catches. Average catch rates were 52 kg h⁻¹ (44 percent of the total catch in this depth region). There were very few catches of groupers, snappers and barracudas. In the depth region between >50 m and 100 m, in the northwestern area, some Carangidae were caught, with catch rates of 13.855 kg h⁻¹. In areas with bottom depths of >100 m to 500 m the seabream *Dentex macrophthalmus* was caught with a catch rate of 97.8 kg h⁻¹, and from the “Others” category some Zeidae were caught, dominated by *Zenopsis conchifer* (catch rate 77.5 kg h⁻¹).

Table 5. Survey effort in number of bottom trawls per depth stratum (>30 m to 50 m, 50 m to <100 m, >100 m to <500 m and >500 m to 1 000 m) by region

Depth stratum (m)	Cabo Verde islands, east	Cabo Verde islands, northwest	Total
30–50	6	2	8
50–100	8	3	11
100–500	5	1	6
500–1 000	2		2
Total	21	6	27

Notes: Right square bracket indicates that the maximum depth is included in the range.

Table 6. Catch rates in tonnes per NM² (per depth stratum by region) for the main resources groups in the survey area

Area	Depth stratum (m)	Demersal	Pelagic	Cephalopods	Shrimps	Sharks	Rays	Other	Total
Cabo Verde islands, east	30–50	68.265	2.722	0.094	0.000	2.430	0.717	43.409	117.637
Cabo Verde islands, east	50–100	0.783	3.698	0.625		11.724	7.116	68.547	92.493
Cabo Verde islands, east	100–500	16.584	0.059	2.812	0.000	0.342	0.240	321.485	341.522
Cabo Verde islands, northwest	30–50	0.443	1.981	0.001		1.688	6.609	9.923	20.645
Cabo Verde islands, northwest	50–100	2.802	13.902	0.350		2.111		6.469	25.634
Cabo Verde islands, northwest	100–500	0.040		0.285	0.001		0.007	2.622	2.955
Total		88.917	22.362	4.167	0.001	18.295	14.689	452.455	600.886

Notes: Right square bracket indicates that the maximum depth is included in the range.

Table 7. Catch rates in tonnes per NM² (per depth stratum by region) for the commercially important demersal resources in the survey area

Area	Depth stratum (m)	Groupers	Seabreams	Snappers	Other	Total
Cabo Verde islands, east	30–50	0.143	52.061	0.019	65.415	117.638
Cabo Verde islands, east	50–100	0.548	0.004		91.941	92.493
Cabo Verde islands, east	100–500		16.572		324.950	341.522
Cabo Verde islands, northwest	30–50	0.416			20.229	20.645
Cabo Verde islands, northwest	50–100	0.123	0.050	0.260	25.201	25.634
Cabo Verde Islands, northwest	100–500		0.040		2.915	2.955
Total		1.230	68.727	0.279	530.651	600.887

Notes: Right square bracket indicates that the maximum depth is included in the range.

3.4.3 Diversity

From the 23 accepted endemic marine fish species of Cabo Verde, seven were caught during this survey, with *Raja herwigii* being the most frequent (7 stations, Table 8).

Within the benthos the most abundant group was Crustacea with 61 species, followed by Mollusca (22 species), Echinodermata (20 species), Porifera (10 species) and Cnidaria (8 species).

Demersal catches were highly diverse in terms of number of species and, in general, stations deeper than 200 m exhibited the highest diversity in the catch with up to 58 different identified fish species and up to 14 different identified benthos species. Overall, 194 different fish species were identified, with the bony fish being the dominant group, followed by 120 identified benthic invertebrate species, 14 identified species of chondrichthyes and one turtle.

Pelagic catches were dominated by mesopelagic fish species mainly from the families Nomeidae, Myctophidae and Gempylidae. Several taxa from the subphylum Crustacea and the class of Cephalopoda were also found in the catches. In a single pelagic trawl 480.1 kg of the surgeonfish *Acanthurus monroviae* were caught.

Table 8. Endemic marine fish species of Cabo Verde, recorded in the surveys with the RV *Dr Fridtjof Nansen* in 2021 and 2011

Species	Occurrence	Weight (kg)	No. individuals
	Survey 2011/Survey 2021	Survey 2011/Survey 2021	Survey 2011/Survey 2021
<i>Chromis lubbocki</i>	1/1	2.64/0.15	41/2
<i>Diplodus prayensis</i>	2/1	9.38/30.75	37/126
<i>Diplodus fasciatus</i>	2/1	27.18/0.75	92/1
<i>Parapercis atlantica</i>	2/1	0.06/0.03	5/4
<i>Parapristipoma humile</i>	2/2	3.78/15.8	14 67
<i>Pegusa cadenati</i>	0/4	0/0.69	0/9
<i>Raja herwigii</i>	8/7	5.12/9.05	14/24

3.5 Microplastics

Microplastic-like items were found in five of the 26 Manta trawl stations.

3.6 Marine debris

Marine debris was registered at 11 trawl stations, mostly with one piece of litter per station. One station (station 38) had three pieces of litter.

3.7 Bottom habitat mapping

Not relevant for this survey.

3.7.1 Seafloor maps

Not relevant for this survey.

3.7.2 Sediment composition

Not relevant for this survey.

3.7.2.1 Granulometry

Not relevant for this survey.

3.7.2.2 Chemical composition

Not relevant for this survey.

3.7.2.3 Contaminants

Not relevant for this survey.

3.7.3 Benthic fauna

Benthic invertebrates were collected in all 27 bottom and two beam trawls deployed during the survey, however, species caught using the beam trawl were used only for taxonomic purposes.

3.7.3.1 Epibenthic fauna

A total of 125 morphospecies were identified on board, almost half of them (61) being crustaceans. Other groups, such as molluscs and echinoderms, were also well represented, with 22 and 20 species, respectively. The phyla Porifera (ten) and Cnidaria (eight) were represented by a low number of species.

Regarding the estimated biomass for the swept area, echinoderms presented the highest weight, with more than half of the estimated biomass, followed by molluscs, crustaceans and porifera. It is noteworthy that the higher biomass of the echinoderms is almost exclusively due to a deep-water species – *cf. Zygothuria lacteal* – caught in only the two deepest stations. Likewise, the estimated biomass of the molluscs is mostly due to *Illex coindetii*, and to a lesser extent, *Sepia officinalis*. Both of these two species should be explored further as they might have potential for commercial exploitation. Among the crustaceans, the highest biomass is also due to a deep-water species, *Aristaeopsis edwardsiana*, an unexplored shrimp species in Cabo Verde. The number of individuals captured in the two deep water stations and the fairly large size of the specimens, suggest that better exploration of the distribution and abundance of this deep-water resource could prove economically important. However, before any conclusions can be drawn, estimation of this resource using other fishing gears would be necessary, since bottom trawls would not be sustainable nor profitable due to the nature of the seabed around the Cabo Verde islands.

3.7.3.2 Benthic infauna

Not relevant for this survey.

3.8 Vulnerable marine ecosystems

Not relevant for this survey.

3.9 Top predators

Not relevant for this survey.

3.10 Food safety

Only two of the pelagic species preselected for nutrition and food safety (microplastics and parasites) analysis, namely *Auxis thazard* and *Selar crumenophthalmus*, were caught in the trawls and collected for this purpose. None of these species were caught at more than one station, thus there are no replicates, as was requested in the Sailing Orders. Additionally, three species with high ecosystem importance were opportunistically sampled at one station each. The collected fish species and the station in which they were caught is provided in Table 9.

Table 9. Overview of samples collected for nutrition and food safety analysis (by fish species, station number and number of individuals preserved)

Species	Station	No. of individuals preserved
<i>Auxis thazard</i>	35	5
<i>Selar crumenophthalmus</i>	24	19
<i>Antigonia capro</i>	26	5
<i>Ariomma melanum</i>	37	15
<i>Maurolicous muelleri</i>	44	150

3.11 Sediment analysis

Not relevant for this survey.

3.12 Additional sampling

3.12.1 Sampling at Cabo Verde Ocean Observatory and Nola Seamount

Oceanography

At the CVOO monitoring station, all sampling was performed successfully, aside from the first water sample being collected at 3 650 m instead of 450 m. All water sample measurements were performed on board except for dissolved nutrients and DIC/TA. The dissolved nutrient samples were to be measured on board in February 2022 and the DIC/TA samples were sent to GEOMAR for analysis with an estimated completion date of March 2022.

The Nola Seamount region in the northwest area of the survey area depicted possible small-scale eddy interactions which resulted in different water characteristics on each seamount. The southwestern seamount showed a shearing effect that induced a downwelling of warmer, more saline waters that were also more oxygenated with a higher fluorescence signal. No downwelling or upwelling was detected in between the seamounts and despite the DOC dropping below 2.0 ml l^{-1} , it was still relatively higher than the $\sim 1 \text{ ml l}^{-1}$ OMZ observed east of the northeastern seamount. No indications of downwelling were observed at the northeastern seamount, as at the southwestern seamount (figures 18, 19 and 20).

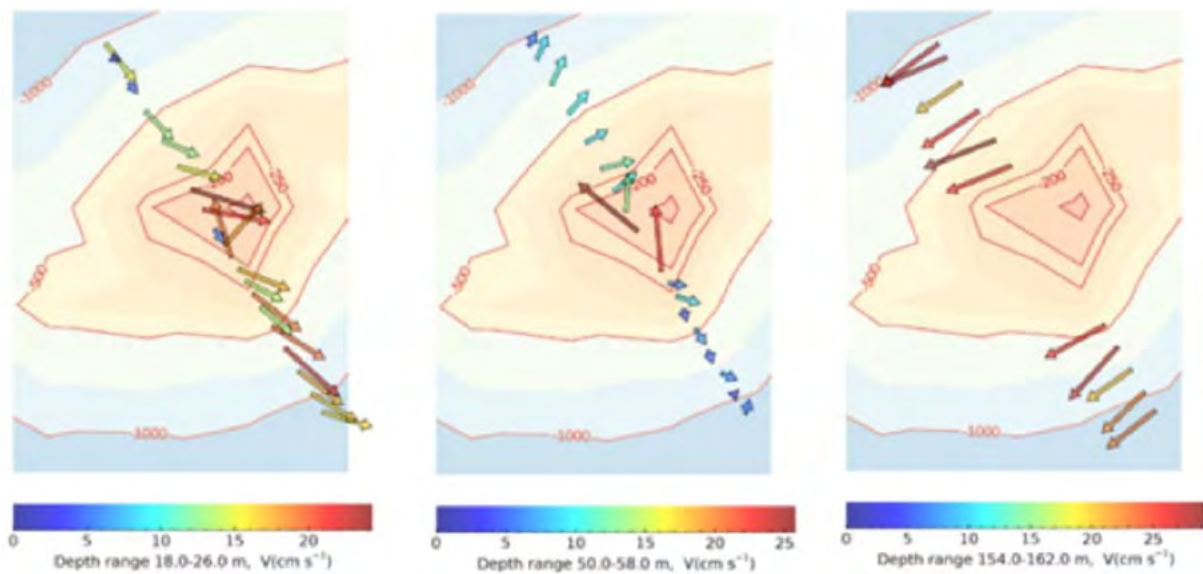


Figure 18. Southwestern Nola seamount current depiction at 18 m to 26 m and 50 m to 58 m depths (currents averaged at 0.1 NM) and a cross-section display of the shearing observed over the southwestern seamount

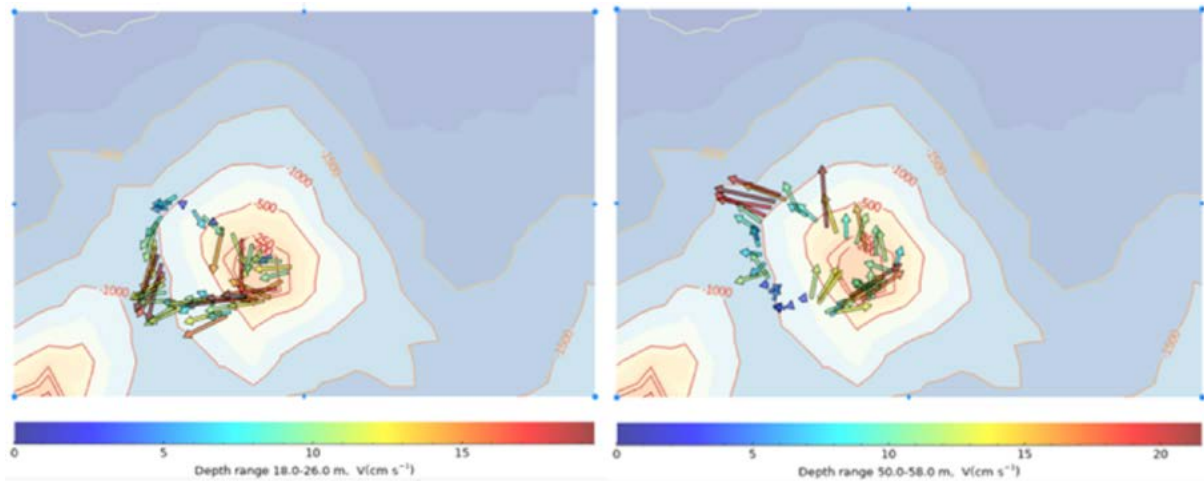


Figure 19. Northwestern Nola Seamount current depiction at 18 m to 26 m and 50 m to 58 m depths (currents averaged at 0.25 NM)

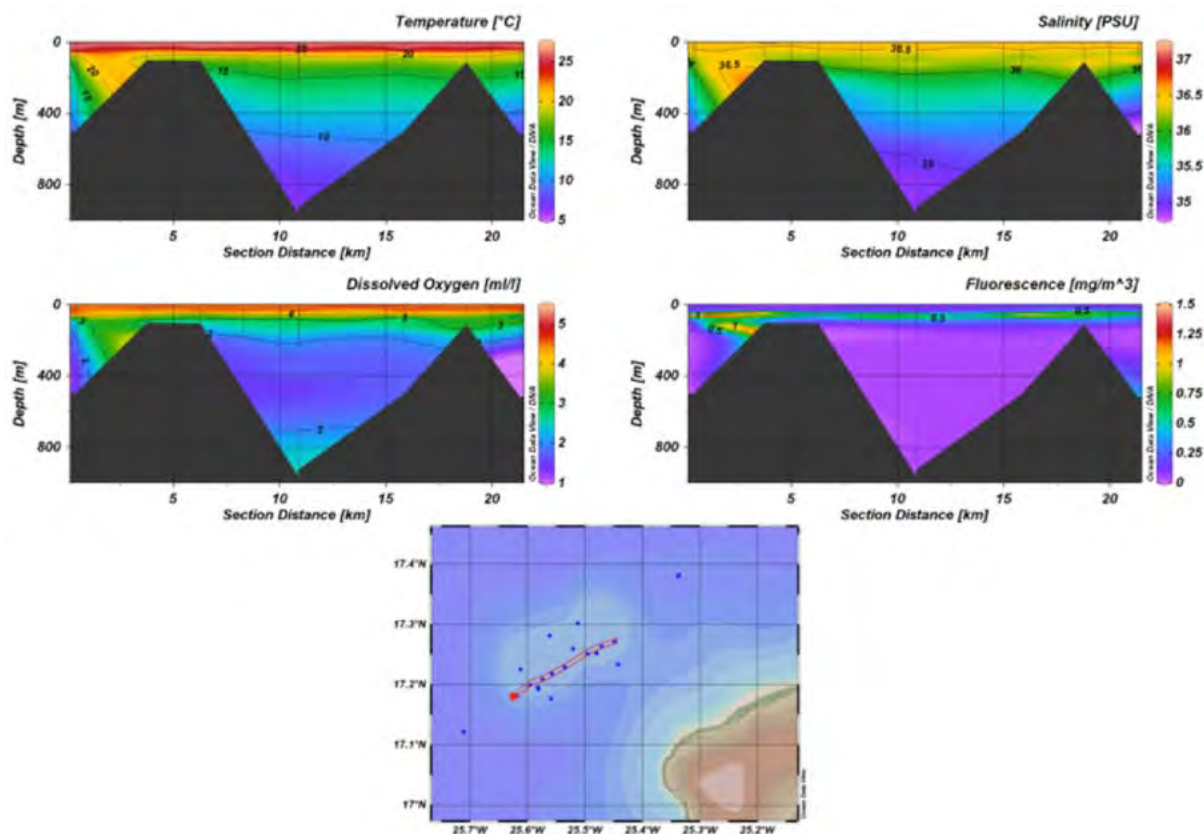


Figure 20. CTD transect over the Nola Seamount depicting temperature, salinity, dissolved oxygen and fluorescence. The two outer CTD stations were removed from the figure to avoid questionable interpolation.

Plankton

A total of five samples were collected with the multinet Midi (180 μ m) at CVOO station.

A total of 161 samples were collected from the 26 stations sampled in Nola Seamount with the multinet Mammoth.

3.12.2 *Sargassum* spp. sampling

Sargassum spp. was found at three trawl stations and three plankton stations, all in the southern part of the archipelago, around the islands of Fogo and Brava. At each station, a few specimens of *Sargassum* spp. were taken aside for close up photography of certain details, and then frozen at -20°C for further analysis on land.

3.13 Capacity building

Capacity building and hands-on training in scientific sampling and reporting were carried out in the fields of survey planning, procedures and standardized survey methodology, species identification, use of state-of-the art scientific equipment, as well as in execution and reporting from the survey. In addition, assistance was given in handling of the data collected, and in conducting preliminary analyses to be used in the survey report. Several scientific meetings were arranged during the survey and an overview of these can be found in Annex 5.

3.14 Data recorded and collected samples

In line with the Nansen Data Policy, the Cruise Leader ensured that each national institution represented on board the vessel received a copy of the draft report and the basic data pertaining to the particular survey and for their national waters, before leaving the vessel.

The cruise participants have signed the Data Policy and agreed to collaborate with scientists from other partner countries and through the EAF-Nansen Science Plan.

An overview of samples collected, people responsible for analysis and status of analysis by the time of the post-survey meeting, can be found in Annex 6.

Persons responsible for samples collected reported on the status of the analysis at the post-survey meeting. A copy of the analysis/results must be sent to IMR (Nansen_data@hi.no). The reason for this is to ensure that IMR, as a data custodian, has an overview of the samples for various purposes, such as:

- data custodian purposes;
- use in capacity building and workshops;
- collaboration and publication through the EAF-Nansen Science Plan;
- as reference libraries and DNA barcoding; and
- for more efficient planning of future surveys in the same area.

IMR has a responsibility to develop and maintain a functioning and updated system for the storage, management and retrieval of all the data collected during the surveys, or produced from activities performed during the surveys, and to make available any part of these data and scientific information as and when required. All requests for data should be sent to FAO (www.fao.org/in-action/eaf-nansen/data-access-requests/en).

Data collected and an overview of procedures for data hand-over to partners can be found in Annex 7.

4 CONCLUDING REMARKS

Leg 2 had multiple objectives, almost all of which were successfully achieved. Samples were collected according to the Sailing Orders. Due to some adverse weather conditions, some stations were dropped and some additional sampling was carried out, to obtain better biological and oceanographic information in specific areas. The information presented in this report summarizes the results of the data analysed during the survey. Some samples and data have been transported to research institutions in the region, to IEO in Spain and to IMR in Norway. These samples will be analysed and reported on later.

The acoustic recordings were low for the whole area covered and, as in 2011, no clupeids were found (Figure 21). On one occasion only the trawl was deployed based on strong acoustic signals. This turned out to be a very dense aggregation of mesopelagic fish. However, the abundance of pelagic fish is assessed to be 6 430 tonnes, exclusively belonging to the PEL2 group. This is more than double the roughly 3 000 tons found during the 2011 survey. The geographical distribution of some of the pelagic fish species did vary slightly between the two surveys, with pelagic fish mainly occurring south of the northern islands in 2021, while they were distributed north of the same islands in 2011. The recordings of PEL2 fish south of Boa Vista occurred in both of the surveys.

Species like mackerel scad, round scad and false scad accounted for the main pelagic species found during the 2011 Nansen survey in Cabo Verde. During the 2021 survey, only a few individuals of round scad were caught and neither of the other two species were caught. However, another very important species (monitored by IMar), *Selar crumenophthalmus*, were found in some of the catches.

The catch rates (density in tonnes per NM²) of the demersal trawls were far higher in 2021 than in 2011 (around 600 tonnes per NM², compared to 95 tonnes per NM²). In fact, the catch rates of the all the main categories (demersal fish, pelagic fish, cephalopods, sharks, rays and “other”, in all depth levels and in both areas, were higher in 2021 or similar to 2011 (Table 10). Shrimps were caught in negligible amounts in both surveys and have been excluded. The highest catch rates of demersal fish were recorded in 2021, in the depth interval between 30 m to 50 m (68.27 tonnes per NM²), while the highest catch rates of demersal fish recorded in 2011 were found in deeper waters, between 100 m and 500 m (18.60 tonnes per NM²). At this depth interval (100 m to 500 m) the catch rates were more or less the same for both surveys. The catch rates of sharks were in total four times higher in 2021 than in 2011. Catch rates for sharks increased markedly in the depth interval 50 m to 100 m. Catch rates for rays were also much higher in 2022 than in 2011. In 2021 only two of the demersal species monitored by IMar were caught (*Pseudupeneus prayensis* and *Lithognathus mormyrus*).

Microplastic-like items were found in only five stations of the 26 in which the Manta net was deployed, indicating low concentrations of microplastics in the surroundings of Cabo Verde.

Marine debris was registered at 11 trawl stations, mostly with one piece of litter per station. One station (station 38) had three pieces of litter. Reference photographs of the individual items were taken at three of the stations. Marine debris, remains of fishing equipment and other evidence of human activities were found in trawl catches as deep as 150 m.

Only two of the pelagic species preselected for nutrition and food safety analysis (microplastics and parasites), namely *Auxis thazard* and *Selar crumenophthalmus*, were caught in the trawls and collected for this purpose. None of these species were caught at more than one station, thus there are no replicates as was requested in the Sailing Orders (Annex 1). Additionally, three species (*Antigonia capro*, *Ariomma melanum* and *Maurolicus* spp.) with high ecosystem importance were opportunistically sampled at one station each.

High biodiversity was found in some of the catches. Two hauls were conducted at 800 m and each of them contained more than 80 different species. Most of the species were identified, others were preserved in ethanol and were sent to experts on land for verification, further investigation and archived for potential new studies. In total, more than 400 species were registered during the survey.

The topography (on land and underwater) in the area of Cabo Verde is complex, with a narrow and steep shelf, from 30 m to 2 000 m within a short distance. Some of the islands are flat (age and erosion) while others are mountain-like with heights of >2 000 m. Previous studies from this area, indicate complex current patterns with frequently occurring mesoscale eddies, a high mixing rate of water masses, areas with high turbidity and other areas with stratified conditions. Cabo Verde is an area with coastal upwelling, high biodiversity and part of the large-scale OMZ band/area. Two monitoring stations have been sampling since 2006. A particle distribution model that has been developed for Cabo Verde can be further tuned based on the collected data from this survey. The model can be used to improve understanding of dispersal patterns of fish eggs and larvae between the islands. So far, this model lacks depth stratified data and information about spawning areas and spawning times for both fish and plankton, information that future analysis of the samples collected during this survey can provide.

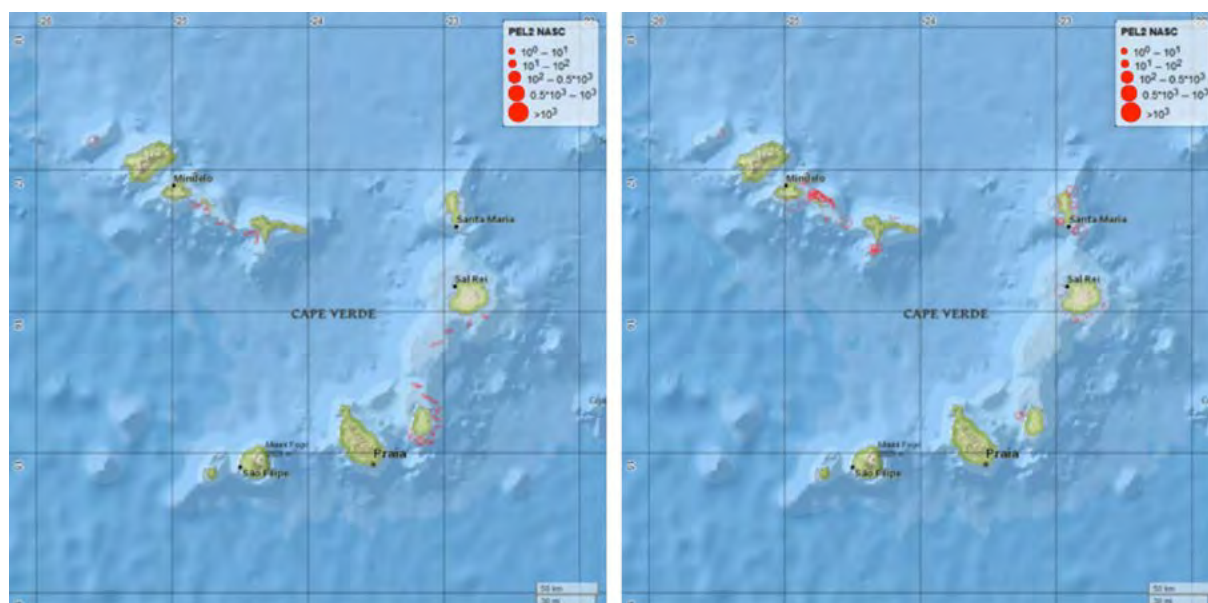


Figure 21. Comparison of NASC in June 2011 (left) and November/December 2021 (right)

Table 10. Catch rates (density in tonnes per NM²) of the demersal trawl hauls in June 2011 and November/December 2021

Year	Area	Depth stratum (m)	Demersal	Pelagic	Cephalopods	Sharks	Rays	Other	Total
2011	Cabo Verde islands, east	30–50	3.01	2.15	0.00	1.73	1.10	28.74	36.73
2011	Cabo Verde islands, east	50–100	0.56	0.37	0.02	0.57	1.99	6.56	10.07
2011	Cabo Verde islands, east	100–500	18.60	0.01	1.48	0.21	0.02	16.14	36.47
2011	Cabo Verde islands, northwest	30–50	0.00	0.36	0.00	0.31	0.00	1.32	1.99
2011	Cabo Verde islands, northwest	50–100]	0.00	0.90	0.02	2.14	4.03	2.41	9.50
2011	Cabo Verde islands, northwest	100–500	0.00	0.00	0.06	0.00	0.00	0.64	0.70
2011	Total		22.17	3.79	1.57	4.96	7.14	55.81	95.45
2021	Cabo Verde islands, east	30–50	68.27	2.72	0.09	2.43	0.72	43.41	117.64
2021	Cabo Verde islands, east	50–100	0.78	3.70	0.63	11.72	7.12	68.55	92.49
2021	Cabo Verde islands, east	100–500	16.58	0.06	2.81	0.34	0.24	321.49	341.52
2021	Cabo Verde islands, orthwest	30–50	0.44	1.98	0.00	1.69	6.61	9.92	20.65
2021	Cabo Verde islands, northwest	50–100	2.80	13.90	0.35	2.11	0.00	6.47	25.63
2021	Cabo Verde islands, northwest	100–500	0.04	0.00	0.29	0.00	0.01	2.62	2.96
2021	Total		88.92	22.36	4.17	18.30	14.69	452.46	600.89

5 REFERENCES

- Becker, S., Aoyama, M., Woodward, E.M.S., Bakker, K., Coverly, S., Mahaffey, C. & Tanhua, T. 2019. *GO-SHIP repeat hydrography nutrient manual: The precise and accurate determination of dissolved inorganic nutrients in seawater, using Continuous Flow Analysis methods. Version 1.1.* Go Ship Program and SCOR. 56 pp. <http://www.go-ship.org/HydroMan.html>
- Clayton, T.D. & Byrne, R.H. 1993. Spectrophotometric seawater pH measurements: Total hydrogen ion concentration scale calibration of m-cresol purple and at-sea results. *Deep-sea Research Part I – Oceanographic Research Papers*, 40: 2115–2129. [https://doi.org/10.1016/0967-0637\(93\)90048-8](https://doi.org/10.1016/0967-0637(93)90048-8)
- Chierici, M., Fransson, A. & Anderson, L.G. 1999. Influence of m-cresol purple indicator additions on the pH of seawater samples: Correction factors evaluated from a chemical speciation model. *Marine Chemistry*, 65: 281–290. [https://doi.org/10.1016/S0304-4203\(99\)00020-1](https://doi.org/10.1016/S0304-4203(99)00020-1)
- Dickson, A.G., Sabine, C.L. & Christian, J.R., eds. 2007. *Guide to best practices for ocean CO₂ Measurements*. PICES Special Publication, 3. IOCCP Report No. 8. Sidney, Canada, North Pacific Marine Science Organization. 191 pp.
- Emery, W. 2001. Water types and water masses. *Encyclopedia of Ocean Sciences*, p. 3179–3187. Academic Press.
- Grasshoff, K., Erhardt, M. & Kremling, K., eds. 1983. *Methods of seawater analysis*. Weinheim, Wiley-VCH.
- Harbitz, A. 2019. A zigzag survey design for continuous transect sampling with guaranteed equal coverage probability. *Fisheries Research*, 213: 151–159.
- Johnsen, E., Totland, A., Skålevik, Å., Holmin, A.J., Dingsør, G.E., Fuglebakk, E. & Handegard, N.O. 2019. StoX: An open source software for marine survey analyses. *Methods in Ecology and Evolution*, 10(9): 1523–1528. <https://doi.org/10.1111/2041-210X.13250>
- Langdon, C. 2010. Determination of dissolved oxygen in seawater by Winkler titration using the amperometric technique. In: E.M. Hood, C.L. Sabine & B.M. Sloyan. *The GO-Ship repeat hydrography manual: A collection of expert reports and guidelines. Version 1*, p. 1–18. IOCCP Report No. 14; ICPO Publication Series No. 134. <https://doi.org/10.25607/OBP-1350>
- Lloris, D., Rucabado, J. & Figueroa, H. 1991. Biogeography of the Macaronesian ichthyofauna (The Azores, Canary Islands, Cape Verde and the African enclave). *Boletim do Museo Municipal do Funchal (Madeira)*, 43(234): 191–241.
- Merino, S.E., Moses, C., Jurado, L.F., Jann, B. & Rolán, E. 2002. *Marine biodiversity in the Cape Verde Islands: importance and needs for conservation*. Poster presented at the 4th Symposium on Fauna and Flora of the Atlantic islands, 9–13 September 2002, Praia, Cape Verde.
- Pelegri, J., Peña-Izquierdo, J., Machín, F. & Meiners-Mandujano, C. & Presas-Navarro, C. 2017. Oceanography of the Cabo Verde Basin and Mauritanian Slope waters. In: A. Ramos, F. Ramil & J.L. Sanz, eds. *Deep-sea ecosystems off Mauritania. Research of marine biodiversity and habitats in the northwest African margin*, p. 119–154. Springer Science and Business Media B.V.
- Peña-Izquierdo, J., Van Sebille, E., Pelegri, J.L., Sprintall, J., Mason, E., Llanillo, P.J. & Machín, F. 2015. Water mass pathways to the North Atlantic oxygen minimum zone. *Journal of Geophysical Research Oceans*, 120(5): 3350–3372.
- Roberts, C.M., McClean, C.J., Veron, J.E.N., Hawkins, J.P., Allen, G.R., McAllister, D.E., Mittermeier, C.G., et al. 2002. Marine biodiversity hotspots and conservation priorities for tropical reefs. *Science*, 295: 1280–1284.
- Toresen, R., Grøseter, H. & de Barros, P. 1998. The acoustic method as used in the abundance estimation of capelin (*Mallotus villosus* Muller) and herring (*Clupea harengus* Linne) in the Barents Sea. *Fisheries Research*, 34: 27–38.

ANNEX 1. SAILING ORDERS

RV Dr Fridtjof Nansen Leg 2 G2 ecosystem survey in Cabo Verde

A1.1 Annual survey plan

The survey plan for 2021 had to be substantially reduced because of the COVID-19 pandemic and only two surveys will be conducted by the *RV Dr Fridtjof Nansen*. The survey programme for 2021 will include a dedicated bottom habitat habitat/environmental monitoring survey outside Mauritania and Senegal and an ecosystem survey of Cabo Verde. The survey in Cabo Verde will cover shallow (about 20 m depth) to the upper waters slope (about 1000 m depth). The survey will start and end in Las Palmas, Spain (Table A1.1).

Table A1.1. Dates, port calls and duration of the survey in Cabo Verde

Survey name	Duration in days	Departure	Arrival	Port of Departure	Port of Arrival
Leg 2 ecosystem survey in Cabo Verde	27	19/11/2021	15/12/2021	Las Palmas	Las Palmas

A1.2 Survey plan for leg 2 ecosystem survey in Cabo Verde

A1.2.1 Survey objectives

The overall objectives of the surveys, as indicated in a message by the Minister of Maritime Economy, Paulo Lima Veiga, to the Coordinator of the EAF-Nansen Programme, Merete Tandstad, dated 31 July 2020, are to obtain an assessment of the demersal and pelagic resources of the continental shelf and upper slope, while determining the distribution and abundance of demersal and pelagic species and communities, and collecting biological samples as specified below. Sampling will also be carried out to determine oceanographic conditions (physical, chemical and biological), as well as sampling for microplastics, recording the occurrence of marine debris and mapping of nutrients and contaminants in commercial fish for the purposes of determining food safety.

Based on the above, the following is proposed (in a prioritized order):

- A combined demersal and pelagic survey for biomass estimation of resources;
- Standard biological sampling of priority species (including length, weight, sex and maturity);
- Fish samples for nutrients, contaminants and the presence of microplastics in the most common species;
- Standard sampling of environmental variables with CTD and water sampling (salinity, temperature, dissolved oxygen, fluorescence, dissolved nutrients, pH, total alkalinity and chlorophyll *a*);
- Measuring currents with an ADCP;
- Sampling of phytoplankton, zooplankton and ichthyoplankton;
- Sampling with manta trawl for the presence of microplastics in surface waters;
- Underway pCO₂ measurements in surface water; and
- CVOO mooring station hydrographic and plankton sampling.

The objectives were further discussed in a meeting held on 29 June 2020 between IMar, FAO and IMR. This document (Sailing Orders) was also discussed at two pre-survey meetings, held on 25 August and 14 September 2020.

A1.2.2 Survey area

The area to be surveyed by the RV *Dr Fridtjof Nansen* in 2021 includes the continental shelf and upper slope of Cabo Verde, from 20 m to 1 000 m (Figure A1.1).

Standard sampling will be carried out, as is carried out on most surveys conducted by RV *Dr Fridtjof Nansen* (e.g. oceanographic conditions, including oceans pH and carbonate system, primary productivity, food safety, microplastics, marine debris, biodiversity, pelagic and demersal fish resources). Sampling protocols are standardized to the extent possible to allow comparability across larger geographic scales.

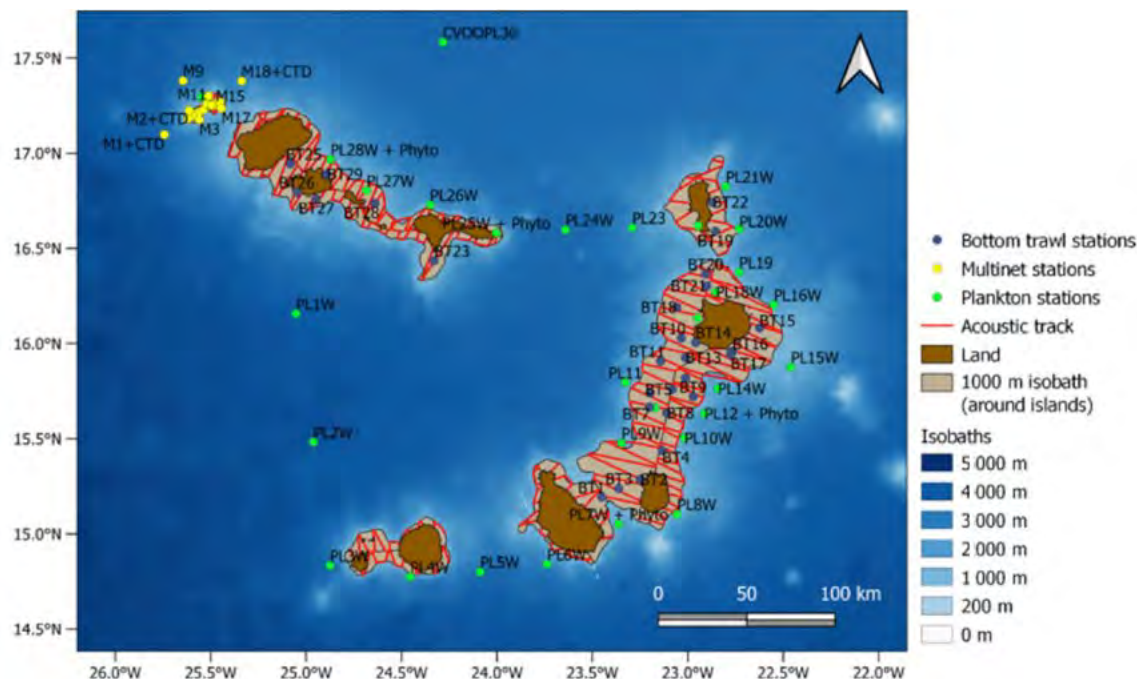


Figure A1.1. Survey area with indication of environmental stations and potential positions for demersal trawling

Notes: CVOOPL30 = mooring station; six plankton stations include additional phytoplankton sampling (+ phyto); five multinet stations include additional CTD sampling (+ CTD).

A1.2.3 Proposed participation in different legs and experience required

Typically, the surveys with the RV *Dr Fridtjof Nansen* include participants from different countries and institutions. A combination of experienced and less experienced scientists and technicians will receive on-the-job training in sampling procedures. An overview of the main types of expertise and number of participants required for Leg 2 is provided in Table A1.2 and the list of planned survey participants is shown in Appendix A1.1 (in Annex 1).

Table A1.2. Main types of expertise required and number of participants (scientific team only)

Leg	Position/ laboratory	DFN core team	Cabo Verde	External experts
Leg 2	Cruise leader	1		
Leg 2	Co-cruise leader	1	1	
Leg 2	Water chemistry	1	3	
Leg 2	Physical oceanography			1
Leg 2	Plankton biology	1	4	
Leg 2	Fisheries biology/taxonomy	2	8	1
Leg 2	Benthos			
Leg 2	Nutrition and fish safety			
Leg 2	Whale/seabird observatory			
Leg 2	Local authority			

A1.3 Scientific rationale

The overall framework and rationale for the surveys carried out by the RV *Dr Fridtjof Nansen* are provided by the EAF-Nansen Programme Science Plan which covers three main pillars (Figure A1.2). Within this framework, the scope and objectives of the surveys in 2021 have been identified with partners, based on national and regional priorities. In addition, data recorded and samples collected during the survey will be analysed by means of collaborations initiated through the various Science Plan themes. An overview of the equipment that is going to be used, data expected to be recorded and samples planned to be collected can be found in Appendix A1.2 (in Annex 1), Annex 6 and Annex 7, respectively.

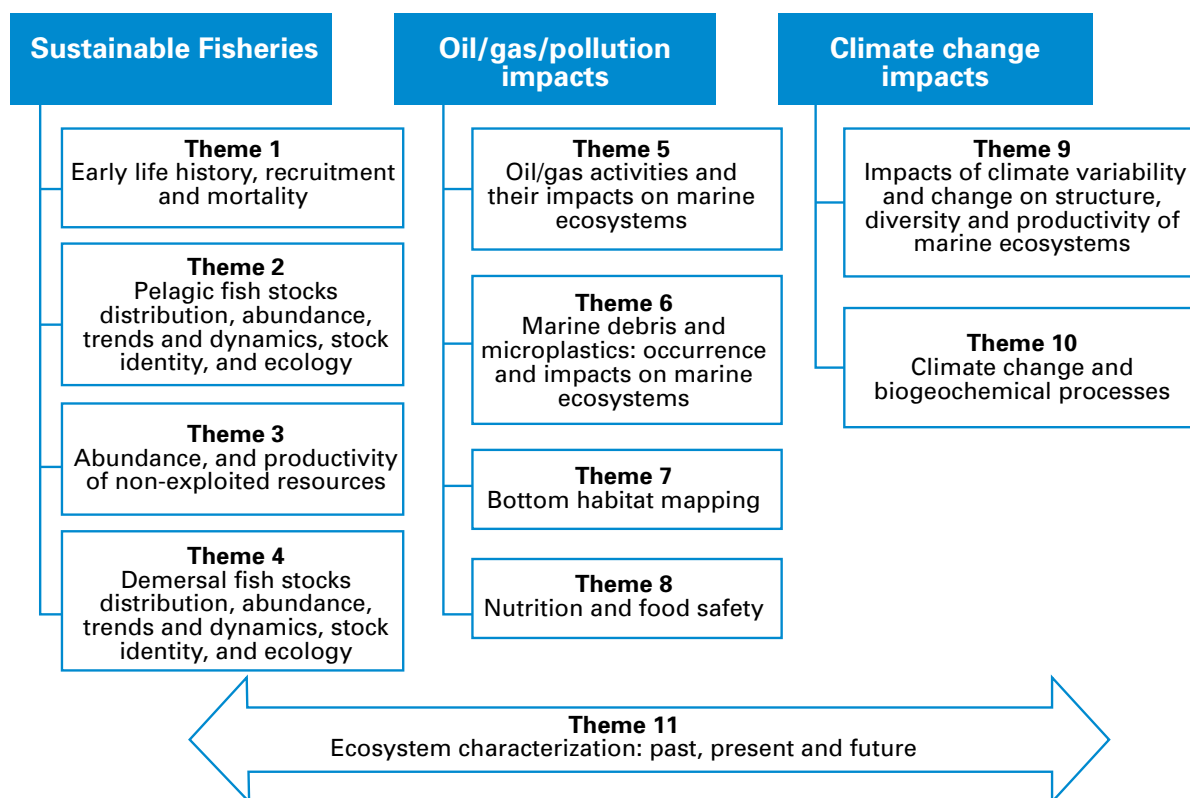


Figure A1.2. Main pillars and research themes of the EAF-Nansen Programme

Specific objectives and scientific questions for the survey

The specific objectives of the survey are listed below; the emphasis will be based on local needs and scientific importance.

Hydrography

- To map the hydrographic and environmental conditions in the survey area (temperature, salinity, dissolved oxygen, chlorophyll-*a*, nutrients, pH, total alkalinity and ocean currents);
- To obtain information on dissolved oxygen concentrations, ocean acidification and the calcium carbonate saturation state relevant for calcifying organisms; and
- To capture OMZ and identify vulnerable or high-risk areas.

Primary productivity, zooplankton, ichthyoplankton and microplastics

- To describe the abundance and biomass patterns of phytoplankton of the region, as potential food for zooplankton and pelagic fish species;
- To describe the abundance and biomass patterns of the meso-zooplankton community, as well as its species composition;
- To provide information on the abundance patterns of ichthyoplankton community (fish eggs and

larvae), at the lowest possible taxonomic level to identify possible spawning areas for commercially valuable fish species;

- Map the occurrence of microplastics;
- Collect phytoplankton samples relevant to harmful algal blooms; and
- Carry out combined CTD and plankton sampling at selected stations on the Nola Seamount.

Demersal resources

- Catch rates, distribution and biological parameters (length, weight, sex and maturity staging) of priority demersal resources by means of demersal trawling; and
- Length–weight relationships for all other species not defined above as priority demersal resources.

Pelagic resources

- Distribution, abundance and biological parameters (length, weight, sex and maturity staging) of priority pelagic resources using hydro acoustics and trawling for target identification and biological sampling to provide survey-based biomass indices;
- Length–weight relationships for all other species not defined above as priority pelagic resources; and
- To collect samples for genetic analysis of selected pelagic species.

Other trawl-related sampling

- Distribution, abundance and biological parameters (length, weight, sex and maturity staging for males) of priority cartilaginous fish. Vulnerable or red listed species will be released and in those cases maturity staging will only be done by visual inspection of male genitals;
- Collection of samples for taxonomic courses in the framework of the EAF-Nansen Programme;
- Record the occurrence of seaweed (*Sargassum* spp.) in fish trawls and the Manta trawl and collect samples for genetic studies;
- Collect samples for levels of nutrients and contaminants in selected commercially important pelagic fish species;
- Record the occurrence of marine debris collected during demersal trawling by using the draft protocol and take photos for future inclusion in the protocol itself;
- Identify epibenthos occurring in the bottom trawl; and
- Collection of samples for taxonomic identification in the framework of the EAF-Nansen Programme.

Other activities

- Hands-on training in the various laboratories; and
- Communication through social media.

A1.4 Sampling methods

In Leg 2 the area will be surveyed for pelagic fish abundance estimation using the acoustic method, and for demersal species catch rates using bottom trawling. Specific environmental transects will be carried out during the survey. In addition, a CTD station will be carried out at each bottom trawl station.

The design for the pelagic fish abundance estimate will consist of transects extending from about 20 m to 1 000 m bottom depth (with acoustic recordings from below the echosounder blind zone to a maximum depth of 600 m). To attempt comparison with the survey in 2011, two main areas have been defined, one comprising the islands in the eastern part (Area 1= São Tiago, Maio, Boa Vista and Fogo) and the islands of the north-western part (Area 2= Santo Antão, São Vicente, Santa Luzia and São Nicolau) and of the Cabo Verde islands complex. Since a standard zig-zag pattern around an island will give areas

close to the coast a higher probability for being sampled than areas further away from the coast, each of the main strata around the island have been split in two (along the horizontal axis), creating 10 new strata (strata 0–9, see Figure A1.1). The abundance of pelagic fish in each of these strata will be summed to represent one value for Area 1 and one for Area 2.

The distance between the acoustic transects has been set using the surveyPlanner (<https://rdrr.io/github/Sea2Data/Rstox/man/surveyPlanner.html>) of the Rstox package. Here the acoustic transects are created based on the time available, the speed of the vessel and the optimal degree of coverage aimed for. This is an application linked to Stox, which will also be used for calculation of the abundance indices (<https://www.hi.no/hi/forskning/prosjekter/stox>).

Acoustic targets will be identified by pelagic and bottom trawling and acoustic densities will be allocated to main species groups according to the list given in Section 2.4 (Table 3). Bottom trawl stations for the estimation of catch rates of demersal resources will be placed at approximately the same positions as during the survey in 2011, to ensure comparability between the two surveys. The possibility of conducting additional bottom trawl stations will be explored and will depend on time availability.

CTD deployments with rosette water column sampling will be performed at each demersal trawl station in addition to the environmental stations. Additional environmental station sampling will also take place at the Cabo Verde monitoring station CVOO (see more details below).

Detailed sampling protocols for what is considered to be standard sampling on *Dr Fridtjof Nansen* can be found in the Nansen manual site.

Other sampling protocols, area specific information and detailed description of the sampling conducted, is provided below, with more detailed information presented in the appendices to this annex.

A1.4.1 Meteorological and hydrographic sampling in surface water

A1.4.1.1 Weather station

Wind direction, wind speed, air pressure, relative humidity, air temperature and solar radiation data are averaged every 60 seconds and logged continuously from the Aanderaa Automatic Weather Station 2700.

A1.4.1.2 Thermosalinograph

A SBE 21 SeaCAT Thermosalinograph ran continuously during the survey, obtaining samples at 4 m depth to measure seawater salinity and temperature every 10 seconds. The 4 m engine cooling water intake is also equipped with a Turner Designs C3 submersible fluorometer with added turbidity detection capability.

A1.4.1.3 Underway pCO₂

Water from the vessel's 4 m intake is pumped through the flow head of a CONTROS HydroC® CO₂ FT sensor for pCO₂ measurements determined via IR absorption spectrometry.

A1.4.1.4 Current speed and direction measurements

Ocean current data is collected with a vessel-mounted Teledyne RDI Ocean Surveyor acoustic Doppler profiler operating at 150 kHz with a depth range of 400 m. The ADCP runs in narrow band mode and averages data in 8 m vertical bins. Heading, pitch, roll and positional data are acquired by a Kongsberg Marine SEAPATH unit. Teledyne's VmDAS software is used to collect the raw current data, whereas the data is processed using software created at IMR.

A1.4.2 Hydrographic sampling in the water column

A1.4.2.1 Conductivity, temperature and depth

A Sea-Bird 911plus CTD with the following sensors is mounted to a 12-bottle rosette water sampler for use at every hydrographic station. All sensor data logging and real-time profiling is performed using

Sea-Bird's Seasave software.

- 2 x SBE 3Plus temperature (T) sensors
- 2 x SBE 4C conductivity sensors for salinity (S)
- Digiquartz pressure (P) sensor
- SBE 43 dissolved oxygen (DO) sensor
- WET Labs ECO-AFL fluorometer (F)
- Satlantic photosynthetically active radiation (PAR) LOG ICSW sensor.

CTD deployments will be carried out at every trawl station as well as at every station where environmental sampling is conducted (“super-station”), in accordance with IMR’s standard depth designations shown in Figure A1.3. Additional sampling devices will be deployed for the collection of plankton (including ichthyoplankton) and microplastics. Extra CTD deployments will be carried out at each multinet station at Nola Bank (to 200 m) and six additional deep stations at Nola Bank. The water samples from the rosette bottles will be used for measurements of pH, total alkalinity, dissolved nutrients (nitrite, nitrate, silicate and phosphate), dissolved oxygen, chlorophyll-*a* and salinity. Detailed sampling protocols for these parameters can be found in the Nansen manual site (http://nansen-surveys.imr.no/doku.php?id=ctd_lab_information).

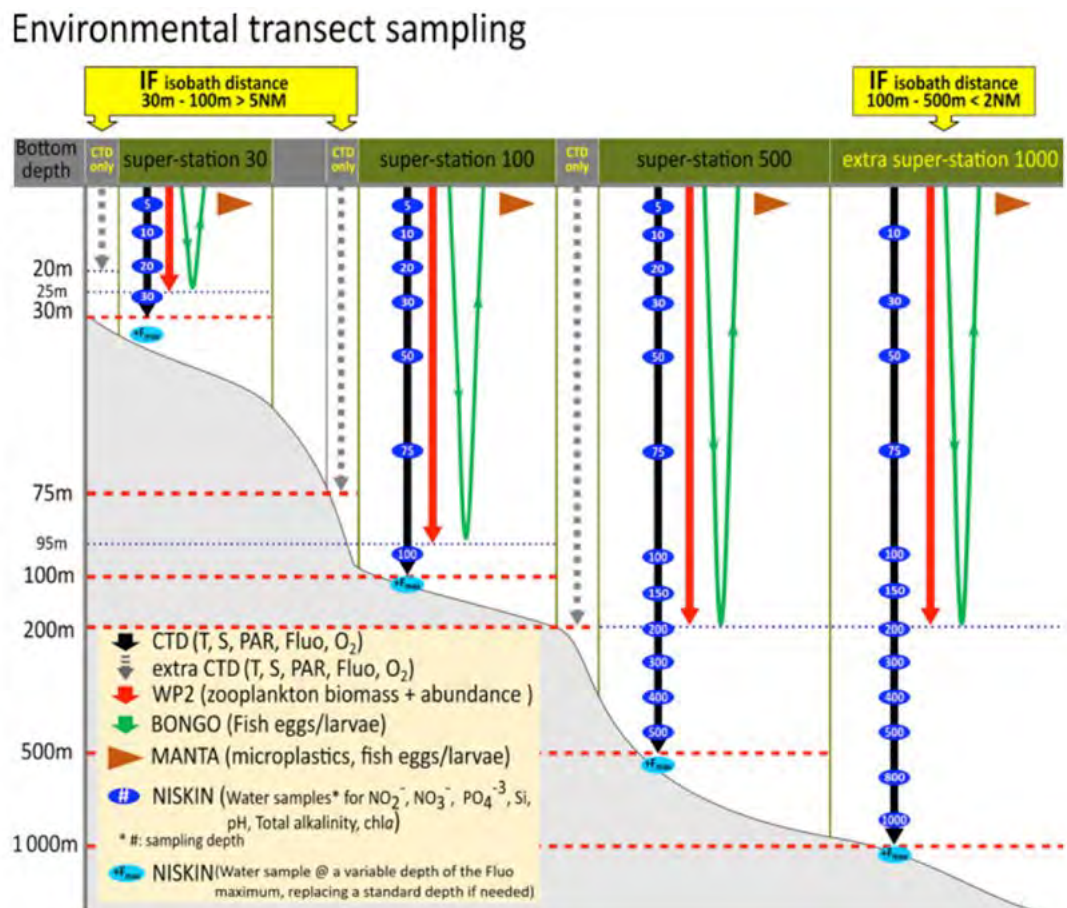


Figure A1.3. Sampling along environmental transects

A1.4.2.2 Ocean acidification parameters – pH and total alkalinity

Seawater samples for the analysis of pH and total alkalinity are collected together in 250 ml borosilicate glass bottles with silicone tubing. Samples are collected from the rosette water sampler in accordance with IMR’s standard depth designations. If oxygen water samples are not being collecting, these pH/AT

samples will be collected first from the rosette bottles due to their gas sensitivity that will alter the pH.

- pH is determined using an Agilent Cary 8454 UV-Vis Diode Array spectrophotometer and a 2-mM m-cresol purple indicator dye solution. (See the pH protocol for further details, in addition to sampling and analysis procedures.)
- Total alkalinity is determined via an open-cell potentiometric titration using a 0.05 M HCl solution with a sodium chloride background as the titrant (Dickson *et al.*, 2007). (See the total alkalinity protocol for further details, in addition to sampling and analysis procedures.)

To assess the ocean acidification state and change in the region, the pH and total alkalinity data will be combined with phosphate, silicate, temperature, salinity and depth data to calculate additional parameters of the carbonate system, such as dissolved inorganic carbon, $p\text{CO}_2$ and the calcium carbonate saturation states for aragonite and calcite.

A1.4.2.3 Dissolved nutrients

Seawater samples for the analysis of nitrite, nitrate, silicate and phosphate are collected in 20 ml polyethylene vials. Samples are collected from the rosette water sampler in accordance with IMR's standard depth designations. Samples are frozen for preservation until analysis in January 2022 with a SEAL QuAAtro39 continuous segmented flow analyser.

A1.4.3 Primary productivity

Seawater samples for the analysis of chlorophyll *a* and phaeopigments are collected in 269 ml high-density polyethylene bottles from depths ranging from 5 m to 200 m. Samples are filtered with 25 mm Munktel MG F filters with 0.7 μm particle retention on a 200 mm Hg vacuum pumped filtration system. The filters are stored in a -20°C freezer until ready for extraction for 15 to 24 hours with 10 ml of 90 percent acetone at 4°C . Samples are then centrifuged and transferred to cuvettes for analysis on a Turner Designs 10AU fluorometer (Welschmeyer, 1994; Jeffrey and Humphrey, 1975). Samples are first measured without acid for chlorophyll *a* determination and then a second time with two drops of 5 percent HCl for phaeopigment determination. The 10AU is calibrated approximately once a year with standards created from a chlorophyll *i* solid. (See the chlorophyll *a* protocol for further details in addition to sampling and analysis procedures.)

Qualitative phytoplankton samples are collected at selected stations in the studied area (Appendix A1.6). A phytoplankton net (35 cm in diameter and mesh-size of 10 μm), is hauled vertically at a speed $<0.1\text{ m s}^{-1}$ from the depth of 30 m to the surface (5 m above bottom at the 30 m stations). Samples will be preserved in 100 ml dark glass bottles with the addition of 2 ml of neutral Lugol solution.

A1.4.4 Zooplankton

Zooplankton samples are collected at 29 selected stations (Appendix A1.6). Samples will be collected by vertical hauls of a WP2 net (56 cm diameter, mesh size 180 μm) at a speed of 0.5 m sec^{-1} . The net will be towed within 5 m from the bottom to the surface, or from 200 m depth to the surface at deep stations. The plankton leader should ensure that this procedure is followed accurately on deck, write the actual depth of the tow (displayed in the monitor), and take flowmeter counts before and after the tow.

In summary, the WP2 samples will be processed as follows:

- The sample will be halved into parts with a Motoda splitter.
- One half will be used for biomass estimation (size fractionation through 2 000 μm , 1 000 μm and 180 μm mesh sizes).
- The second half will be preserved in 4 percent borax buffered formaldehyde solution and will be processed onboard through the FlowCam-Macro.

The exact procedure to be followed is described in detail in the Nansen Plankton Guidelines (see https://nansen-surveys.imr.no/doku.php?id=plankton_lab_information).

A1.4.5 Ichthyoplankton

Ichthyoplankton is collected at all zooplankton stations (Appendix A1.6), with double oblique tows of a Bongo net equipped with 405 µm nets. The Bongo will be towed obliquely within 5 m from the bottom or a maximum depth of 200 m to the surface at deep stations. Wire speed and the vessel speed should strictly follow the Nansen Plankton Guidelines (see https://nansen-surveys.imr.no/doku.php?id=plankton_lab_information).

Once the Bongo net is on board, the samples will be transferred in the lab and will be processed as follows:

- The sample of the left net (V) will be preserved directly in 4 percent borax buffered formaldehyde solution (especially made for ichthyoplankton).
- The sample of the right net (H) will be used for estimation of the zooplankton displacement volume (details in the Nansen Plankton Guidelines). Afterwards, the sample will be examined under the microscope and all fish larvae (and eggs if possible) will be sorted out. Plankton examination should be done thoroughly and in small amounts; during this process the rest of the sample should be kept cool (by the use of ice packs, fridge etc.). When sorting has finished, the bulk sample should be preserved in 96 percent ethanol (in bottles labelled as “sorted”). The sorted ichthyoplankton will be photographed and preserved for genetic analysis in 96 percent ethanol in small, labelled scintillation vials indicating clearly which net was used for sorting, the preservative, station, etc. In case of a lack of time or bad weather conditions, an alternative approach is to fix the total sample for ichthyoplankton without separating larvae from the sample onboard. However, to the extent possible this method should be avoided. More details on the sample processing can be found in the Nansen Plankton Guidelines (see https://nansen-surveys.imr.no/doku.php?id=plankton_lab_information).

If Bongo deployment is not possible due to net malfunction etc., ichthyoplankton will be sampled by multinet (Midi or Mammoth) according to the instructions in the Nansen Plankton Guidelines (see https://nansen-surveys.imr.no/doku.php?id=plankton_lab_information).

A1.4.6 Sampling at Cabo Verde Ocean Observatory and Nola Seamount

A1.4.6.1 Cabo Verde Ocean Observatory

The CVOO is located at 17N 36.40' – 24W 14.98. The following sampling is to be carried out:

Plankton

Stratified zooplankton sampling will be conducted at night by the use of a Midi multinet equipped with fine nets (180 µm mesh size). The net will be towed vertically at a speed 0.5 m sec⁻¹ and five depth strata (0 m to 100 m; 100 m to 200 m; 200 m to 300 m; 300 m to 600 m; and 600 m to 1 000 m) will be sampled. The entire samples will be preserved in 4 percent borax buffered formaldehyde solution (no biomass estimation will be done).

Hydrography

A CTD rosette water sampler will be deployed to approximately 3 630 m to obtain the full sensor profile of the water column. Beginning at 450 m, water samples will be collected according to Table 3. Dissolved oxygen and chlorophyll *a* will be analysed on board during the survey. Nutrients will be analysed on board in January 2022. Dissolved inorganic carbon/total alkalinity samples will be sent to GEOMAR after the survey for later analysis.

Table A1.3. Except for samples taken at 250 m and 150 m, all depths correspond to the R/V *Dr Fridtjof Nansen*'s standard sampling depths

Bottle	Depth	Oxygen	DIC TA	Nutrients	Chlorophyll <i>a</i>
11	450	1	2	1	
2	350	2	1	3	
3	250	1	1	1	
4	200	1	1	1	1
5	150	2	1	3	1
6	120	1	2	1	1
7	100	1	1	1	1
8	80	1	1	1	1
9	60	1	1	1	1
10	40	1	1	1	1
11	20	1	1	1	1
12	10	1	1	1	1

Notes: CTD to go down to 3 650 m but water bottle sampling will begin at 450 m.

A1.4.6.2 *Nola* Seamounts

Ichthyoplankton stratified sampling will be conducted at selected stations in the area of *Nola* Seamounts (Appendix A1.7) using oblique tows of a Mammoth multinet equipped with nets of 405 µm mesh size (downward speed 30 m min⁻¹, retrieval: 30 m min⁻¹). The exact sampling strata will be defined based on station depth and can be a maximum of six (i.e. 0 m to 25 m; 25 m to 50 m; 50 m to 75 m; 75 m to 100 m; 100 m to 150 m; and 150 m to 200 m). Nets with a mesh size of 405 µm will be mounted from the second to seventh position in the Mammoth frame (N2–N7). One additional net of 180 µm will be mounted at the first position (N1) and will be opened during the downcast for the collection of mesozooplankton.

Samples from nets N2–N7 (405 µm) will be preserved in 96 percent ethanol, while the samples from N1 (180 µm) will be split with a Motoda splitter. One half will be preserved in 4 percent formaldehyde solution and the other half in 96 percent ethanol.

A1.4.7 Microplastics

A1.4.8 Biological sampling for fisheries resources

Once the catch is on deck, it is assessed and, if necessary, subsamples are taken. At all trawl hauls, the catch is sorted according to species, and length and weight measurements are taken for all fish species using an electronic fish meter connected to a customized data acquisition system (Fish2Data and Biotic Editor) running on the server of the vessel. In addition, further biological sampling will be done for pre-agreed priority species.

Appendix A1.8 and Appendix A1.9 provide guidelines on the detailed sampling protocols. As preparation for the work in the fish lab, a training course will be provided on the use of the electronic fish meter, Fish2Data and Biotic Editor software.

A1.4.8.1 *Demersal* fish

Conduct length–weight measurements, length frequencies and standard biological sampling for all priority species at all stations where they are caught (Appendix A1.10). Length–weight measurements of 30 individuals from all other fish species found in the catches are also taken.

A1.4.8.2 *Pelagic* fish

Length–weight measurements, length frequencies and standard biological sampling for priority species at all stations where they are caught will be carried out (Appendix A1.10). Length–weight measurements will also be taken from 30 individuals from for all other species found in the catches. Genetic samples will be taken from selected species (Appendix A1.10) based on the standard sampling protocol for genetics (https://nansen-surveys.imr.no/doku.php?id=biological_sampling_procedures).

A1.4.8.3 Other trawl-related sampling

The protocol that will be followed for cartilaginous fish will be the same as for demersal and pelagic resources described in Appendix A1.10. Specimens for taxonomic courses will be collected following the protocol in Appendix A1.11. Species that cannot be identified should be preserved in formalin or frozen and sent to taxonomic experts for identification following the protocol in Appendix A1.14.

IMar is interested in individuals of crustaceans and molluscs (except for cephalopods) for curation in long-term storage (see Appendix A1.14). With respect to fish, cephalopods and sponges, the samples should be sent to IEO. However, in some cases IMar would also like to have samples of specific fish taxa, listed in Appendix A1.14 (preserved in formalin or ethanol). These samples will be off-loaded in Las Palmas, taken care of by IEO and then sent to Cabo Verde. The frozen samples will have to be sent together with other frozen samples (sea freight from Las Palmas to Bergen and then reshipped and sent to Cabo Verde).

In addition, IEO is interested in samples of all individuals of fish and crustaceans that those working in the fish lab have difficulty identifying. The protocol for preservation and labelling can be found in Appendix A1.14. IEO will be responsible for taking care of the samples when the vessel arrives in Las Palmas.

In cases where more than two partners are interested in the same individual, then IMar should have the samples.

For *Sargassum* spp. specimens, the protocol in Appendix A1.12 is to be followed.

For nutrition and food safety sampling, the protocol in Appendix A1.13 is to be followed.

Samples of selected commercially important pelagic species will be taken and analysed for nutrients profiles, contaminants and microplastics. Samples will be taken at different geographical coordinates.

A total of 25 specimens at three different stations will be frozen and later analyzed for determination of nutrients and food safety parameters.

Marine debris will be registered and classified at each station according to the categories listed in the sampling protocol for marine litter (https://nansen-surveys.imr.no/doku.php?id=biological_sampling_procedures). All pieces of litter should be weighed and counted, and when possible, photographs should be taken of the individual pieces for a reference catalogue. The data and the reference photographs should be recorded using Fish2Data and imported into the Biotic Editor. One person from each shift will be responsible for recording this data.

For epibenthos sampling, specimens in the trawl catches will be identified and pictures of all epibenthos (one picture per trawl, all species included, see Figure A1.4) should be taken.



Figure A1.4. Example of a photograph with epibenthos to be taken at each demersal trawl station

A1.4.8.4 Demersal fish catch rates

Due to the limited number of stations available for demersal trawling, only catch rates will be provided because the sampling cannot be considered representative of the whole surveyed area.

A1.4.8.5 Pelagic fish biomass estimation – acoustics

The biomass of pelagic fish will be estimated using the acoustic method as in previous surveys (https://nansen-surveys.imr.no/doku.php?id=biomass_calculations), using the 38KHz frequency transducer of the EK80 scientific split beam echosounders. The StoX [StoX (stoxproject.github.io)] software will be used for the biomass indices calculations.

Training of regional scientists in the processing of the collected acoustic data will be carried out during the survey. Table 3 (on page 19 and 20) shows the species listed within acoustic categories that will be used when scrutinizing the echograms.

A1.5 Data collected and standard procedure for handing over samples to partners

In line with the Nansen Data Policy, the cruise leader is to ensure that each national institution represented in the survey receives a copy of the draft report and the basic data (e.g. catch data) pertaining to the particular survey, and for national waters, before leaving the vessel, and document what data and samples have been shared and with whom (Appendix A1.3; Appendix A1.4).

The cruise participants are expected to sign and conform to the Data Policy of the EAF-Nansen Programme

Persons responsible for samples need to report on the status of analysis at the post-survey meeting, and send a copy of the analysis/results to IMR (nansen_data@hi.no). IMR, as custodian of the EAF-Nansen Programme, is required to have an overview of the samples/analysis at any point in time to:

- ensure that data life-cycle management is carried out in the expected way;
- use for capacity building purposes (e.g. workshops, training courses, etc.);
- advance collaboration and publishing through the EAF-Nansen Science Plan;
- maintain reference libraries (e.g. for taxonomic purposes) and DNA barcoding; and
- Efficiently plan future surveys in the same area.

A1.6 BIBLIOGRAPHY

A1.6.1 References

- Clayton, T.D. & Byrne, R.H.** 1993. Spectrophotometric seawater pH measurements: Total hydrogen ion concentration scale calibration of m-cresol purple and at-sea results. *Deep-sea Research Part I – Oceanographic Research Papers*, 40: 2115–2129. [https://doi.org/10.1016/0967-0637\(93\)90048-8](https://doi.org/10.1016/0967-0637(93)90048-8)
- Dickson, A.G., Sabine, C.L. & Christian, J.R., eds.** 2007. *Guide to best practices for ocean CO₂ Measurements*. PICES Special Publication, 3. IOCCP Report No. 8. Sidney, Canada, North Pacific Marine Science Organization. 191 pp.
- Jeffrey, S.W. & Humphrey, G.F.** 1975. New spectrophotometric equations for determining chlorophylls a, b c1 and c2 in higher plants, algae and natural phytoplankton. *Biochemie und Physiologie der Pflanzen*, 167(2): 191–194.

Reksten, A.M., Bøkevoll, A., Frantzen, S., Lundebye, A.-K., Kögel, T., Kolås, K., Aakre, I. & Kjellefold, M. 2020. Sampling protocol for the determination of nutrients and contaminants in seafood. EAF-Nansen Programme MethodsX 7, 101063. Rome, FAO.

Welschmeyer, N.A. 1994. Fluorometric analysis of chlorophyll *a* in the presence of chlorophyll *b* and phaeopigments. *Limnology and Oceanography*, 39(8): 1985–1992.

A1.6.2 Further reading

Chierici, M., Fransson, A. & Anderson, L.G. 1999. Influence of m-cresol purple indicator additions on the pH of seawater samples: Correction factors evaluated from a chemical speciation model. *Marine Chemistry*, 65: 281–290. [https://doi.org/10.1016/S0304-4203\(99\)00020-1](https://doi.org/10.1016/S0304-4203(99)00020-1)

Marques V., Peliz A., Lopes P., Moniz E., Morais A., Rosa T. L. & Almada E., eds. 1997. Campanha de Oceanografia e Avaliação de Pequenos Pelágicos na ZEE de Cabo Verde – NI “Capricornio” [Oceanographic assessment campaign in the EEZ of Cabo Verde – NI “Capricornio”]. Relatório Científico e Técnico [Scientific and Technical Report]. Fisheries and Sea Research Institute Cooperation Series No. 4. Lisbon. 99 pp.

Stramma, L., Brant, P., Schafstall, J., Schott, F., Fisher, J. & Kortzinger, A. 2008. Oxygen minimum zone in the North Atlantic south and east of the Cape Verde Islands. *Journal of Geophysical Research*, 113(4).

Stramma, L. & Schott, F. 1998. The mean flow field of the Tropical Atlantic Ocean. Deep-Sea Research, part II, Topical Studies in Oceanography, 46(1–2): 279–303.

Strømme, T. 1992. NAN-SIS: Software for fishery survey data logging and analysis. User’s manual. FAO Computerized Information Series (Fisheries), No. 4. Rome, FAO. 103 pp.

Zenk, W., Klein, B. & Schroder, M. 1991. Cabo Verde frontal zone. *Deep-Sea Research, part A. Oceanographic Research Papers*, 38(S1): S505–S530.

APPENDICES TO ANNEX 1

Appendix A1.1 Planned list of participants for Leg 2

Table A1.4. List of planned participants, role, e-mail address, gender, affiliation and country of origin

No	LEG	PARTICIPANT	ROLE	E MAIL	SEX	AFFILIATION	COUNTRY
1	Leg 2	Kathrine Michalsen	Cruise leader	kathrine@hi.no	F	IMR	Norway
2	Leg 2	Sarah Ann Bruck	Fish Team leader	sarah.ann.bruck@hi.no	F	IMR	Norway
3	Leg 2	Diana Zaera-Perez	Fish Team leader	diana.zaera-perez@hi.no	F	IMR	Norway
4	Leg 2	Sara Zamora Terol	Plankton Team leader	sara.zamora.terol@hi.no	F	IMR	Norway
5	Leg 2	David Cervantes	Chemical oceanography	david.cervantes@hi.no	M	IMR	Norway
6	Leg 2	Olaf J. Sørås	Chief instruments engineer	Olaf.soeraas.hi.no	M	IMR	Norway
7	Leg 2	Jori Neteland-Kyte	Instruments engineer		F	IMR	Norway
8	Leg 2	Vito Ramos	Co-cruise leader	vito.melo@imar.gov.cv	M	IMar	Cabo Verde
9	Leg 2	Sandra Correia	Fish Team	sandra.correia@imar.gov.cv	F	IMar	Cabo Verde
10	Leg 2	Alciany Luz	Fish Team	alciany.luz@imar.gov.cv	F	IMar	Cabo Verde
11	Leg 2	Carla Santos	Fish Team	limacarla92@gmail.com	F	IMar	Cabo Verde
12	Leg 2	Ailton Rocha	Fish Team	ailton.rocha@imar.gov.cv	M	IMar	Cabo Verde
13	Leg 2	Anibal Medina	Chemical/physical oceanography	anibal.medina@yahoo.com	M	IMar	Cabo Verde
14	Leg 2	Kateline Delgado	Fish Team	katecruz06@gmail.com	F	IMar	Cabo Verde
15	Leg 2	Rui Freitas	Fish Team	rfreitas@uta.cv	M	UTA	Cabo Verde
16	Leg 2	Valéria Lopes	Fish Team	vhfortes@uta.cv	F	UTA	Cabo Verde
17	Leg 2	Pérciles Silva	Chemical/physical oceanography	pericles.silva@oscm.cv	M	IMar	Cabo Verde
18	Leg 2	Ivanice Silva	Chemical/physical oceanography	ivanice.monteiro@oscm.cv	F	IMar	Cabo Verde
19	Leg 2	Dario Évora	Chemical/physical oceanography	dario.evora@imar.gov.cv	M	IMar	Cabo Verde
20	Leg 2	Elizandro Rodrigues	Biological oceanography	elizandro.rodrigues@oscm.cv	M	IMar	Cabo Verde
21	Leg 2	Keider Neves	Biological oceanography	biokeider2012@hotmail.com	M	Biosfera I	Cabo Verde
22	Leg 2	Marcia Costa	Biological oceanography	marcia.costa@imar.gov.cv	F	IMar	Cabo Verde
23	Leg 2	Chrislainne Alves	Biological oceanography	chrislainne.alves@outlook.com	F	IMar	Cabo Verde
24	Leg 2	Francisca A. Salmeron Jimenez	Fish Team; taxonomist	paqui.salmeron@ieo.es	F	FAO expert	Spain
25	Leg 2	Benjamin N'Guessan	Physical oceanographer	nguessan.k.benjamin@gmail.com	M	CRO	Côte d'Ivoire
26	Leg 2	Maik Tiedemann	Training (Cruise leader)	maik.tiedemann@hi.no	M	IMR	Norway

Appendix A1.2. Overview of sampled components and equipment to be used

Overview of the gear codes to be used in the Toktlogger system, detailed information about the rigging of the equipment as well as technical drawings and other specifications of the various equipment can be found in the Nansen wiki site (<https://nansen-surveys.imr.no/doku.php>).

Table A1.5. Overview of sampled components and equipment to be used

Component sampled	Equipment
Physical – air	Weather station (wind speed, direction, temperature, light sensor)
Physical – water	CTD, fluorescence sensor, PAR sensor; 150kHz ADCP; thermosalinograph (T,S,Fl); turbidity sensor
Chemical – water	Water samples and chemical analysis, pH, AT, pCO ₂ , nutrient analyser; fluorometer instrument, salinity, oxygen
Phytoplankton	Phytoplankton net (non-quantitative), water samples (quantitative), chlorophyll analysis from fluorometer instrument
Zooplankton	WP2 (180 µm), Bongo net (180 µm), multinet Midi (180 µm), echo sounders (EK 80), flow cam
Ichthyoplankton and jellyfish	Multinet Midi (180 µm), multinet Mammoth (180 µm and 405 µm), Bongo net (180 µm), CUFES (continuous underwater fish egg sampler)
Fish (incl. mesopelagic fish)	Pelagic (Åkra trawl, MultPelt trawl) and demersal trawls (Gisund Super), echo sounders (EK 80), omni-directional fisheries sonar (SH 90)
Benthos	Demersal trawl (Gisund)
Plastic	Manta trawl (180 µm), demersal and pelagic trawl
Mammals and seabirds	
Bathymetry	Olex (with input from EM 710)

Appendix A1.3. Data collected and overview of procedures for data hand-over to partners

See Annex 7.

Appendix A1.4. Samples to be collected, preservation and follow-up work

See Annex 6.

Appendix A1.5. Allocation of acoustic densities to species groups

The updated table is presented in Section 2.4 (Table 3).

Appendix A1.6. Plankton sampling locations for WP2, Bongo, Manta and phytoplankton sampling

Table A1.6. Plankton sampling locations for WP2, Bongo, Manta and phytoplankton sampling

Station	Longitude	Latitude	WP2	Bongo	Manta	Phytoplankton	Bottom depth (m)
PL1W	-25.053	16.157	x	x	x		3 900
PL2W	-24.962	15.483	x	x	x		4 120
PL3W	-24.875	14.833	x	x	x		3 480
PL4W	-24.455	14.774	x	x	x		2 140
PL5W	-24.088	14.800	x	x	x		3 360
PL6W	-23.737	14.840	x	x	x		2 580
PL7W + Phyto	-23.362	15.051	x	x	x	x	1 700
PL8W	-23.056	15.103	x	x	x		1 340
PL9W	-23.347	15.477	x	x	x		1 380
PL10W	-23.018	15.503	x	x	x		1 620
PL11	-23.326	15.796	x	x	x		2 060
PL12 + Phyto	-22.912	15.632	x	x	x	x	1 900
PL13W	-23.171	15.662	x	x	x		80
PL14W	-22.841	15.763	x	x	x		1 740
PL15W	-22.459	15.876	x	x	x		1 300
PL16W	-22.549	16.203	x	x	x		1 580
PL17	-22.948	16.134	x	x	x		28
PL18W	-22.854	16.270	x	x	x		43
PL19	-22.730	16.374	x	x	x		1 420
PL20W	-22.730	16.604	x	x	x		1 360
PL21W	-22.800	16.824	x	x	x		1 660
PL22W + Phyto	-22.946	16.619	x	x	x	x	20
PL23	-23.291	16.609	x	x	x		2 600
PL24W	-23.641	16.598	x	x	x		3 460
PL25W + Phyto	-24.003	16.581	x	x	x	x	1 060
PL26W	-24.350	16.729	x	x	x		1 720
PL27W	-24.684	16.804	x	x	x		20
PL28W + Phyto	-24.873	16.968	x	x	x	x	960
PL29W	-25.547	17.295	x	x	x		1 380
PL30W	-25.051	16.786	x	x	x		2

Appendix A1.7. Multinet stations over the Nola Seamounts

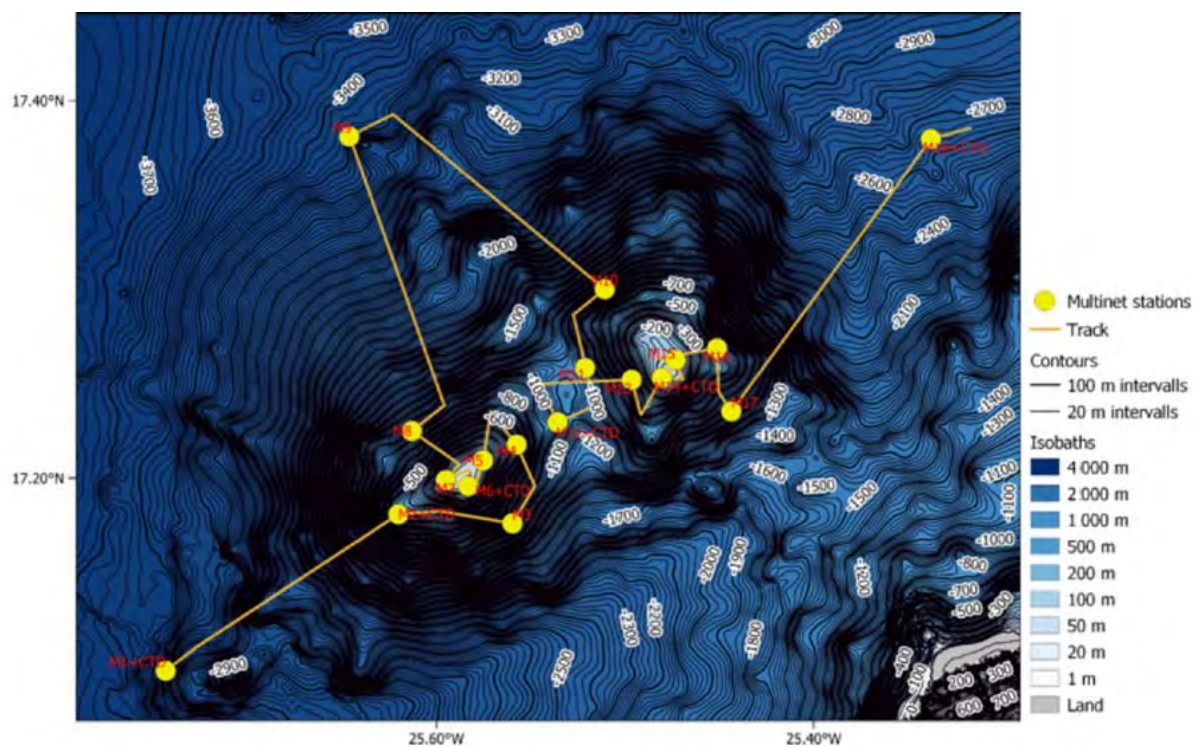


Figure A1.5. Map of selected multinet stations over the Nola Seamounts

Table A1.7. Multinet stations over Nola Seamounts

Station	Latitude	Longitude	Bottom depth (m)
M1 + CTD	17.098	-25.744	3 300
M2 + CTD	17.181	-25.620	500
M3	17.176	-25.560	1 000
M4	17.218	-25.558	500
M5	17.209	-25.575	100
M6 + CTD	17.196	-25.583	60
M7	17.199	-25.595	100
M8	17.225	-25.613	1 000
M9	17.352	-25.693	3 300
M10	17.300	-25.511	1 000
M11	17.258	-25.521	1 000
M12 + CTD	17.230	-25.536	1 000
M13	17.252	-25.497	500
M14 + CTD	17.253	-25.481	100
M15	17.263	-25.473	100
M16	17.269	-25.451	500
M17	17.235	-25.444	1 000
M18 + CTD	17.380	-5.338	2 640

Appendix A1.8. Overview of sampling procedures in the fish laboratory

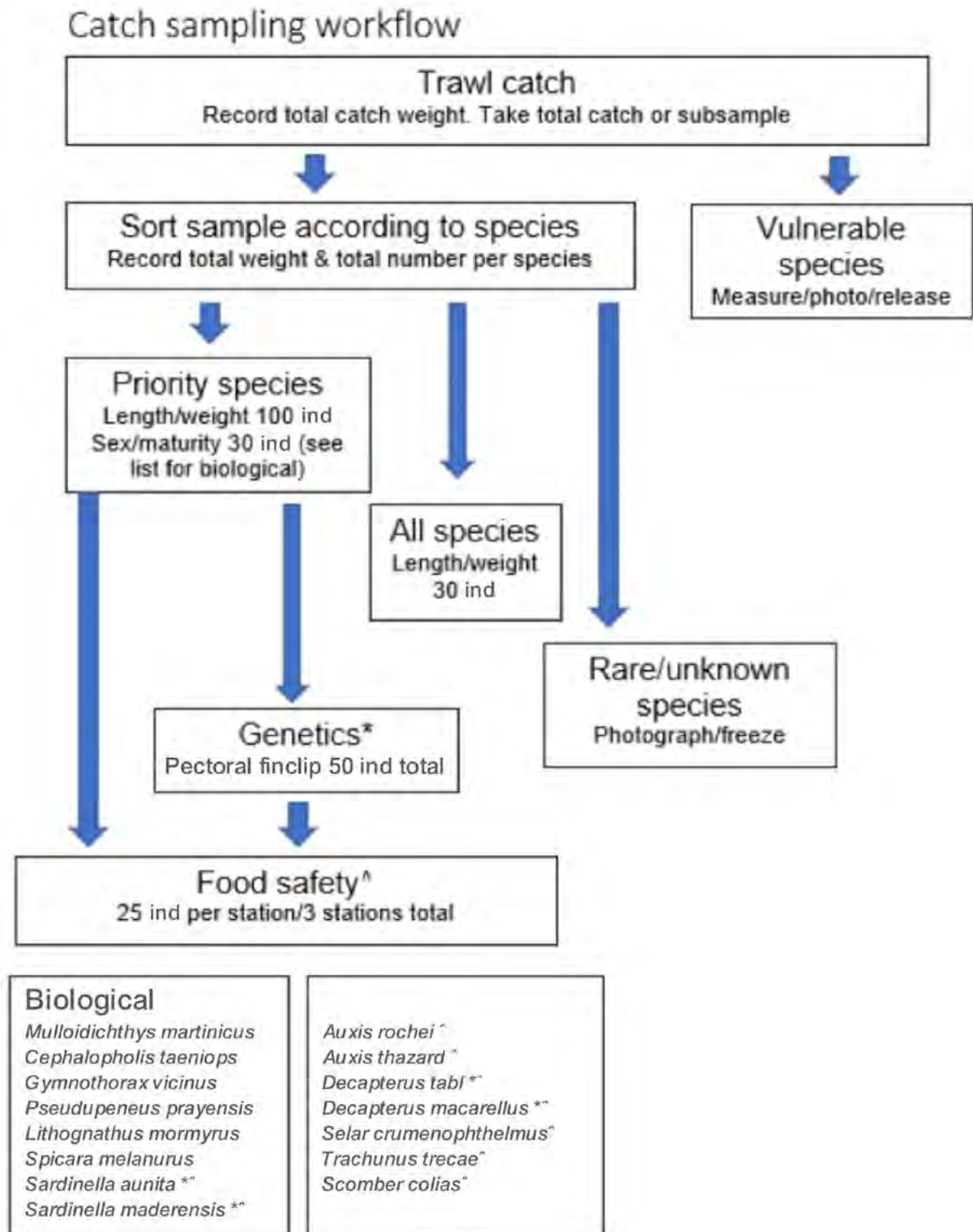


Figure A1.6. Diagram of sampling procedures in the fish laboratory

Notes: The priority species are listed in Appendix A1.10

Appendix A1.9. Biological scales used for maturity stages

Table A1.8. Biological scales used for maturity stages

Stage	State	Description
I	Immature	Ovary and testis about 1/3rd length of body cavity. Ovaries pinkish, translucent, testis whitish. Ova not visible to naked eye.
II	Maturing virgin and recovering spent	Ovary and testis about 1/2 length of body cavity. Ovary pinkish, translucent, testis whitish, symmetrical. Ova not visible to naked eye.
III	Ripening	Ovary and testis about 2/3 length of body cavity. Ovary pinkish-yellow with granular appearance, testis whitish to creamy. No transparent or translucent ova visible.
IV	Ripe	Ovary and testis from 2/3 to full length of body cavity. Ovary orange-pink with conspicuous superficial blood vessels. Large transparent, ripe ova visible. Testis whitish-cream, soft.
V	Spent	Ovary and testis shrunken to about 1/2 length of body cavity. Walls loose. Ovary may contain remnants of disintegrating opaque and ripe ova, darkened or translucent. Testis bloodshot and flabby.

Appendix A1.10. Biological sampling protocol for priority fish species

Table A1.9. Biological sampling protocol for priority fish species

Species	Catch weight number	Measured specimen lengths weights	Sex maturity	Genetic FINCLIPS	Whole specimens	Photograph	Nutrition and food safety sampling
All species	Every station	Length/weight of all abundant species			Species that are difficult to ID	Species that are difficult to ID	
PRIORITY SPECIES							
Demersal							
<i>Mulloidichthys martinicus</i>	Every station	Max 100/ station	Max 30/ station				
<i>Cephalopholis taeniops</i>	Every station	Max 100/ station	Max 30/ station				
<i>Gymnothorax vicinus</i>	Every station	Max 100/ station	Max 30/ station				
<i>Pseudupeneus prayensis</i>	Every station	Max 100/ station	Max 30/ station				
<i>Lithognathus mormyrus</i>	Every station	Max 100/ station	Max 30/ station				
<i>Spicara melanurus</i>	Every station	Max 100/ station	Max 30/ station				
Pelagic							
<i>Sardinella aurita</i>	Every station	Max 100/ station	Max 30/ station	50 individuals in total			Max 25 per station/3 stations
<i>Sardinella maderensis</i>	Every station	Max 100/ station	Max 30/ station	50 individuals in total			Max 25 per station/3 stations
<i>Auxis rochei</i>		Max 100/ station	Max 30/ station				Max 25 per station/3 stations
<i>Auxis thazard</i>		Max 100/ station	Max 30/ station				Max 25 per station/3 stations
<i>Decapterus tabl</i>		Max 100/ station	Max 30/ station	50 individuals in total			Max 25 per station/3 stations
<i>Decapterus macarellus</i>		Max 100/ station	Max 30/ station	50 individuals in total			Max 25 per station/3 stations
<i>Selar crumenophthalmus</i>		Max 100/ station	Max 30/ station				Max 25 per station/3 stations
<i>Trachurus spp.</i>		Max 100/ station					Max 25 per station/3 stations
<i>Trachurus trecae</i>		Max 100/ station	Max 30/ station	50 individuals in total			Max 25 per station/3 stations
<i>Scomber colias</i>		Max 100/ station	Max 30/ station				Max 25 per station/3 stations

Species	Catch weight number	Measured specimen lengths weights	Sex maturity	Genetic FINCLIPS	Whole specimens	Photograph	Nutrition and food safety sampling
<i>Trachurus picturatus</i>		Max 100/ station					
<i>Decapterus punctatus</i>		Max 100/ station					
<i>Caranx crysos</i>		Max 100/ station					
<i>Caranx hippos</i>		Max 100/ station					
<i>Caranx latus</i>		Max 100/ station					
<i>Caranx lugubris</i>		Max 100/ station					
<i>Elagatis bipinnulata</i>		Max 100/ station					
<i>Trachinotus ovatus</i>		Max 100/ station					
<i>Trachinotus goreensis</i>		Max 100/ station					
<i>Trichionotus teraia</i>		Max 100/ station					
<i>Selene dorsalis</i>		Max 100/ station					
<i>Seriola carpenteri</i>		Max 100/ station					
<i>Seriola dumerili</i>		Max 100/ station					
<i>Seriola rivoliana</i>		Max 100/ station					
<i>Katsuwonus pelamis</i>		Max 100/ station					
<i>Euthynnus alletteratus</i>		Max 100/ station					
<i>Apsilus fuscus</i>		Max 100/ station					
<i>Acanthocybium solandri</i>		Max 100/ station					
<i>Orcynopsis unicolor</i>		Max 100/ station					
<i>Sarda sarda</i>		Max 100/ station					
<i>Scomberomorus tritor</i>		Max 100/ station					
<i>Thunnus alalunga</i>		Max 100/ station					
<i>Thunnus albacares</i>		Max 100/ station					
<i>Thunnus obesus</i>		Max 100/ station					
<i>Sphyraena barracuda</i>		Max 100/ station					
<i>Sphyraena guachancho</i>		Max 100/ station					
<i>Sphyraena viridensis</i>		Max 100/ station					
Elasmobranchs		Max 100/ station	Max 30/ station (sex only)				
Vulnerable species	Every station	All	All/no				
Benthic epifauna	Every station				Unidentified species in formalin or ethanol	One picture per station – all species included	
Plastic/rubbish/ fishing gear	Every station						

Appendix A1.11. Protocol for sampling for taxonomic collections/courses

1. Fish and macroinvertebrate samples for regional taxonomy course (date and venue to be defined)

Representatives of selected families will be retained from the trawl catches to be used for the practical sessions of the taxonomy training course. The list is intended to provide collectors with a broad selection of taxa that might be available in the catches with the goal of having enough variety of species/families as well as sufficient specimens.

For the purpose of the training course, two to three selected specimens in good condition should be:

- placed in clearly labelled plastic bags;
- preserved in frozen condition, making sure the individuals are separated from each other at least until they freeze (consider separate bags for delicate taxa); and
- stored in clearly marked styrofoam boxes in the large walk-in freezer.

A log sheet is to be kept with an overview of the collected specimens. The log should contain the following information:

- species name
- station number
- number of individuals preserved.

The log is to be sent to peter.psomadakis@gmail.com at the end of the survey, together with trawl station information data.

Representative specimens from selected families/genera will be collected from the list provided (species_capeverde_species for collection IMR.xlsx)

2. Fish samples for IMR internal courses and training

Representative specimens from selected families/genera will be frozen in clearly labelled ziplock bags.

One or two specimens of each species from the list provided (species_capeverde_species for collection IMR.xlsx) are to be preserved. Specimens in good condition should be chosen. These samples are to be sent to IMR and preserved for training purposes.

If deep-water hauls are carried out, specimens are also to be preserved for training purposes and included to the list for IMR. A log sheet is to be kept (with survey number, station number and a comment with the photograph name if a photograph is taken).

Appendix A1.12. Protocol for seaweed sampling

Opportunistic samples collected in the trawl and/or manta net will be photographed and stored following the protocol below:

- collect sample (bulk);
- take photograph of sample collected (bulk);
- clean/rinse thalli of *Sargassum* spp. with clean sea water to remove attached organisms and adherent debris such as shells, sand, mud; and remove epiphytes by scraping gently with tissue paper. This is for molecular/DNA barcoding;
- Take a close-up photograph of single thallus of *Sargassum* spp. to show characteristic features: (i) branching pattern, of main axis, (ii) leaf-like appendage form/shape, (iii) margin of leaf-like appendage, (iv) air bladder, (with or without spine), (v) look for spines on main axis (stem);

- FOR MOLECULAR [DNA barcoding] preserve a small quantity (a small branch of *Sargassum* spp. sample, with leaf-like structures, which should cover an adult palm) from each collection (of the cleaned thalli in bullet 4); and
- preserve the above at –20 °C in a plastic bag with appropriate labelling (same information as for the samples from the sampling gear used).

Place remaining bulk sample (bullet 2) in plastic bag and store at –20 °C; label appropriately.

Appendix A1.13. Protocol for nutrition and food safety sampling

Sample selected pelagic species listed in Appendix A1.10 for analysis of nutrients and contaminants. N=25 of each species from 3 stations. If small catches, a minimum of five fish from each station. The fish should be stored whole in the freezer pending shipment to IMR. IMR will be responsible for analysing the samples (Reksten *et al.*, 2020). The fish will be stored in plastic bags marked with species, station and date. The trawl form with station information should follow the samples. An electronic copy should be sent to marian.kjellevold@hi.no.

Appendix A1.14. Protocol for nutrition and food safety sampling

Fish and macroinvertebrates samples for post-survey taxonomic studies

Specimens included in a list of relevant/problematic taxa (Cabo Verde survey fish taxa to be collected post taxonomy studies.xlsx) will be retained for post-survey taxonomy studies.

For this activity, the collectors will follow a standard step-by step sampling procedure (Sampling procedures onboard the R.V. *Dr. F. Nansen.docx*)

Specimens will be:

- measured and weighed;
- photographed in fresh condition if the species is unknown;
- tissue sampled if the species is unknown;
- individually labelled/tagged; and
- preserved either in frozen condition (large specimens) or fixed in formalin (small, medium-size specimens).

Benthic invertebrates samples

Benthic specimens of special interest or that are difficult to identify will be sampled according to a given protocol (Methods for sampling, processing, photographing, fixing and preserving benthic invertebrates' samples).

Specimens will be:

- photographed in fresh condition (label with station information and field collection number visible in photographs);
- fixed and preserved according to taxa specific methodology described in above mentioned document; and
- Stored in clearly labelled containers.

For both fish/macroinvertebrates and benthos, a log sheet is to be kept with an overview of the collected specimens.

ANNEX 2. DESCRIPTION OF INSTRUMENTS AND FISHING GEAR

Acoustic instruments

The Simrad EK80/18, 38, 70, 120 and 200 kHz scientific sounder was run during the survey. Scrutinizing was done in LSSS using the data from the 38-kHz transducer. The standard sphere calibrations were checked on 26 November 2021 in Flamenco Bay, São Vicente, Cabo Verde, using Cu64 for the 18 kHz. Due to some technical problems and increasing amount of plankton in the water as daylight decreased, another calibration took place on 2 December 2021, southwest of Maio for the Cu60, 38 kHz, WC38.1, 70, 120 and 200 kHz. The details of the settings for the 38-kHz echo sounder were as follows:

Table A2.1. Echosounder settings used during the survey

Transceiver2 menu 38kHz			
Transducer depth	6.84 m	SA correction	-0.09 dB
Absorption coeff.	9.7 dB/km	Angle sensitivity	18
Pulse duration	medium (1.024ms)	3 dB beamwidth	6.22° along ship
Bandwidth	34-38kHz		6.29 athwart ship
Max power	2000 Watt	Alongship offset	-0.05°
2way beam angle	0	Athwardship offset	-0.06°
gain	26.98 dB	Bottom detection menu	Minimum level -50dB

Fishing gear

The vessel has one small four-panel Åkrahamn pelagic trawl (Figure A2.1), one MultPelt 624 trawl (new in 2017)(Figure A2.2) and one “Gisund super bottom trawl” (Figure A2.3). All trawls were used during the survey. The smallest pelagic trawl has an 8 m to 12 m vertical opening under normal operation, whereas the MultPelt 624 trawl has a 25 m to 35 m opening.

The bottom trawl has a 31 m headline and a 47 m footrope fitted with 12” rubber bobbins. The codend has 24 mm mesh. The vertical opening is about 5.5 m. The distance between the wing tips is about 18 m during towing. The sweeps are 40 m long. The trawl doors are Thyborøen combi, 8 m² and weigh 2 000 kg. The door spread is about 45 m when using restraining rope. Trawling was conducted for species identification only and therefore no restraining rope was used during the survey.

The SCANMAR system was used during all trawl hauls. This equipment consists of sensors, a hydrophone, a receiver, a display unit and a battery charger. Communication between sensors and ship is based on acoustic transmission. The doors are fitted with sensors to provide information on their interdistance and angle, while a height sensor is fitted on the bottom trawl to measure the trawl opening and provide information on clearance and bottom contact.

All trawls are equipped with a trawl eye that provides information about the trawl opening and the distance of the footrope from the bottom. A pressure sensor is used to show the depth on the headline.

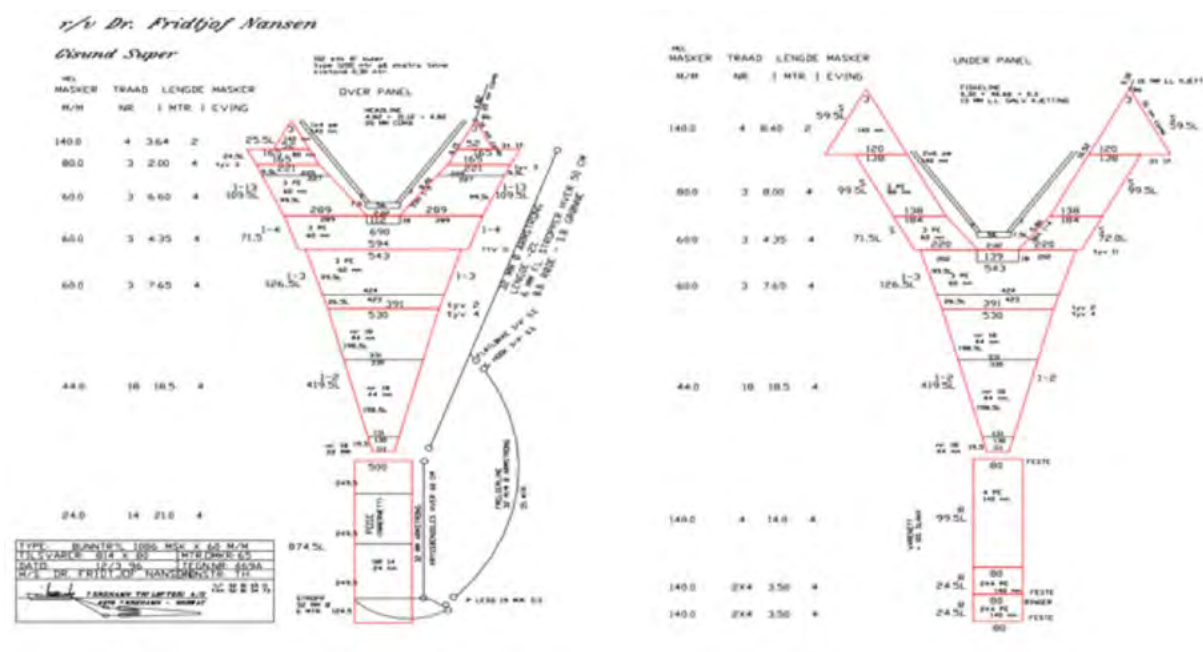


Figure A2.3. Technical drawing of the Gisund Super trawl

CTD sensors

Table A2.1. Quality assurance of the CTD sensors showing the serial number, model, calibration date and station number in which it was used

Type	Serial number	Model	Calibration date	Stations
Deck unit	11-1082	SBE 11plus		0443–0543
Pressure sensor	127957	DigiQuartz	21.09.2013	0443–0543
Underwater unit	1160	SBE 9plus 6800m	20.10.2018	0443–0543
Water sampler	32-0972	SBE 32 6800m		0443–0543
Conductivity sensor	3848	SBE 4C 6800m	08.07.2021	0443–0543
Conductivity sensor	2798	SBE 4C 6800m	22.04.2021	0443–0543
Oxygen sensor	3527	SBE 43 7000m	18.06.21	0443–0479
Oxygen sensor	3115	SBE 43 7000m	18.06.21	0480–0543
Submersible pump	52147	SBE 5T	2014	0443–0543
Submersible pump	54196	SBE 5T		0443–0543
Temperature sensor	1527	SBE 3plus 6800m	07.05.2021	0443–0543
Temperature sensor	1166	SBE 3plus 6800m	13.07.2021	0443–0543
Fluorometer	4892	FLRTD G4 0-125 µg	08.11.2017	0443–0543
Sonar Altimeter	71815	Benthos PSA-916	01.04.2018	0443–0483
Altimeter	76741	Valeport VA500	06.04.2021	0484–0543
PAR sensor	1123	PAR-LOG ICSW	12.10.2017	0443–0543

Thermosalinograph sensors – 4 m water intake

Table A2.2. Quality assurance of the thermosalinograph sensors (4 m water intake) showing the serial number, model, calibration date and dates it was used

Type	Serial number	Model	Calibration date	Usage dates
Thermosalinograph	3418	SBE21	07.07.2020	19.11.21–09.12.211
Conductivity sensor	3418	SBE21	07.07.2020	19.11.21–09.12.212
Temperature sensor (Int)	3418	SBE21	07.07.2020	19.11.21–09.12.213
Temperature sensor (Ext)	903	SBE38	05.02.2020	19.11.21–09.12.214
Fluorometer	WS1S-257S	Wet Labs	20.04.2015	19.11.21–03.12.2021
Turbidity sensor	2300402	Turner	2019	Did not work

Notes: TSG 3418 needed maintenance on 4 December 2021. No data were recorded on these days.

Thermosalinograph sensors – 6 m drop keel water intake

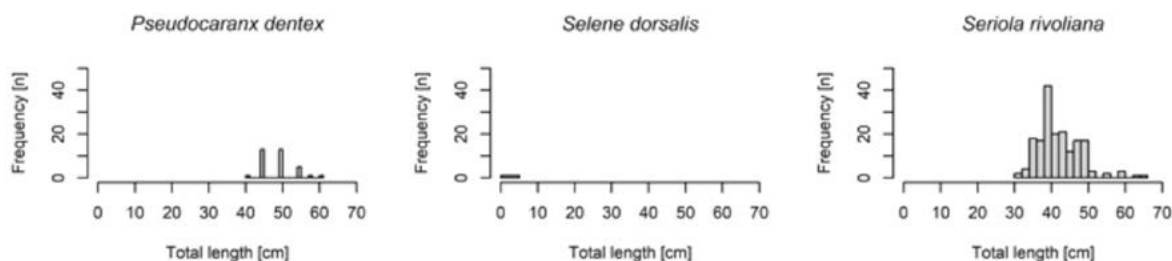
Table A2.3. Quality assurance of the thermosalinograph sensors (6 m water intake) showing the serial number, model, calibration date and dates it was used

Type	Serial number	Model	Calibration date	Usage dates
Thermosalinograph	3419	SBE21	09.03.2021	20.11.21-09.12.211
Conductivity sensor	3419	SBE21	09.03.2021	20.11.21-09.12.212
Temperature sensor (int.)	3419	SBE21	09.03.2021	20.11.21-09.12.213
Temperature sensor (ext.)	880	SBE38	26.11.2020	20.11.21-09.12.21
Fluorometer	WS1S-257S	Wet Labs	20.04.2015	03.12.21-09.12.21
Turbidity sensor	2300402	Turner	2019	Did not work

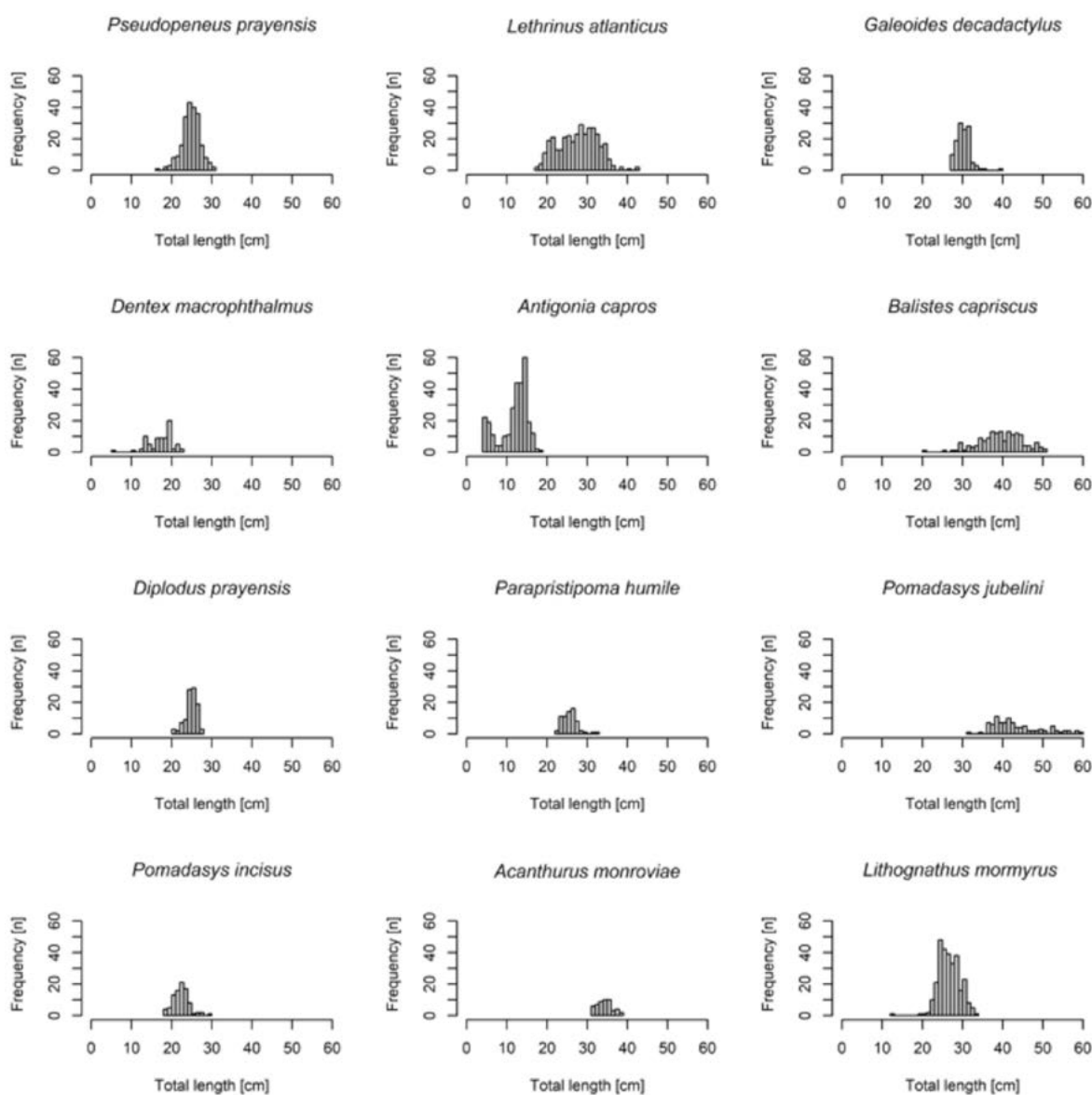
Notes: TSG 3419 needed repair from 23 November to 26 November 2021. No data were recorded on these days.

ANNEX 3. LENGTH DISTRIBUTION OF MAIN SPECIES

Pelagic species



Demersal species



ANNEX 4. MATURITY AND SEX

Table A4.1. Sex and maturity information for priority taxa

Sex	Female	3	4	5	6	7	Male	9	10	11	12	13
Maturity	1	2	3	4	5	Total	1	2	3	4	5	Total
<i>Apristurus laurussonii</i>	nd						2	2				4
<i>Auxis thazard</i>	3	8				11						
<i>Centrophorus uyato</i>			3			3		4				4
<i>Centroscymnus crepidater</i>	nd						1					1
<i>Centroscymnus owstonii</i>	2					2						
<i>Dasyatis</i> spp.	nd							2				2
<i>Leptocharias smithii</i>		2				2		10				10
<i>Lithognathus mormyrus</i>	3	30	60	144		237	2	6	21	120	5	154
<i>Mulloidichthys martinicus</i>	nd							8	3	4		15
<i>Mustelus mustelus</i>	nd						14	66				80
<i>Paragaleus pectoralis</i>	nd						4	6	3			13
<i>Pseudopeneus prayensis</i>	5	30	57	176	5	273	10	28	75	52		165
<i>Raja herwigi</i>	nd						2	8				10
<i>Rhizoprionodon acutus</i>	nd						3	4			5	12
<i>Selar crumenophthalmus</i>	1					1	1					1
<i>Taeniura grabata</i>	nd							4				4
<i>Torpedo</i> spp.	nd						1					1

Notes: nd = not determined.

ANNEX 5. CAPACITY BUILDING

Table A5.1. Overview of the presentations given during the survey

Date	Title	Name
19/11/2021	Safety briefing and daily life on board	Kathrine Michalsen
20/11/2021	Presentation of the Sailing Orders	Kathrine Michalsen
21/11/2021	Briefing and preparations for the work in the various labs	Team leaders
22/11/2021	Briefing and preparations for the work in the various labs continues	Team leaders
23/11/2021	Summing up of the benthos results from 2011	Keider Neves
23/11/2021	Presentation about Cabo Verde (history, daily life, economy, culture)	Anibal Medina and Rui Freitas
27/11/2021	Ex-post evaluation of fishery policy interventions on <i>Decepaterus macarellus</i> stock recovery in northern Cabo Verde	Katelenne Delgado
01/12/2021	Biodiversity in Cabo Verde	Rui Freitas
04/12/2021	Ocean circulation and hydrodynamic connectivity between the Cabo Verde Islands	Anibal Medina
06/12/2021	Guided tour on the bridge – Captain	Aron Håpoldøy
07/12/2021	Presentation of IMar and the institute's statistics	Vito Ramos
08/12/2021	Guided tour in the engine room	Jostein Møvik
09/12/2021	Presentation of IEO	Francisca A. Salmeron Jimenez
10/12/2021	Presentation of otolith work for <i>Spicara melanurus</i>	Alciany Luz
11/12/2021	Guided tour in the acoustic lab	Olaf Sørås
12/12/2021	Presentation of the CVOO	Péricles Silva
13/12/2021	Summing-up meeting with presentations of work on board: plankton lab; CTD lab; physical oceanography; fish lab; acoustics; comparison with the survey results from 2011	Various teams
14/12/2021	Survey report	All

ANNEX 6. SAMPLE OVERVIEW

Table A6.1: Samples collected, preservation and follow-up work

Sample number	Sample type	Analysis	Preservation	Quantity	Port of offloading	Transport method	Receiving country	Receiving institution	Contact person	Status by 17/02
2021407-001	Water samples	Dissolved inorganic carbon/total alkalinity	None	14	Las Palmas	Private to IEO, further arrangements by FAO	Cabo Verde	Imar	Pericles Silva (pericles.silva@oscm.cv)	Will be sent from LP to CV by fishing vessel. Will be analysed in Kiel. Results will be shared with the Nansen programme afterwards (Pericles to David)
2021407-002	Water samples	Nutrients	Frozen	432	On DFN	N/A	N/A	IMR	David. Cervantes@hi.no	To be analysed on DFN by 23.02. Will not be included in the survey report
2021407-003	Phytoplankton	Phytoplankton community composition	Lugol solution	5x100 ml amber glass bottles	Las Palmas	Air freight	Norway	IMR, Algae Laboratory v / Lars Naustvoll	Lars Naustvoll (larsjn@hi.no)	Samples have been received. Results should be communicated to Stamatina and Vito
2021407-004	Zooplankton – WP2	Biomass estimation	Dried and then frozen	101 x aluminum trays	Las Palmas	Sea freight	Norway	IMR, Bergen	Stamatina Isari	Samples have been analysed at the zooplankton lab in Bergen. Maps can be included in the survey report
2021407-005	Zooplankton – WP2	Community identification	4% formaldehyde	35 x 100 ml bottles	Las Palmas	Private to IEO, further arrangements by FAO	Spain/Cabo Verde	Taken care of by IEO, then reshipped to Imar	Elizandro	Waiting for the samples to arrive from Tenerife
2021407-006	Zooplankton – Midi	Community identification	4% formaldehyde	5 x 100 ml bottles	Las Palmas	Private to IEO, further arrangements by FAO	Spain/Cabo Verde	Taken care of by IEO, then reshipped to Imar	Elizandro	Waiting for the samples to arrive from Tenerife
2021407-007	Zooplankton – Mammoth 180 µm (Nola)	Genetics	Absolute ethanol	26 (14x100 ml + 12x250 ml bottles)	Las Palmas	Air freight	Norway	IMR, Bergen	Stamatina Isari	Samples stored in Bergen. Date for analysis will depend on decision on future work
2021407-008	Zooplankton – Mammoth 180 µm (Nola)	Community identification	4% formaldehyde	26 (19x100 ml + 7x250 ml)	Las Palmas	Air freight	Norway	IMR, Bergen	Stamatina Isari	Same as above
2021407-009	Ichthyoplankton – Mammoth 405 µm (Nola)	Community identification	Absolute ethanol	161 (152x100 ml + 9x250 ml)	Las Palmas	Air freight	Norway	IMR, Bergen	Stamatina Isari	Same as above

Sample number	Sample type	Analysis	Preservation	Quantity	Port of offloading	Transport method	Receiving country	Receiving institution	Contact person	Status by 17/02
2021407-010	Ichthyoplankton – Mammoth 405 µm (replacement Bongo)	Community identification	Absolute ethanol	3 (3x250 ml)	Las Palmas	Air freight	Norway	IMR, Bergen	Stamatina Isari	Same as above
2021407-011	Ichthyoplankton – Mammoth 405 µm (replacement Bongo)	Community identification	4% formaldehyde	3 (2x250 ml + 1x500 ml)	Las Palmas	Air freight	Norway	IMR, Bergen	Stamatina Isari	Same as above
2021407-012	Ichthyoplankton – Bongo V	Community identification	4% formaldehyde	32 (8x100 ml + 21x250 ml + 3x500 ml)	Las Palmas	Air freight	Norway	IMR, Bergen	Stamatina Isari	Same as above
2021407-013	Ichthyoplankton – Bongo H	Community identification	Absolute ethanol	32 (8x100 ml + 22x250 ml + 2x500 ml)	Las Palmas	Air freight	Norway	IMR, Bergen	Stamatina Isari	Same as above
2021407-014	Ichthyoplankton – Bongo H	Species identification, genetics	Absolute ethanol	6x20 ml scintillation vials	Las Palmas	Air freight	Norway	IMR, Bergen	Stamatina Isari	Same as above
2021407-015	Ichthyoplankton – Manta	Species identification, genetics	Absolute ethanol	26x20 ml scintillation vials	Las Palmas	Air freight	Norway	IMR, Bergen	Stamatina Isari	Same as above
2021407-016	Neuston	Community identification	Absolute ethanol	26 (17x100 ml + 8x250 ml + 1x500 ml)	Las Palmas	Air freight	Norway	IMR, Bergen	Stamatina Isari	Same as above
2021407-017	Microplastics	Abundance and chemical composition	In freshwater	5x2 ml eppendorf	Las Palmas	Air freight	Norway	IMR, Bergen	Bjørn Einar Grøsvik	All samples sorted on board and there was only one item found. No further analysis
2021407-018	Microplastics	Detection of microplastics in filtered seawater in plankton lab	Seawater	2x500 ml	Las Palmas	Air freight	Norway	IMR, Bergen	Bjørn Einar Grøsvik	Same as above
2021407-019	Whole fish	Taxonomic identification – problematic spp. according to list	Frozen	11 ziplock bags	Las Palmas	Road/private to IEO, further arrangements by FAO	Spain	IEO	Pepe/Peter	Stored at IEO in Tenerife. Will be used for post-survey workshop. Date for workshop is discussed. Results of the work will be communicated to IMR for updates of the Nansen db (Peter)

Sample number	Sample type	Analysis	Preservation	Quantity	Port of offloading	Transport method	Receiving country	Receiving institution	Contact person	Status by 17/02
2021407-020	Whole fish	Taxonomic identification – unknown/problematic spp.	Frozen/10% formalin	67 ziplock bags	Las Palmas	Private to IEO, further arrangements by FAO	Spain	IEO	Pepe	Will be used as additional material for the workshop described above. Samples will be stored at IEO
2021407-021	Whole fish	Taxonomic identification – Post survey training course	Frozen	84 ziplock bags	Las Palmas	Private to IEO, further arrangements by FAO	Spain	IEO	Pepe	Will be used as additional material for the workshop described above. Samples will be stored at IEO
2021407-022	Whole fish	Taxonomic identification – IMR training course	Frozen	43 ziplock bags	Las Palmas	Sea freight	Norway	IMR, Bergen	Sarah/Ines/Diana	Shipped from LP 04.05 to Bergen. No date for the course is set at the moment
2021407-023	Whole fish	Nutrition and food safety	Frozen	5 ziplock bags	Las Palmas	Sea freight	Norway	IMR, Bergen	Mariann Kjellevoid	Shipped from LP 04.05. Will be analysed as soon as possible and communicated through deliveries to the RC and the science Plan
2021407-024	Whole fish	Museum collection	20% formalin	50 ziplock bags	Las Palmas	Private to IEO, further arrangements by FAO	Spain/Cabo Verde	Taken care of by IEO, then reshipped to IMar	Alciany Luz	Sent from LP to CV by fishing vessel. Will them be stored at IMar as a museum collection
2021407-025	Invertebrates	Taxonomic identification/museum collection	4% buffered formalin		Las Palmas	Private to IEO, further arrangements by FAO	Spain/Cabo Verde	Taken care of by IEO, then reshipped to IMar	Keider Neves	Sent from LP to CV by fishing vessel. Will be identified as soon as possible and will be stored at IMar. When the final id is done Keider will communicate this with Diana
2021407-025	Sargassum	Genetics/morphometrics	Frozen	3 ziplock bags + 2x20 ml scintillation vials (ethanol) (from plankton sampling)	Las Palmas	Sea freight	Norway	Min of Fisheries and Aquaculture Development	Stamatina Isari/ Maame Esi Bordah Quayson	Shipped from LP 04.05 to Bergen. Then reshipped to Ghana

ANNEX 7. DATA COLLECTED AND OVERVIEW OF PROCEDURES FOR DATA HAND-OVER TO PARTNERS

Table A7.1. Data collected and overview of procedures for data hand-over to partners (all collected data will be stored on the IMR server)

Survey no 2021402	Data	After the survey	At the post survey meeting	Upon request	Not collected/ stored	Analyzed by partner country	Analyzed through the Science Plan
Data types							
Acoustic data	EK80 narrowband (CW)		x				
Acoustic data	EK80			x			
Acoustic data	MS70				x		
Acoustic data	ME70				x		
Acoustic data	SU90			x			
Acoustic data	SH90				x		
Acoustic data	SBP300				x		
Acoustic data	EM302				x		
Acoustic data	EM710				x		
Physics	CTD probe	x					
Physics	CTD underway				x		
Physics	ADCP 75kHz				x		
Physics	ADCP 150kHz	x					
Physics	LADCP				x		
Physics	Thermosalinograph	x					
Physics	Nutrients		x				
Physics	pH	x					
Physics	Total alkalinity	x					
Physics	PCO ₂	x					
Physics	Chlorophyll	x					
Biology	Trawl catch data	x	x				
Biology	Zooplankton biomass		x				
Biology	Phytoplankton		x				
Pollution	Microplastics						x
Observation platforms	VAMS				x		
Observation platforms	WBAT				x		
Observation platforms	Deep vision				x		
Observation platforms	Scanmar			x			
Observation platforms	Activity diary (cruise logger)	x					
Observation platforms	Nansis survey backup			x			

